



TxOne Stellar Bootcamp

Stellar 3.x

TXOne Global Training Team

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Wifi:

Agenda

- Industry Best Practices, OT Challenges, TxOne Zero Trust for OT (recap)
- Installation and Sizing
- **Stellar Installation and upgrade labs**
- Stellar Main Features
- **Stellar Configuration labs**
- Troubleshooting (Stellar / Edge / Element)

(The windows VMs for this training are provided without a MS license You need to add your own Microsoft license)

Industry Best Practices, OT Challenges, TxOne Zero Trust for OT



Installation and sizing



Download the StellarOne VM from <https://my.txone.com>

txOne
networks

MyTXOne

Overview

Resources

Product License

Software

Pattern

Document

Product Line
Endpoint Protection

Product SKU
All

Component Type
All

Reset Filters

Software Installer

StellarOne

2023-07-07

StellarOne HyperV Image

2.2.1149

File Size: 4GB

Download

StellarProtect StellarProtect Legacy Mode

2023-07-08

StellarProtect Installer

2.2

File Size: 259MB

Download

StellarOne

2023-07-08

StellarOne VMWare Image

2.2.1149

File Size: 4GB

Download

Software Patch

StellarProtect Legacy Mode

2023-07-07

StellarProtect Legacy Mode Patch

1.5

File Size: 70MB

Download

StellarOne

2023-07-07

StellarOne Patch Image

2.2.1149

File Size: 2GB

Download

StellarProtect

2023-07-08

StellarProtect Patch

2.2

File Size: 80MB

Download

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StellarOne Installation

- **Installation into a Virtual Machine via VMware ESXi**
 - Import StellarOne OVA to ESXi (version 6 or above)
 - Configure VM Resources
 - Attach Second Disk at least **50GB [Important!]**
- **The Device is Now Ready for Configuration**
 - Access the StellarOne CLI
 - Configure Network Settings: `iface update eth0 --method static --address 192.168.23.40 --netmask 255.255.255.0 --gateway 192.168.23.2`
 - Setup DNS
 - `resolv mode custom`
 - `resolv add 8.8.8.8`
 - ping command to test
 - `ping www.google.com` or `dx au` to test connectivity with TxOne update server
 - Check deployment:
 - `env ls`

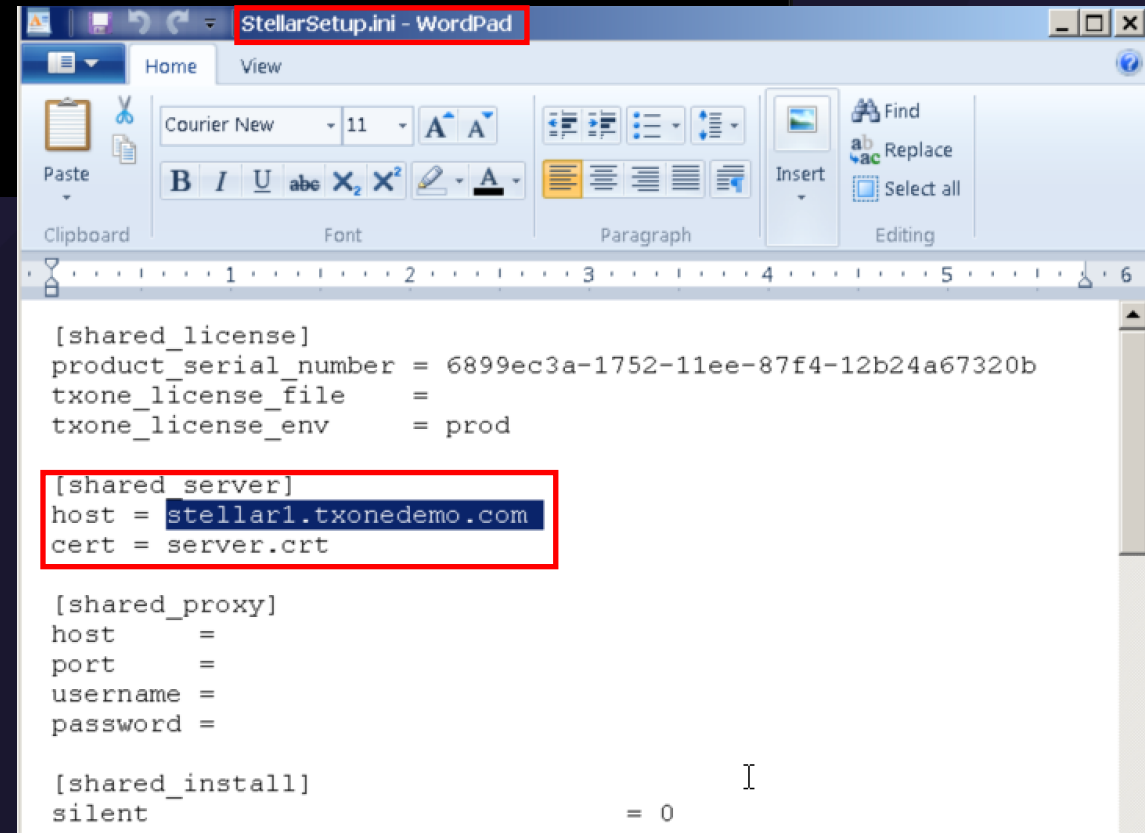
StellarOne boot process:

env ls

1. INIT: CHECK SYSTEM INTERFACE'S IP
2. INIT: CHECK DOCKER INTERFACE'S IP
3. INIT: GET SYSTEM BEST RESOURCE
4. INIT: SYSTEM START
5. INIT: DB INITIALIZATION
6. RUNNING

Deploying Stellar by DNS name ("advertise-addr")

```
(root) $ env advertise-addr mystellarone.local
vShell: unable to resolve address mystellarone.local or it's an invalid IP. Try to add --force after
the address to force the setup.
(root) $ env advertise-addr mystellarone.local --force
Successfully set up advertise address: mystellarone.local.
Type 'service reload' to restart the service for reloading the specified configuration.
(root) $ service reload
Start to reload services...
(root) $ env ls
Hostname: ODC
Status: RUNNING
```



StellarOne Installation - Sizing

- StellarOne CPU and Memory

Agents	CPU	Memory
500	4 cores	8 GB
1,000	4 cores	16 GB
5,000	8 cores	16 GB
10,00	8 cores	16 GB
15,000	8 cores	16 GB
20,000	8 cores	16 GB
30,000	10 cores	24 GB

- StellarOne Second harddisk

#of Logs	Disk
5,000,000	50 GB
10,000,000	150 GB
50,000,000	300 GB
100,000,000	500 GB

txso_3.0_sp1_ig.pdf

StellarOne upgrades

Before upgrading to StellarOne 3.3: first upgrade to **version 3.2.5**.



The screenshot shows the StellarOne Administration web interface. The top navigation bar includes the StellarOne logo, a user profile for 'admin (admin)', and the txOne networks logo. The main navigation menu has tabs for Dashboard, Agents, Logs, Administration (selected), and About. The 'FIRMWARE' section is active, displaying a table with firmware details. An error message is shown at the top of the content area: 'An unexpected error occurred. If the issue persists, contact your support provider.' The table lists the current version as 3.1.3142, with a release date of 2024-08-28T05:23:09+08:00. The description field also contains '3.1.3142'. Below the table is an 'Import' button with an upload icon.

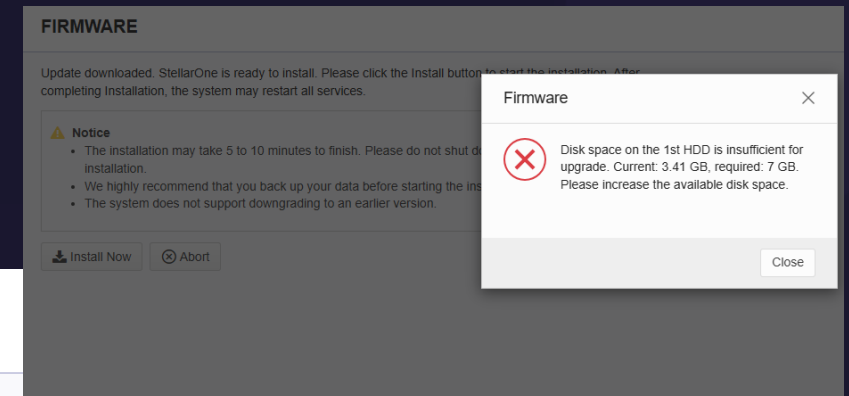
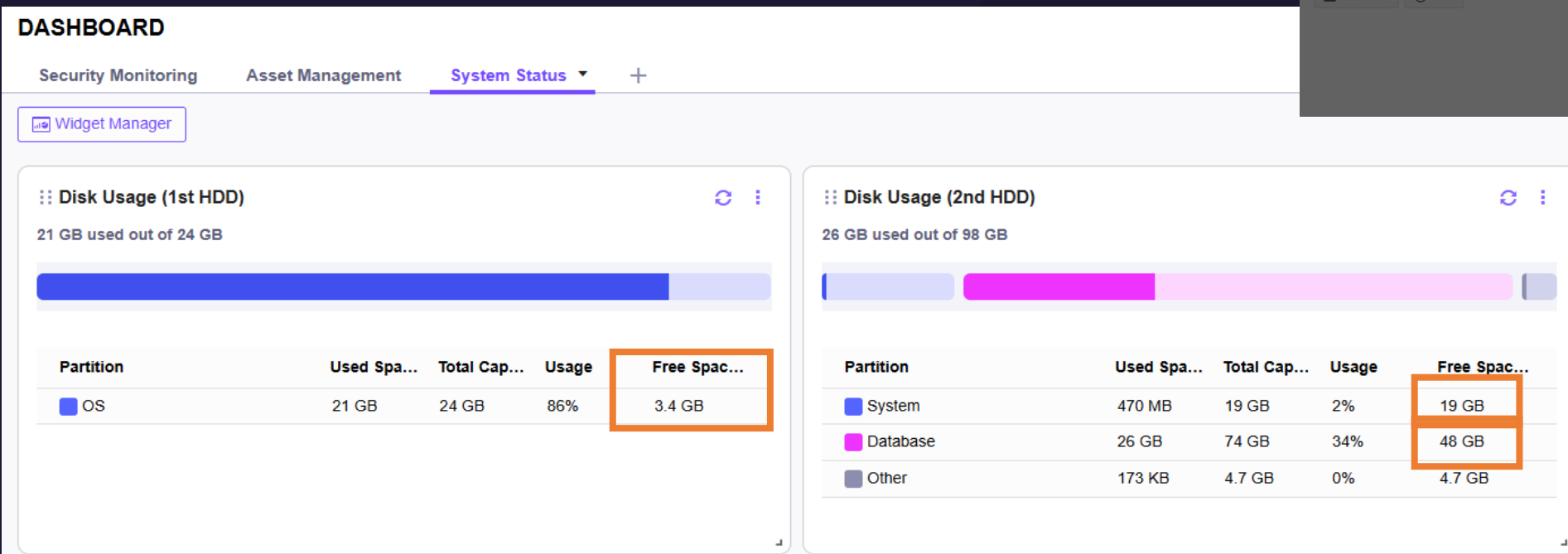
Version	Release Date	Description
3.1.3142	2024-08-28T05:23:09+08:00	3.1.3142

[Import](#)

Direct upgrade to Stellar 3.3 from older versions is not supported, first upgrade to version 3.2.5 (3.2 Service Pack 1 Patch 2)

Disk Space Requirements for upgrade

- **1st HDD (OS disk)**
 - At least **15 GB** of free space is required.
 - Total space : **35GB** is required.
- **2nd HDD**
 - System partition: At least **5GB** of free space is required.
 - Database partition: At least **30%** of free space required



Lab exercises – Stellar installation

Stellar Installation Hands-on Labs

(301StellarLabGuideTxOneSpecialistV3.x.....PDF)

- Lab1: StellarOne VM Deployment
Install: **TXOne-S1-3.1.1264.ova**;
it is in the **VMs and VMs_Updates_and_Patches** folder,
then upgrade to **TXOne-S1-acus.fw_3.2.5035.acf** (in the labkit folder),
then **upgrade to TXOne-S1-acus.fw_3.3.1431.acf** (also in the labkit folder)
- Lab2: Basic configuration of StellarOne
- Lab3: Stellar Agent Management using StellarOne
- Lab4: StellarProtect Agent installation
- (Lab5: StellarProtect Agent Silent installation)
- Lab6: StellarOne and StellarProtect upgrade
(use **TXOne-S1-acus.fw_3.2.4123.acf**
and **txsp-3.2-agent-b1191-patch-en.zip**)

You can ignore the other labs in this labguide.
They are repeated, but with more detail in the 401 document

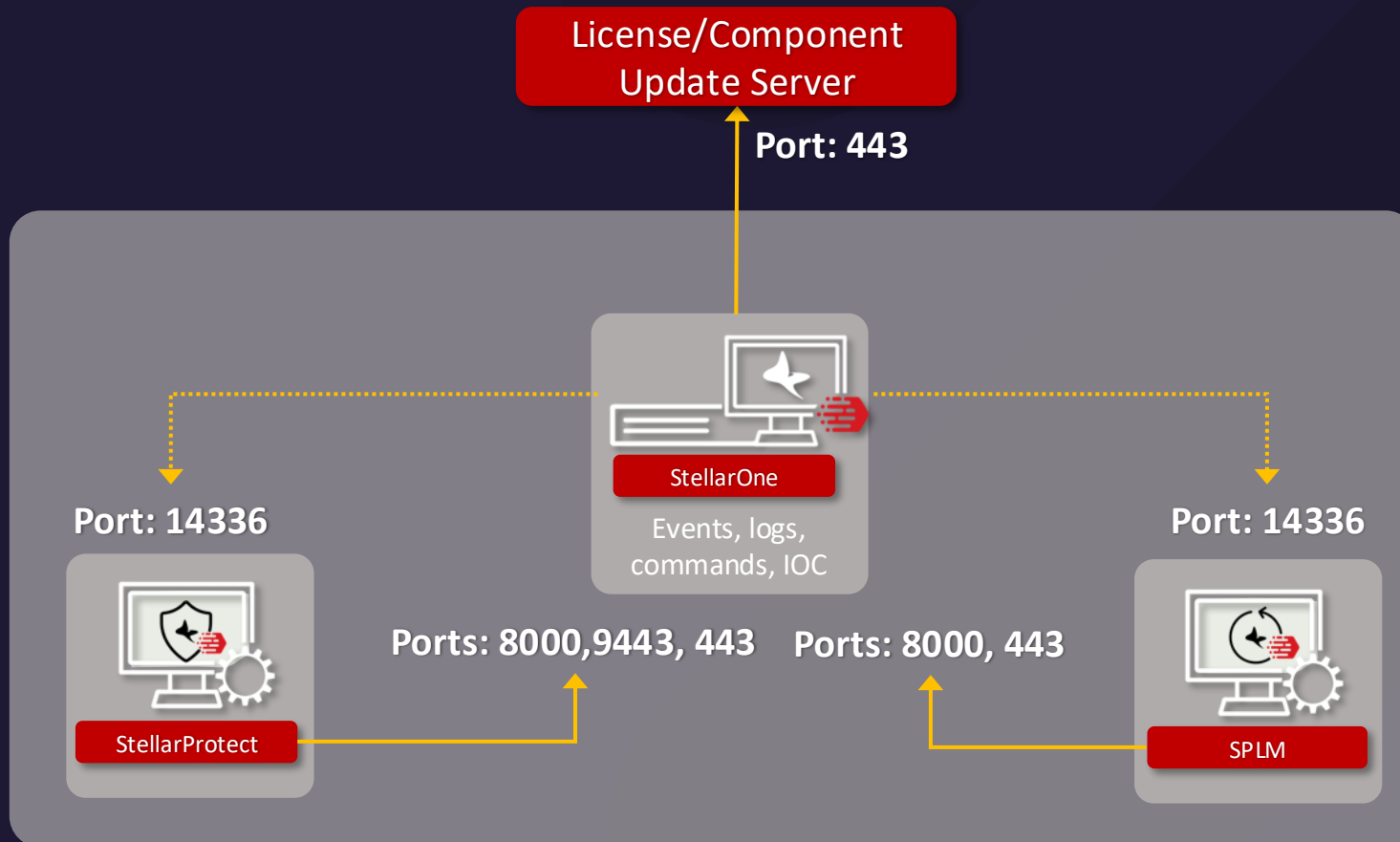
For the Stellar labs, please connect **all VMs via the NAT interface.**

Do NOT UPGRADE VMWARE TOOLS

The licenses are in the file
txonekeys2602xx.md

StellarOne Installation – Ports and URLs

- Firewalls or proxies may impact the communication between StellarOne and the Stellar Agents or between StellarOne and the Update Center



Stellar 2.1 and later use the TXOne Activation Servers (!)

<https://odc.cs.txone-networks.com>

<https://ttau.cs.txone.com/protect/>

<https://ttau.cs.txone.com/enforce/>

Stellar 2.0 and previous version

<https://licenseupdate.trendmicro.com>

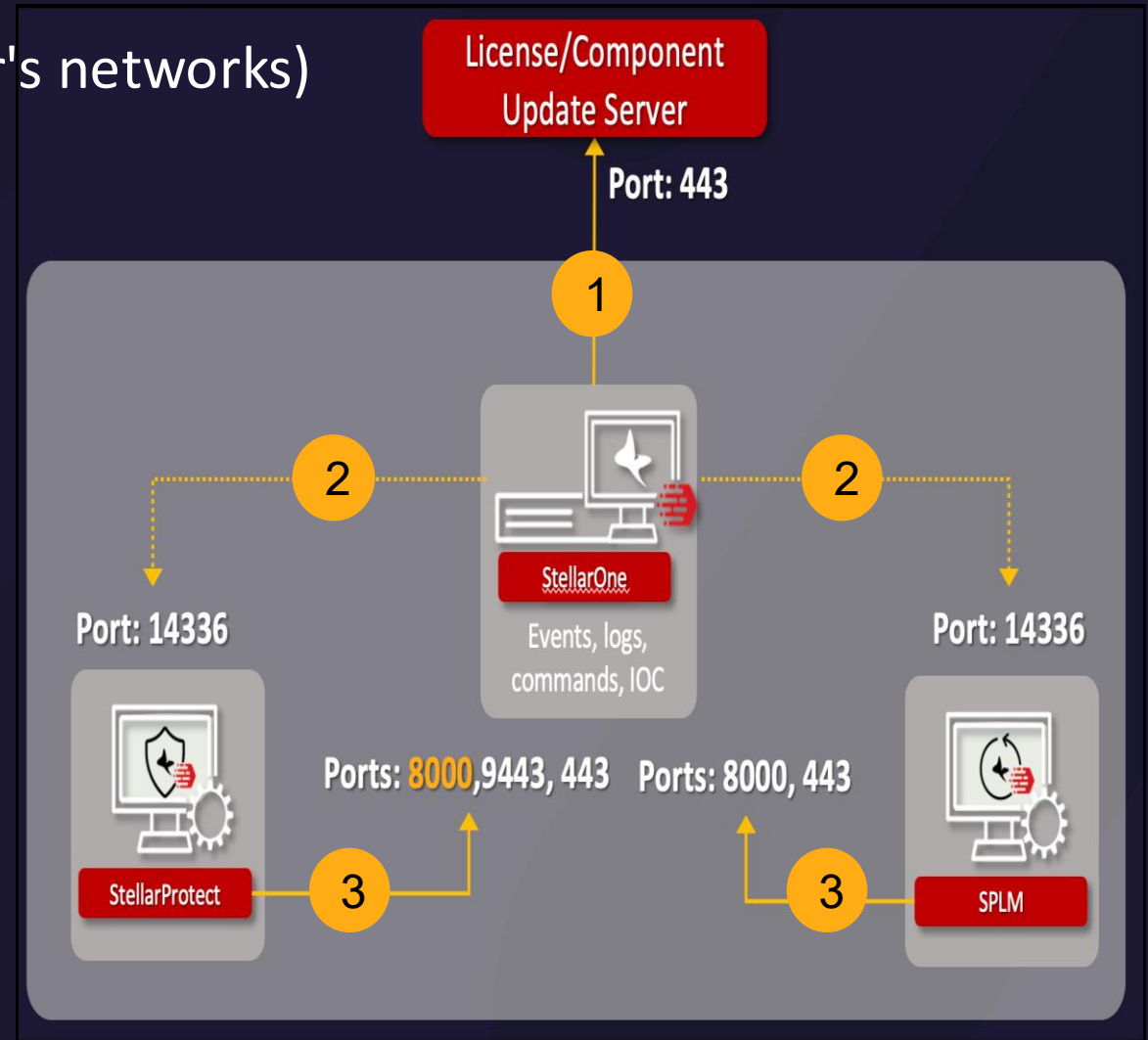
<https://txsp-p.activeupdate.trendmicro.com/activeupdate>

<https://txse-p.activeupdate.trendmicro.com/activeupdate>

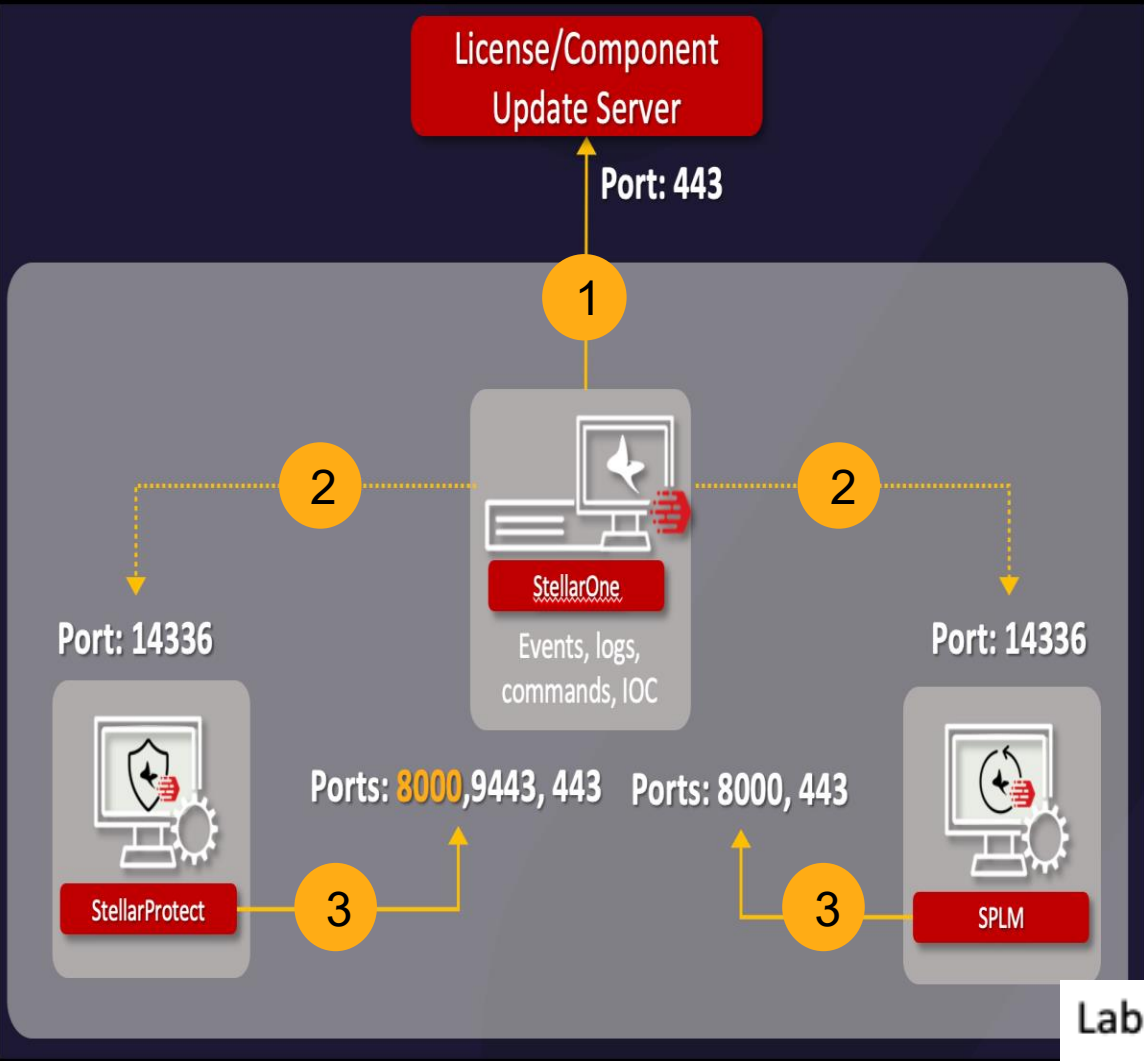
Configure Proxies (if required by customer's networks)

- Three different proxy locations possible:
 1. Proxy from StellarOne to the Internet
 2. Proxy from StellarOne (server) to Agents
 3. Proxy from Agents to Stellarone (this setting will be baked-in to the agent **when it is downloaded from your StellarOne after the Proxy was configured**)

Note: Configure Proxies in StellarOne before downloading the Agent Packages (!)



Configure Proxies



The screenshot shows the "Proxy" configuration page in the StellarOne Administration interface. The page has a navigation bar with "Dashboard", "Agents", "Logs", "Administration", and "About". The "Proxy" section is titled "Proxy Settings for StellarOne to internet" and "Proxy Settings for StellarOne to Agent communications". The "Proxy Settings for StellarOne to internet" section has a toggle switch labeled "1" and "Proxy Settings for StellarOne to internet". The "Proxy Settings for StellarOne to Agent communications" section has a toggle switch labeled "2" and "Proxy Settings for StellarOne to Agent communications". Below this, there are radio buttons for "HTTPS" and "HTTP" (selected), and input fields for "Server Address*" and "Port*". There is also a toggle switch labeled "3" and "Proxy server requires authentication". At the bottom, there are "Save" and "Cancel" buttons.

Lab19: Using a proxy for StellarOne Updates	
Lab20: Using a proxy for Agent to StellarOne communication	

Stellar Agent - Supported Operation Systems

- Operating System requirements are listed in the table below
 - Some security features only work on certain operation systems, please refer to the administration guide for details

Client OS	Server OS	Windows embedded	Linux
• Windows 2000 (SP4) [Professional] (32bit) • Windows XP (SP1/SP2/SP3) [Professional/Professional for Embedded Systems] (32bit) • Windows Vista (NoSP/SP1/SP2) [Business/Enterprise/Ultimate] (32bit) • Windows 7 (NoSP/SP1) [Professional/Enterprise/Ultimate/Professional for Embedded Systems/Ultimate for Embedded Systems] (32/64bit) • Windows 8 (NoSP) [Pro/Enterprise] (32/64bit) • Windows 8.1 (NoSP) [Pro/Enterprise/with Bing] (32/64bit) • Windows 10 [Pro/Enterprise/IoT Enterprise] (32/64bit), LTSC 2015, Anniversary Update, LTSC 2016, Creators Update, Fall Creators Update, April 2018 Update, October 2018 Update*, LTSC 2019, May 2019 Update, November 2019 Update, May 2020 Update, October 2020 Update, May 2021 Update, November 2021 Update, LTSC 2021, 2022 Update • Windows 11 (NoSP) [Pro/Enterprise] (64bit) 2022 Update, 2023 Update • Windows Embedded POSReady 2009 (32bit) • Windows Embedded Standard 7 (NoSP/SP1) (32/64bit) • Windows Embedded POSReady 7 (NoSP) (32/64bit) • Windows Embedded 8 Standard (NoSP) (32/64bit) • Windows Embedded 8 Industry (NoSP) [Pro/Enterprise] (32/64bit) • Windows Embedded 8.1 Industry (NoSP) [Pro/Enterprise/Sideload] (32/64bit)	- Windows 2000 Server SP4* (32-bit) - Windows Server 2003 SP1/SP2 (32-bit) - Windows Server 2003 R2 No-SP/SP2 (Standard/Enterprise/Storage) (32-bit) - Windows Server 2008 SP1/SP2 (32-bit and 64-bit) - Windows Server 2008 R2 No-SP/SP1 (64-bit) - Windows Server 2012 No-SP (64-bit) - Windows Server 2012 R2 No-SP (64-bit) - Windows Server 2016 (Standard) (64-bit) - Windows Server 2019 (Standard) (64-bit) - Windows Server 2022 (Standard) (64-bit) - Windows Storage Server - Windows Storage Server 2012 Standard (64-bit) - Windows Storage Server 2012 R2 Standard (64-bit) - Windows Storage Server 2016	- Windows (Standard) XP Embedded SP1*/SP2 (32-bit) - Windows Embedded Standard 2009 (32-bit) - Windows Embedded Standard 7 (32-bit and 64-bit) - Windows Embedded Standard 8 (32-bit and 64-bit) - Windows Embedded 8 Standard No-SP (32-bit and 64-bit) - Windows Embedded Standard 8.1 (32-bit and 64-bit) - Windows Embedded 8.1 Standard (Professional/Industry Pro) (32-bit and 64-bit) - Windows Embedded POSReady: - Windows Embedded POSReady (32-bit) - Windows Embedded POSReady 2009 (32-bit) - Windows Embedded POSReady 7 (32-bit and 64-bit) - Windows Embedded Enterprise: - Windows Embedded Enterprise XP SP1*/SP2/SP3 (32-bit) - Windows Embedded Enterprise Vista (32-bit) - Windows Embedded Enterprise 7 (32-bit and 64-bit) - Windows Embedded Server: - Windows Embedded Server 2003 SP1/SP2 (32-bit) - Windows Embedded Server 2003 R2 (32-bit) - Windows Embedded Server 2008 (32-bit and 64-bit) - Windows Embedded Server 2008 R2 (64-bit) - Windows Embedded Server 2012 (64-bit) - Windows Embedded Server 2012 R2 (64-bit)	- CentOS 6 - CentOS 7 - CentOS 8 - CentOS 9 - Debian 10 - Debian 11 - Red Hat Enterprise Linux 6 - Red Hat Enterprise Linux 7 - Red Hat Enterprise Linux 8 - Red Hat Enterprise Linux 9 - Rocky 8.4 - Rocky 9 - Rocky 10 - Ubuntu Linux 18 - Ubuntu Linux 20 - Ubuntu Linux 22

Not-Supported Operating Systems – StellarProtect Legacy Mode

- The following windows versions are **not** supported by SPLM
- Most of them require an additional Service Pack
 - Windows NT
 - Windows 2000 all versions earlier than SP4
 - Windows XP earlier than SP1
 - Windows Server 2003 earlier than SP1
- Any windows Starter/Home editions
- Windows XP 64bit
- ~~Windows Server Core (2012,2016,2019,2025)~~ *Is supported from Stellar 3.3 onwards*

Agent installation (windows)

- Install StellarProtect or StellarProtect (Legacy Mode) agent directly via installer (StellarSetup.exe) interactively and do the initial setup
 - Agent Password
 - Device Profile
 - Pre-Scan / Approved List Creation



Agent-Installer-en.zip



```
[shared_license]
product_serial_number = 849e64e4-
txone_license_file    =
txone_license_env     = prod

[shared_server]
host = 192.168.23.19 5
cert = server.crt

[shared_proxy] 6
host =
port =
username =
password =

[shared_install] 4
silent = 0
password =
user_password =
enable_shell_integration = 1

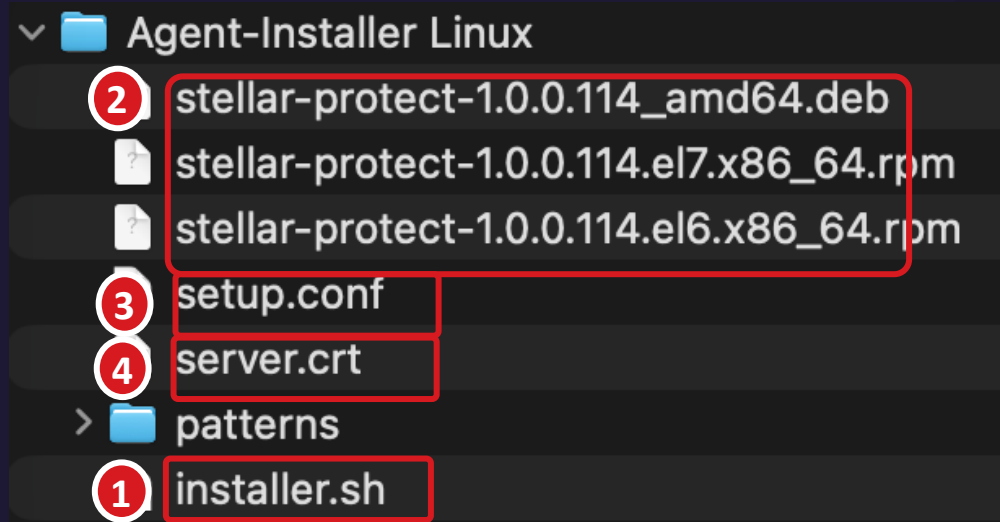
[shared_debug]
enable_debug_log = 0
enable_engine_debug_log = 0
```

Agent installation (Linux)

1. Run **installer.sh**
2. Installer checks the platform (deb or el6/7 rpm)
3. Installer reads **setup.conf**
4. Agent uses **server.crt** to register to StellarOne



Agent-
Installer.tgz



setup.conf

```
asset:
  description: null
  location: null
  model: null
  vendor: null
license:
  serial_number: 547e3fae-cc35-11f0-8d5e-0affc
proxy:
  host: null
  password: null
  port: null
  username: null
server:
  cert: server.crt
  ip: 172.31.27.177
  port: 8000
```

Stellar Agent *Silent* Installation

Install the StellarProtect or StellarProtect (Legacy Mode) agent silently without GUI interaction

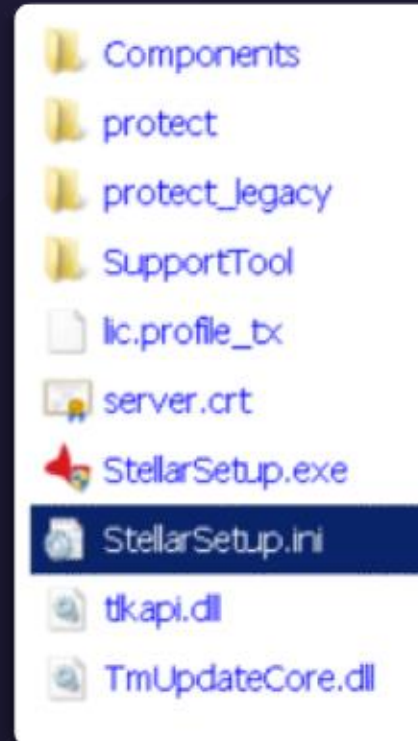
- Command: `StellarSetup.exe -s`
- Update “StellarSetup.ini” before install

```
[shared_install]
```

```
silent = 1
```

```
password = somePassword
```

- The password is in clear-text in the StellarSetup.ini file, but as soon as the agent synchronizes with StellarOne, this password will be overwritten by the password that is defined in the Agent Policy



```
[shared_license]
product_serial_number = 849e64e4-
txone_license_file    =
txone_license_env      = prod

[shared_server]
host = 192.168.23.19
cert = server.crt

[shared_proxy]
host      =
port      =
username  =
password  =

[shared_install]
silent          = 1
password        =
user_password   =
enable_shell_integration = 1

[shared_debug]
enable_debug_log      = 0
enable_engine_debug_log = 0
```

Organize Agents: Create Groups (for agents/endpoints)

admin (admin) txOne networks

DashboardAgentsLogsAdministrationAbout

AGENTS All (2)

+ Add GroupOrganizeProtectionUpdateImport/Export

1 / 1 < > [table icon] [refresh icon]

☐ Name ↑	IP Address	Protection	Policy Inheritance	Approved List	Agent Version	Last Connectio...	Function Type	Action:
☐ FAB12 (0)	-	-	Customized	-	-	-	-	⋮
☐ FAB18 (0)	-	-	Inherited	-	-	-	-	⋮
☐ QQ-STELLAR	192.168.23.24		● Inherited	39682	2.2.1044	2023-07-12T17:5...	StellarProtect	⋮
☐ QQ-STELLARLM	192.168.23.28		● Inherited	39102	1.5.1053	2023-07-12T17:5...	StellarProtect (Legacy Mode)	⋮

"Policy Inheritance" and "Inheritance Exceptions"

stellarOne

admin (admin)

Dashboard Agents Logs Administration About

← FAB12 (0)

Function Type: StellarProtect

Policy Inheritance: ☒ Inherit from parent group: All

Self-management: ☐

Policy

Application Lockdown

When Application Lockdown is turned on, the endpoint will only be able to access applications in the Approved List. If required files are not in the Approved list, there is a risk that this endpoint will be able to restart or log on.

- ☐ Detect
When an application not in the Approved List launches, it is allowed and the user will receive a notification.
- ☐ Enforce
When an application not in the Approved List launches, it is blocked and the user will receive a notification.
- ☒ Disable
Application Lockdown mode is disabled.

Multi-Method Threat Prevention

OT root of trust and advanced threat scan secure the assets without interruption on the operation.

- ☒ Real-Time Protection
 - ☒ Real-Time Scan

Custom Exceptions StellarProtect

The selected options will become customizable. The unselected options remain inherited from the parent group.

Application Lockdown

- ☐ Excluded Paths
- ☐ Trusted Hash Values

Multi-Method Threat Prevention

- ☐ Real-Time Scan Exceptions
- ☐ Scheduled Scan Exceptions

OT Application Safeguard

- ☐ File/Folder
- ☐ Authorized Processes

Operations Behavior Anomaly Detection

- ☐ Policy-based Watchlist
- ☐ Policy-based Approved Login Accounts
- ☐ Policy-based Approved Applications
- ☐ Policy-based Approved Network Access

External Device Management

- ☐ Trusted USB Device

Stellar Policy Inheritance

Global Policy

Group policy

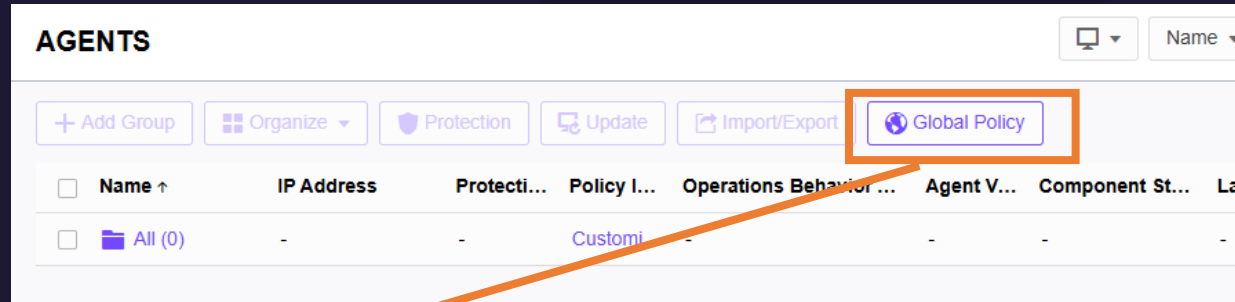
Agent Policy

Global Policy

The Global Policy only contains:

- **Trusted Certificates**
- **User-Defined Suspicious Objects**

It applies these settings automatically to all groups and agents, **except agents in Self-Management mode.**



← GLOBAL POLICY

Function Type: StellarProtect

Policy

Save Changes Discard Changes

User-Defined Suspicious Objects (0)

Update new threat information to protect against suspicious objects not yet identified on your network.

+ Add

Hash/File Path ↓	Type	Notes	Action...
No data to display			

Trusted Certificates (0)

The trusted certificates defined by users will be bypassed during scanning and will not be blocked by Application Lockdown.

Import

Issued To ↓	Issued By	Hash	Action...
No data to display			

"Policy Inheritance" and "Inheritance Exceptions"

The screenshot displays the stellarOne management console. At the top, the navigation bar includes 'Dashboard', 'Agents', 'Logs', 'Administration', and 'About'. The user 'admin (admin)' is logged in. The main content area shows the configuration for a policy named 'FAB12 (0)'. The 'Function Type' is set to 'StellarProtect'. The 'Policy Inheritance' section is highlighted with a red box, showing a toggle switch set to 'Inherit from parent group: All'. The 'Self-management' toggle is also visible. Below this, the 'Policy' dropdown is expanded, showing 'Custom Exceptions' with a 'StellarProtect' button, also highlighted with a red box. The 'Custom Exceptions' panel is open, displaying a list of settings that can be customized or inherited. The settings include:

- Application Lockdown**
 - ☐ Detect: When an application not in the Approved List launches, it is allowed and the user will receive notification.
 - ☐ Enforce: When an application not in the Approved List launches, it is blocked and the user will receive notification.
 - ☒ Disable: Application Lockdown mode is disabled.
- Multi-Method Threat Prevention**
 - ☒ Real-Time Protection
 - ☒ Real-Time Scan
- Custom Exceptions** (StellarProtect)
 - The selected options will become customizable. The unselected options remain inherited from the parent group.
 - Application Lockdown**
 - ☐ Excluded Paths
 - ☐ Trusted Hash Values
 - Multi-Method Threat Prevention**
 - ☐ Real-Time Scan Exceptions
 - ☐ Scheduled Scan Exceptions
 - OT Application Safeguard**
 - ☐ File/Folder
 - ☐ Authorized Processes
 - Operations Behavior Anomaly Detection**
 - ☐ Policy-based Watchlist
 - ☐ Policy-based Approved Login Accounts
 - ☐ Policy-based Approved Applications
 - ☐ Policy-based Approved NetworkAccess
 - External Device Management**
 - ☐ Trusted USB Device

Agents can be configured from StellarOne or from the Agent Console

The image displays two interfaces for configuring agents. The top interface is the StellarOne web dashboard, showing the 'Agents' tab for a specific agent named 'QQ-STELLAR'. It includes settings for 'Policy Inheritance' (set to 'Inherit from parent group: FAB12') and 'Self-management' (turned off). The 'Policy' tab is active, showing 'Application Lockdown' settings. The bottom interface is the TXOne StellarProtect agent console, which mirrors the web settings. A yellow callout bubble points to the 'Self-management' toggle in the StellarOne interface, stating: 'If this is not turned on, then any changes made **LOCALLY** on the agent **WILL BE REVERTED** at the next agent synchronization'. A red box highlights the 'Application Lockdown' section in both interfaces, showing options to 'Detect', 'Enforce', or 'Disable' lockdown. Another red box highlights the 'Set policy refresh interval' dialog box, which allows specifying how often the StellarOne policy is applied to the agent/group, with a dropdown menu set to '20 Minutes'. The 'Save' button is visible in the dialog.

stellarOne

Dashboard Agents Logs Administration About

← QQ-STELLAR

StellarProtect Policy Inheritance: ☒ Inherit from parent group: FAB12 | Self-management: ☐

General Info Policy

Application Lockdown

When Application Lockdown is turned on, the endpoint will only be able to run applications that are in the Approved List. If required files are not in the Approved List, this endpoint will not be able to restart or log on.

- ☐ Detect
When an application not in the Approved List launches, it is allowed to run and receive a notification.
- ☐ Enforce
When an application not in the Approved List launches, it is blocked from running and receive a notification.
- ☒ Disable
Application Lockdown mode is disabled.

[Exception Paths \(0\)](#)
To configure lockdown exclusion settings for Approved List.

[Trusted Hash Values \(0\)](#)
To configure trusted hash values for Approved List.

TXOne StellarProtect

stellarProtect

parent group: All | Self-management: ☐

Overview OT Applications OT Certificates Approved List Password Operations Settings About

Application Lockdown

- ☐ Detect: When an application not in the Approved List launches, it is allowed to run and receive a notification.
- ☒ Enforce: When an application not in the Approved List launches, it is blocked from running and receive a notification.
 - ☒ DLL/Driver Lockdown
 - ☒ Script Lockdown
 - ☒ Intelligent Runtime Learning
- ☐ Disable: Application lockdown is disabled.

Industrial-Grade Next-Generation Antivirus

- ☒ Real-time Malware Scan

Set policy refresh interval

Specify how often StellarOne policy is applied to agent/group. [Learn More](#)

Refresh Interval

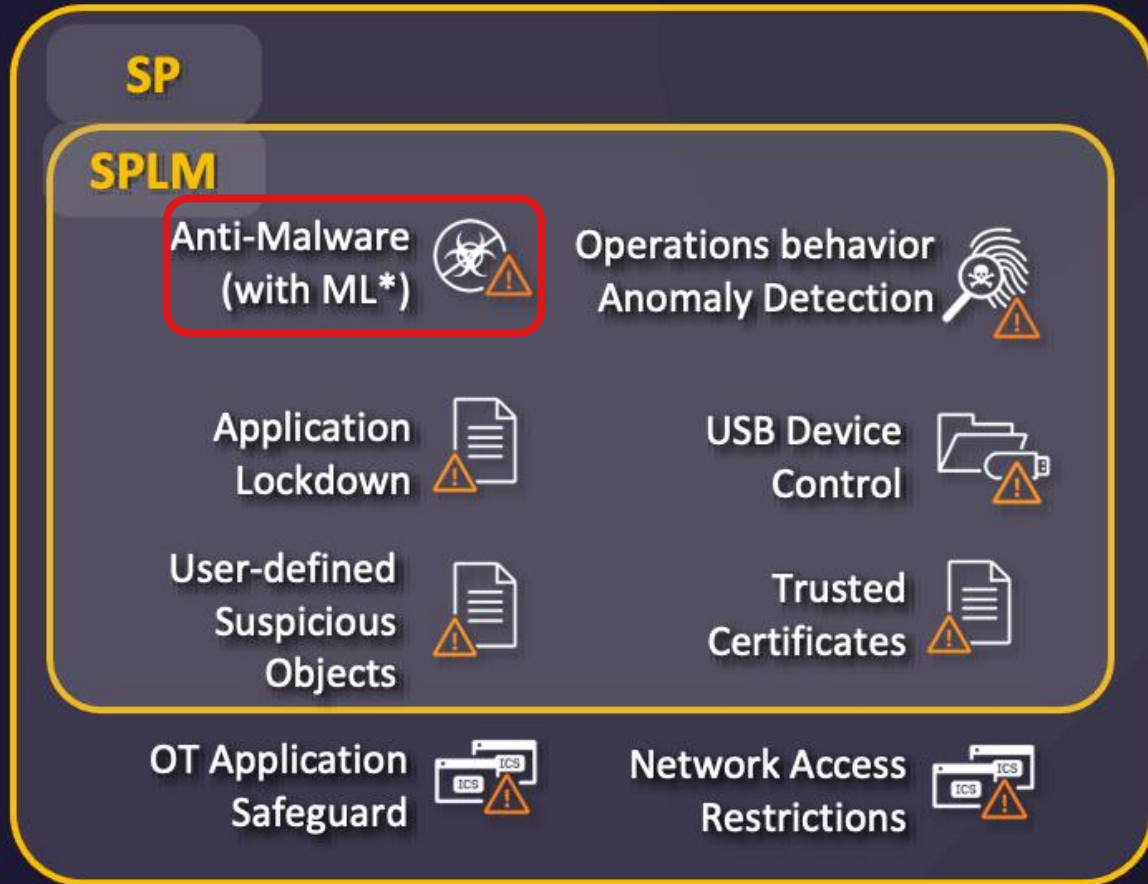
[Save](#) [Cancel](#)

Agent administrator-password versus Agent user-password

Agent Password -> Function v	Administrator	User
Overview Main UI	All operations	Can turn on / off realtime scan and lockdown Can't renew license
Approved List Main UI	All operations	View only and search
Password Main UI	All operations	Can't access
Operations Main UI	All operations	All operations
Settings Main UI	All operations	Can't change
CLI	All operations	Can't use CLI
Diagnostic	All operations	Can only collect debug logs/ delete old debug logs

Stellar Security Controls

Security Controls: 1/8 Anti-Malware



Anti-Malware

- Real-time / Scheduled Scanning is available in both agents (SP & SPLM)
- Lightweight scanner that is tuned for OT environments
- Advanced Threat Scanning with Machine Learning in Stellar Protect agent

(*) Machine Learning option of Anti-Malware is only available in the Stellar Protect Agent

Anti-Malware Ransomware Behavior Monitoring

SP

Multi-Method Threat Prevention

OT root of trust and advanced threat scan secure the assets without the interruption on the operations. ⓘ

☒ Real-Time Protection

- ☒ Real-Time Scan
 - ☒ Predictive Machine Learning
 - > Advanced Options
 - ☒ Ransomware Behavior Monitor

☐ Scheduled Scan [Schedule](#)

> Advanced Options

[Real-Time Scan Exclusions \(0\)](#)
To configure scan exclusion settings for real-time scan.

[Scheduled Scan Exclusions \(0\)](#)
To configure scan exclusion settings for scheduled scan.

SPLM

Threat Prevention

Persistently scan new and changed files, along with system memory, to provide security assessment for maximum protection against malware. ⓘ

☒ Real-Time Protection

- ☒ Real-Time Scan
 - > Advanced Options
 - ☒ Ransomware Behavior Monitor

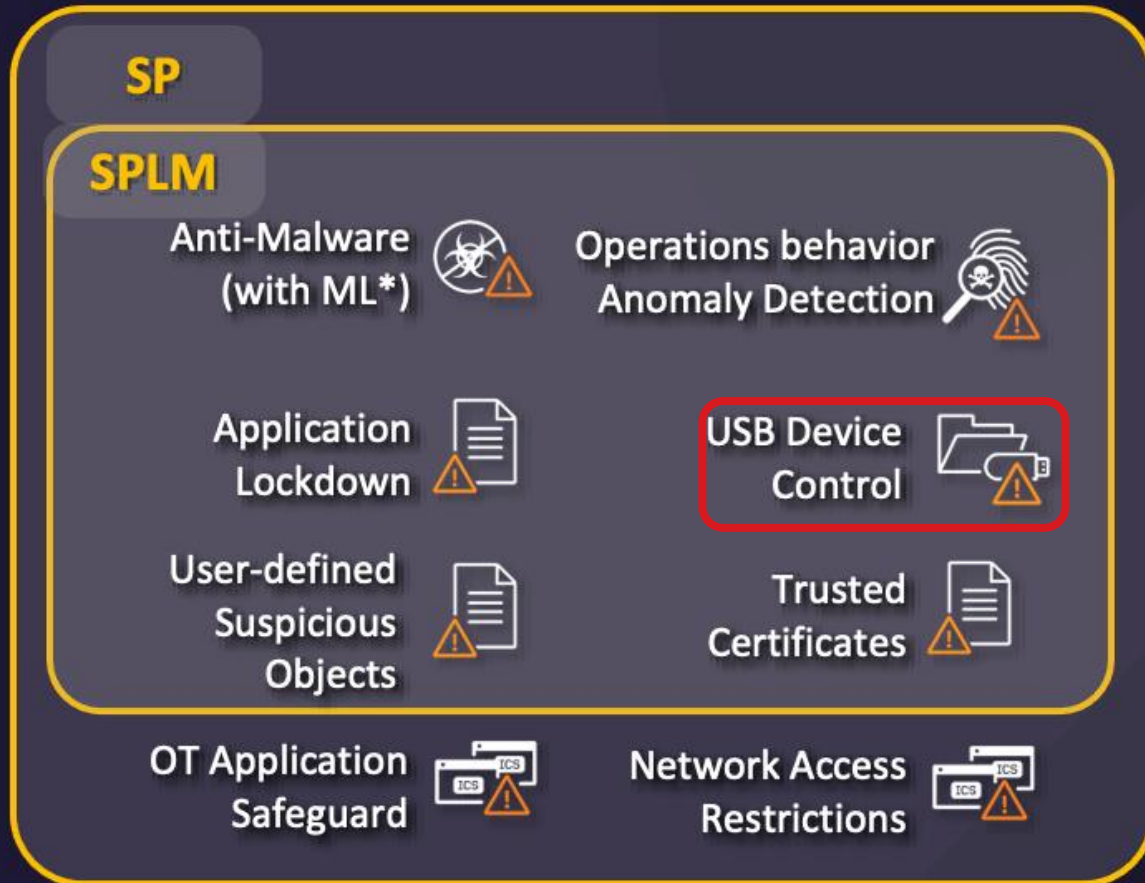
☐ Scheduled Scan [Schedule](#)

> Advanced Options

[Real-Time Scan Exclusions \(0\)](#)
To configure scan exclusion settings for real-time scan.

[Scheduled Scan Exclusions \(0\)](#)
To configure scan exclusion settings for scheduled scan.

Security Controls: 2/8 USB Storage Device Control



USB Storage device Control

- Control access to USB storage devices
- Use "**USB Access Tracking**" to build the list of USB devices that are in use. (including the systems on which they are used and the frequency)

External Device Management

- ☐ Device Control
 - Block USB mass storage devices.
 - ☐ Block USB optical disc drives (ODD), including CDs, DVDs, and Blu-ray devices.
- [Trusted USB Device \(0\)](#)

The USB storage devices connected will be granted permanent access to managed endpoints.
- ☐ USB Auto Scan
 - Scan USB storage devices immediately after they are connected to managed endpoints.
- ☐ Block USB AutoRun
 - Prevent connected USB storage devices from automatically launching programs or applications on managed endpoints.
- ☐ USB Access Tracking
 - Send event logs when USB storage devices are connected to or disconnected from managed endpoints.

SP&SPLM: (USB storage) Device Control

- VID (Vendor ID), PID (Product ID), and SN (Serial Number) can be used as conditions for USB vector control approval

Device Control (0)

☒ Block external device access for USB drives on managed endpoints. To allow access for the following trusted USB devices.

+ Add

Vendor ID	Product ID	Serial Number
No data to display		

Add Trusted USB Device

Specify at least one of the following information for the trusted USB device.

Vendor ID:

Product ID:

Serial number:

Note: You can use one of the following methods to get the information of a connected device to an endpoint:

- (1) Open the Device Manager on the agent endpoint
- (2) Use `opcmd.exe -p usb info -d <drive_letter>` command on the agent endpoint

SP&SPLM: (USB storage) Device Control

- A simple way to find the VID (Vendor ID), PID (Product ID), and SN (Serial Number) is to just insert a new USB stick in a protected system and read the log file.
- From the log file, then the new USB stick can be added
- One-time approval feature is **now available in StellarProtect and Legacy Mode (Stellar 3.1)**

The screenshot illustrates the workflow for adding a USB device to the trusted list in StellarProtect. It features three main components: a main event log window, a confirmation dialog, and a detailed device information window.

Main Event Log Window:

- Action:** Includes a 'Print' button and an 'Allow 1-time USB device access' button. A red arrow points from this button to the 'Confirm Action' dialog.
- Event Information:**
 - Time: 2022-01-27T12:43:56+08:00
 - Level: Warning
 - Event ID: 5377
 - Event: USB Access Blocked: E:\
 - Detail: Access Image Path: C:\Windows\explorer.exe, Access User: twrtl, Vendor ID: 951, Product ID: 1666

Confirm Action Dialog:

Do you want to add the blocked process to the Trusted USB List?

Buttons: OK, Cancel. A red arrow points from the 'OK' button to the 'Trusted USB Device Added' event in the log.

Trusted USB Device Added Event:

☐	Time	Level	Event
☐	2022-01-27T12:50:35+08:00	Information	1281 Trusted USB Device Added

Device Information Window:

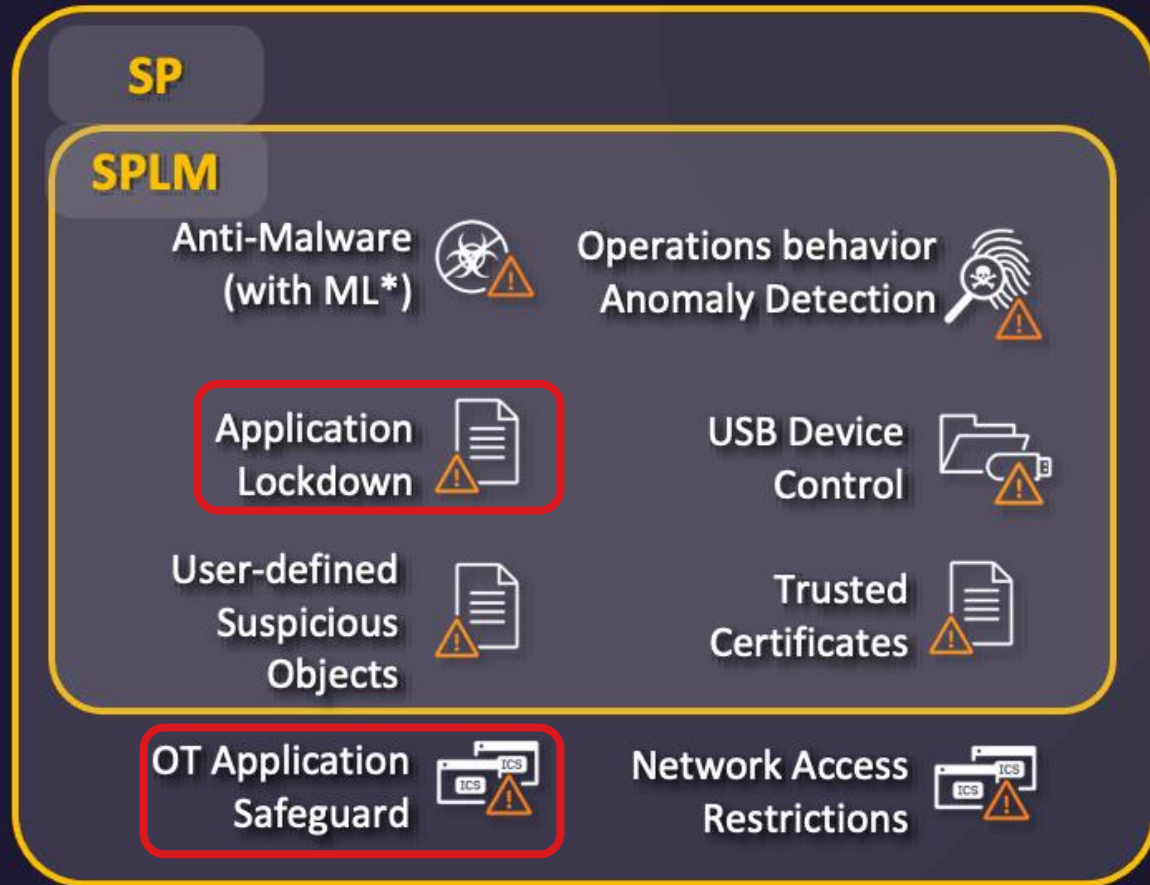
A USB device has been blocked by StellarProtect

Information

- Product ID: 0x1666
- Vendor ID: 0x0951
- Serial Number: E0D55EA574DE1560D840055A

Buttons: Approved this device

Security Controls: 3+4/8 Application Lockdown & OT Application Safeguard

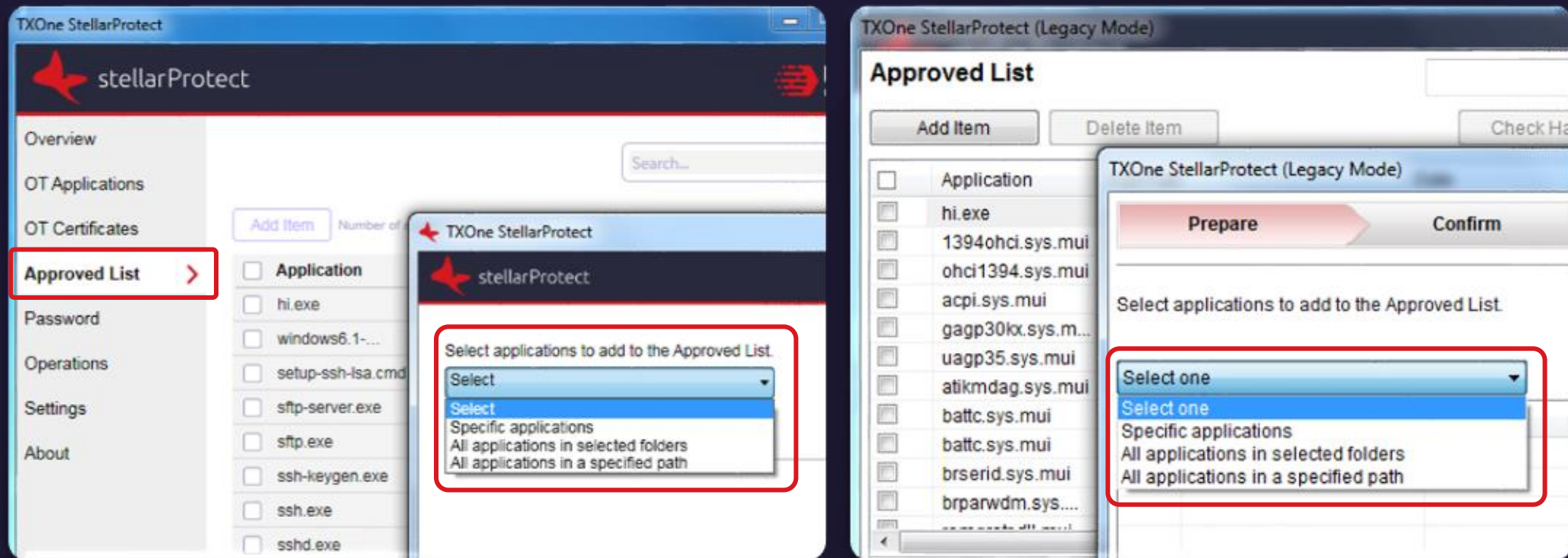


1. "**Application Lockdown**": Builds a whitelist of executable files (*.exe, *.dll,...) that are allowed to *run*. Executables of which the hash does not match (or does no longer match) are not (/no longer) allowed to run. Those could be either new executable files or altered ones.
2. "**OT Application Safeguard**": Auto-recognition and protection of OT applications. Makes all executables (*.exe, *.dll,...) of OT applications *read-only*.

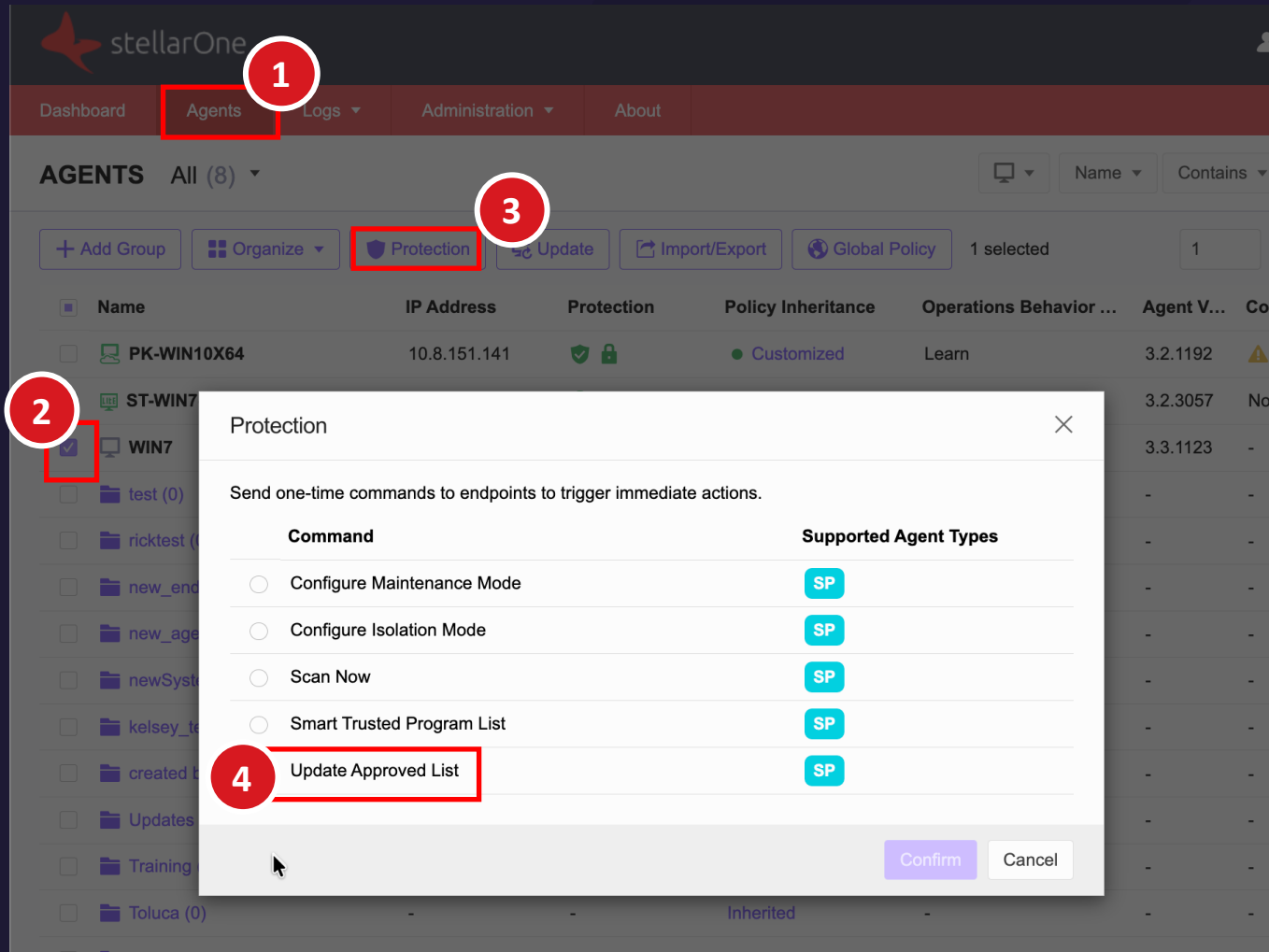
Security Controls: 3/8 Application Lockdown

List of detected Applications by Application Lockdown (Agent-side)

- The list is called the "**Approved List**".
- Only present at the Agent side; not in StellarOne
- Manually browse and select files from the Stellar Agent Console
 - Local command (CLI) is also available for advanced usage



List of detected Applications by Application Lockdown (in StellarOne)



To entirely rebuild the Approved List (list of detected Applications)

1. select the Agents tab
2. select the Agent(s)
3. Protection
4. Update Approved List
5. Confirm

(the Approved List is automatically updated during maintenance mode)

Security Controls: 4/8 OT Application Safeguard

OT Application Safeguard (SP only)

- OT Application Safeguard is industrial-based change control protection
 - ICS application are recognized in the scan phase of the installation.
 - Executable components of the installed ICS applications are made Read-Only (*.exe, .dll,..)
 - Supported vendors include:
 - ABB
 - Acronis
 - Autonics
 - Baker Hughes
 - Bürkert
 - Codesys
 - Delta electronics
 - Druck limited
 - Emerson
 - Epson Robots
 - Fanuc
 - Fisher controls
 - Fuji
 - GE
 - Idec
 - Keyence
 - Hitachi
 - Honeywell
 - Lovato electric
 - Micromeritics
 - Mitsubishi
 - Motorola
 - National instruments
 - Omron
 - Pactware
 - Phoenix contact
 - Red lion
 - Roboticsware
 - Rockwell
 - Seiko Epson
 - Schneider
 - Siemens
 - Verif-I
 - YASKAWA
 - Yokogawa
 - Yamaha

List of detected OT Applications



Dashboard

Agents

Logs ▾

Administration ▾

About

←  EURODEMO1

StellarProtect

Policy Inheritance: ☐ Inherit from parent group: All

Self-management: ☐

General Info

Policy

Baseline

Situational Awareness

OT Applications457

OT Certificates17

Approved Script Behaviors6

Approved Login Accounts5

Approved Applications640

Software ▾

Automatically learn and allow the approved OT applications to run on the endpoints.

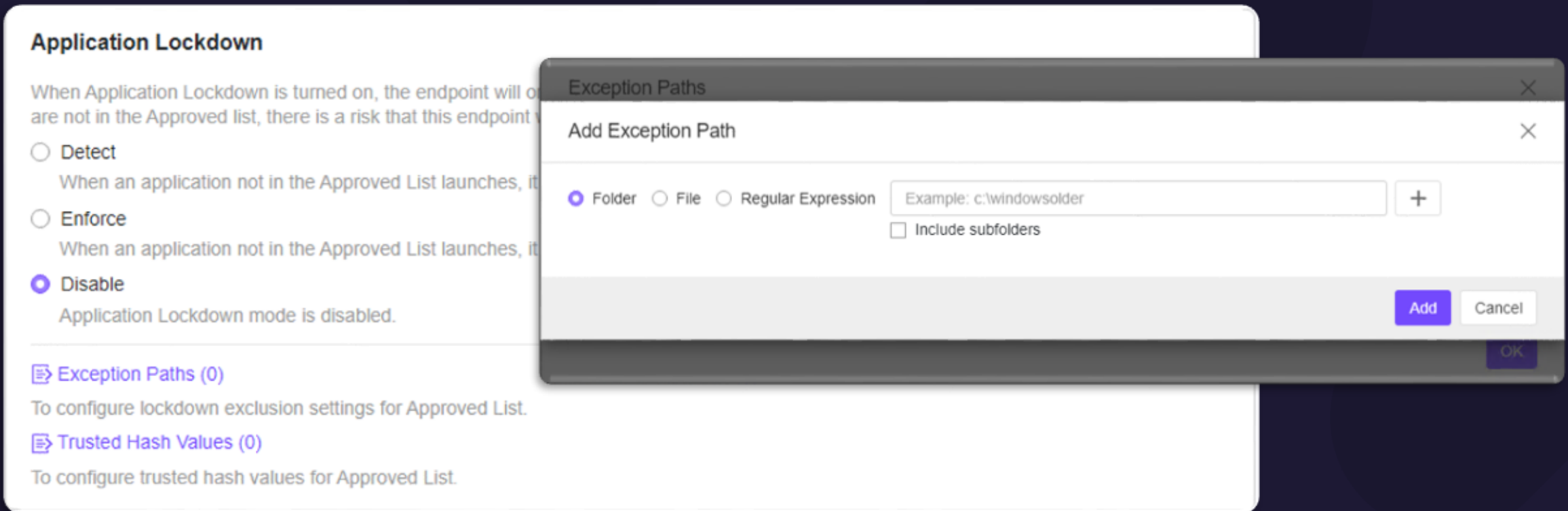
Software ↑	Vendor	Version	Installation Path
Erlang 20.2.0	National Instruments	20.2.0	-
FANUC LADDER-III	FANUC	1.00.000	C:\Program Files (x86)\FANUC PMC
LabVIEW NXG 4.0 RTE Manag...	National Instruments	19.50.49152	-
Math Kernel Libraries	National Instruments	17.01.49152	-
Math Kernel Libraries (64-bit)	National Instruments	17.01.49152	-
Microsoft Silverlight 5.1	National Instruments	5.19.49152	-
NI .NET Framework 4.8.0	National Instruments	23.80.49180	-

To rebuild the list of detected OT Applications:

- from StellarOne:
enable and then disable Maintenance Mode
- on the Agent:
"opcmd -p appinv make"

SP&SPLM: Application Lockdown

- Define exceptions for specific files or folders if needed
 - Regular expression is also supported for advanced use cases



Application Lockdown & OT Application Safeguard: Operations

- Block events from StellarOne
 - When executed file is not part of approved list

The screenshot displays the StellarOne web interface. The top navigation bar includes 'Dashboard', 'Agents', 'Logs', 'Administration', and 'About'. The 'AGENT EVENTS' section is active, showing a table of events. An 'Event Details' modal is open, displaying information for a specific event.

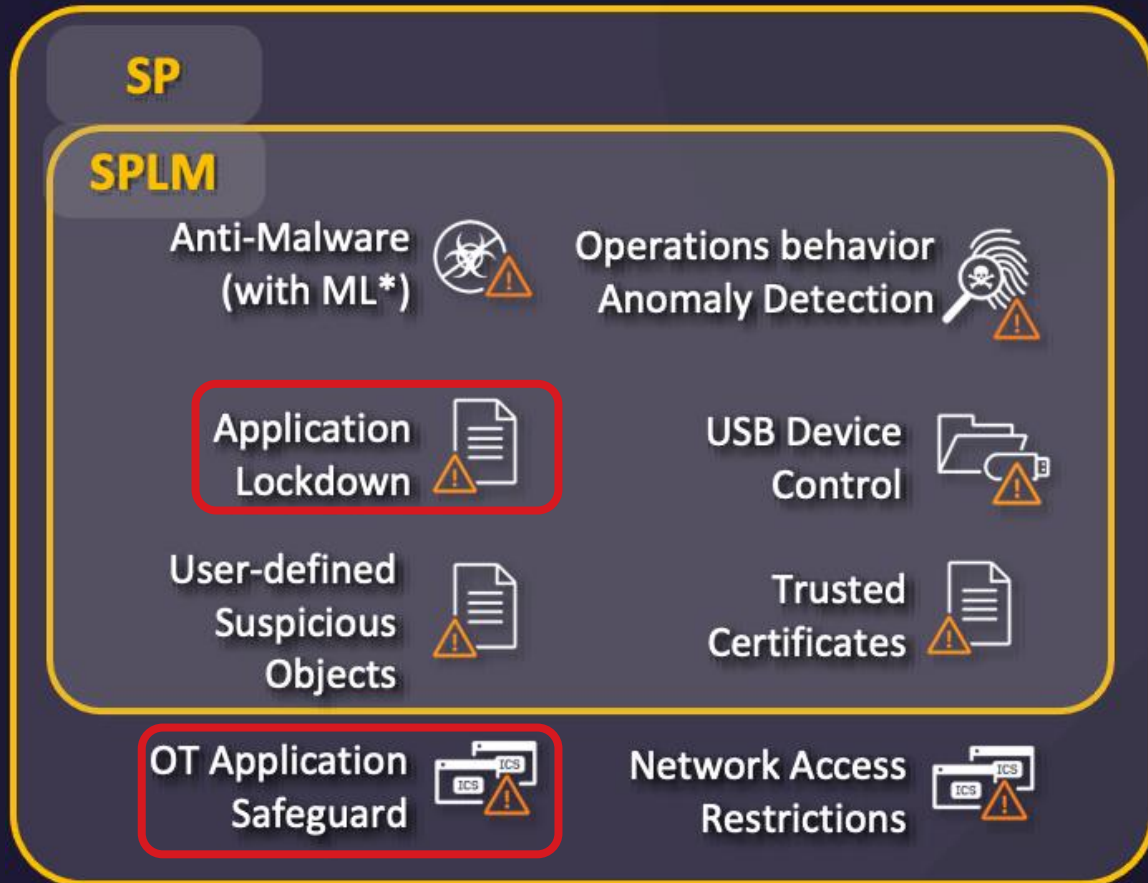
Event Details Modal:

- Action:** Print, Ignore, Delete, Quarantine, **Add to Approved List** (highlighted with a red box).
- Event Information:**
 - Time: 2023-06-05T23:09:44+08:00
 - Level: Warning
 - Event ID: 5889
 - Event Detail: File access has been blocked: C:\Users\vmuser\Desktop\Applications\Applications\hfs-demo.exe
 - Mitre ID: T0865 T0847 T0883 T0864 T0889 T0839 T0873 T0857 T0849 T0867 T0843

Event Log Table:

Time	Level	Event	Endpoint
2023-06-05T23:09:44+08:00	Warning	5889 File access has been blocked: C:\Users\vmuser\Desktop\Applications\Applications\hfs-demo.exe	TXSP-WIN10-A01

Security Controls:-/8 Maintenance Mode (Operations)



- Updates the whitelist of "Application Lockdown" and "OT Applications Safeguard" with any new applications or versions that were installed during the maintenance mode.
- Maintenance mode only has an effect on "Application Lockdown" and "OT Applications Safeguard" (!)
- Configure on-demand maintenance period to align with existing change/modification processes.
- After the maintenance mode a malware scan is automatically initiated to quarantine any possible threats that were detected during the maintenance period

SP&SPLM: Maintenance mode (Server Side)

The screenshot displays the stellarOne web interface with several numbered callouts (1-6) indicating the steps to configure Maintenance Mode:

- 1**: Select the checkbox for the agent **PC-STELLAR** in the agents list.
- 2**: Click the **Protection** button in the top toolbar.
- 3**: Click the **Configure Maintenance Mode** button in the Protection sidebar.
- 4**: Click the **Confirm** button at the bottom of the configuration dialog.
- 5**: Select the **Enable** radio button for Maintenance Mode.
- 6**: Click the **Schedule** radio button and use the **Time Select** icon to set the start time.

Agents List:

Name	IP Address	Protection	Policy Inheritance	Approved List	Agent Version	Last Connection	Function Type
☒ PC-STELLAR	192.168.23.24		Int				
☐ QQ-STELLARLM	192.168.23.28		Int				

Configure Maintenance Mode (1) Dialog:

To allow file updates on endpoints during a specified period, please enable Maintenance Mode. Be sure to run applications from trusted sources for security reasons. Notice: If users change policy settings during maintenance period, part of the policy settings will not be enforced until the maintenance period ends.

☒ Enable ☐ Disable

Maintenance Mode start time:

☒ Schedule ☐ Start now

-- Time Select --

Maintenance Mode will be ended after hour(s)

Application Lockdown: Operations

- Block events from StellarOne
 - When executed file is not part of approved list

The screenshot displays the StellarOne web interface. The top navigation bar includes the 'stellarOne' logo and tabs for 'Dashboard', 'Agents', 'Logs', 'Administration', and 'About'. Below this, the 'AGENT EVENTS' section is active, showing a table of events. An 'Event Details' modal is open, displaying information for a specific event. The modal has a title bar with a close button and a timestamp '2023-06-05T23:09:44+08:00'. The 'Action' section contains buttons for 'Print', 'Ignore', 'Delete', 'Quarantine', and 'Add to Approved List' (which is highlighted with a red box). The 'Event Information' section provides details about the event, including its time, level, ID, and a description of the blocked file access. The background table shows a single event with a 'Warning' level and a description of a blocked file access.

Time	Level	Event	Endpoint
2023-06-05T23:09:44+08:00	Warning	5889 File access has been blocked: C:\Users\vmuser\Desktop\Applications\Applications\hfs-demo.exe	TXSP-WIN10-A01

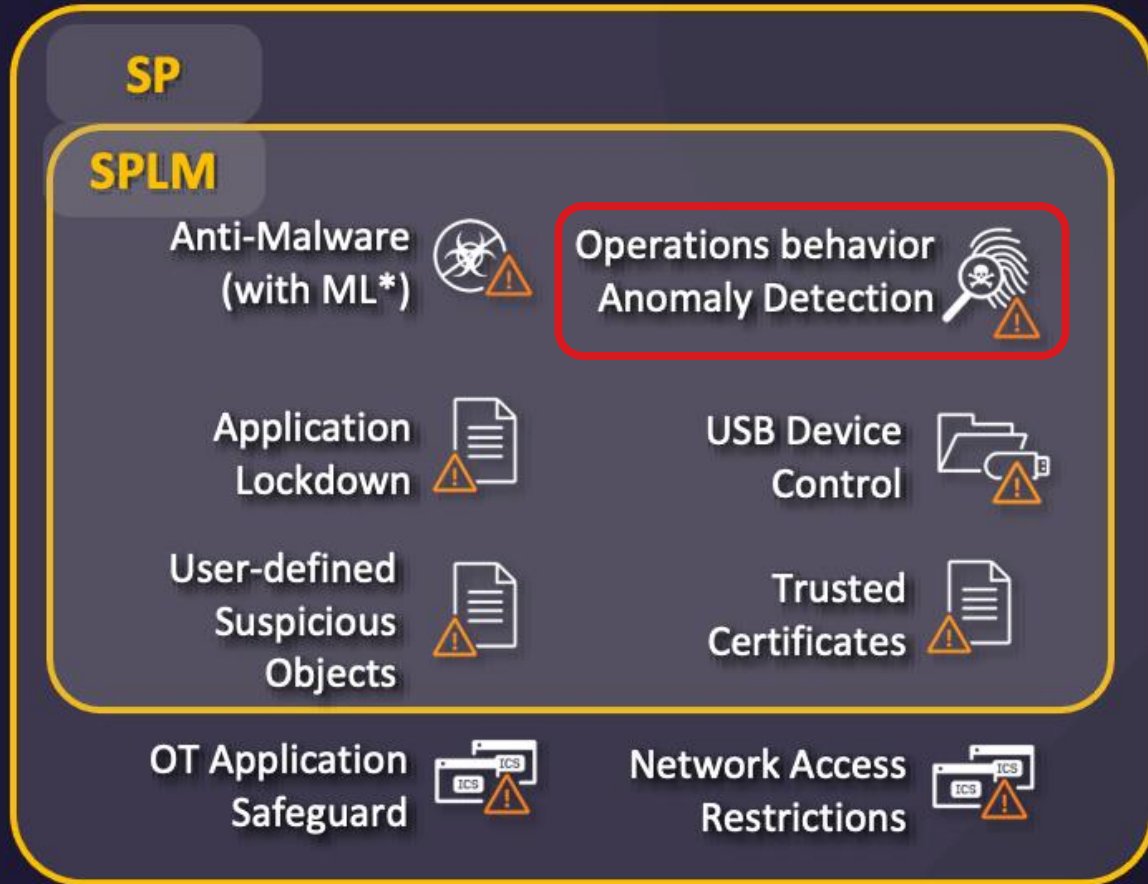
Event Details Modal:

Action: Print, Ignore, Delete, Quarantine, **Add to Approved List**

Event Information:

- Time: 2023-06-05T23:09:44+08:00
- Level: Warning
- Event ID: 5889
- Event: File access has been blocked: C:\Users\vmuser\Desktop\Applications\Applications\hfs-demo.exe
- Detail: Access Image Path: C:\Windows\explorer.exe
Access User: TXSP-WIN10-A01\vmuser
Mode: Enforce
Reason: Not in Approved List
File hash blocked: e678899d7ea9702184167b56655f91a69f8a0bdc9df65612762252c053c2cd7c
- Mitre ID: T0865 T0847 T0883 T0864 T0889 T0839 T0873 T0857 T0849 T0867 T0843

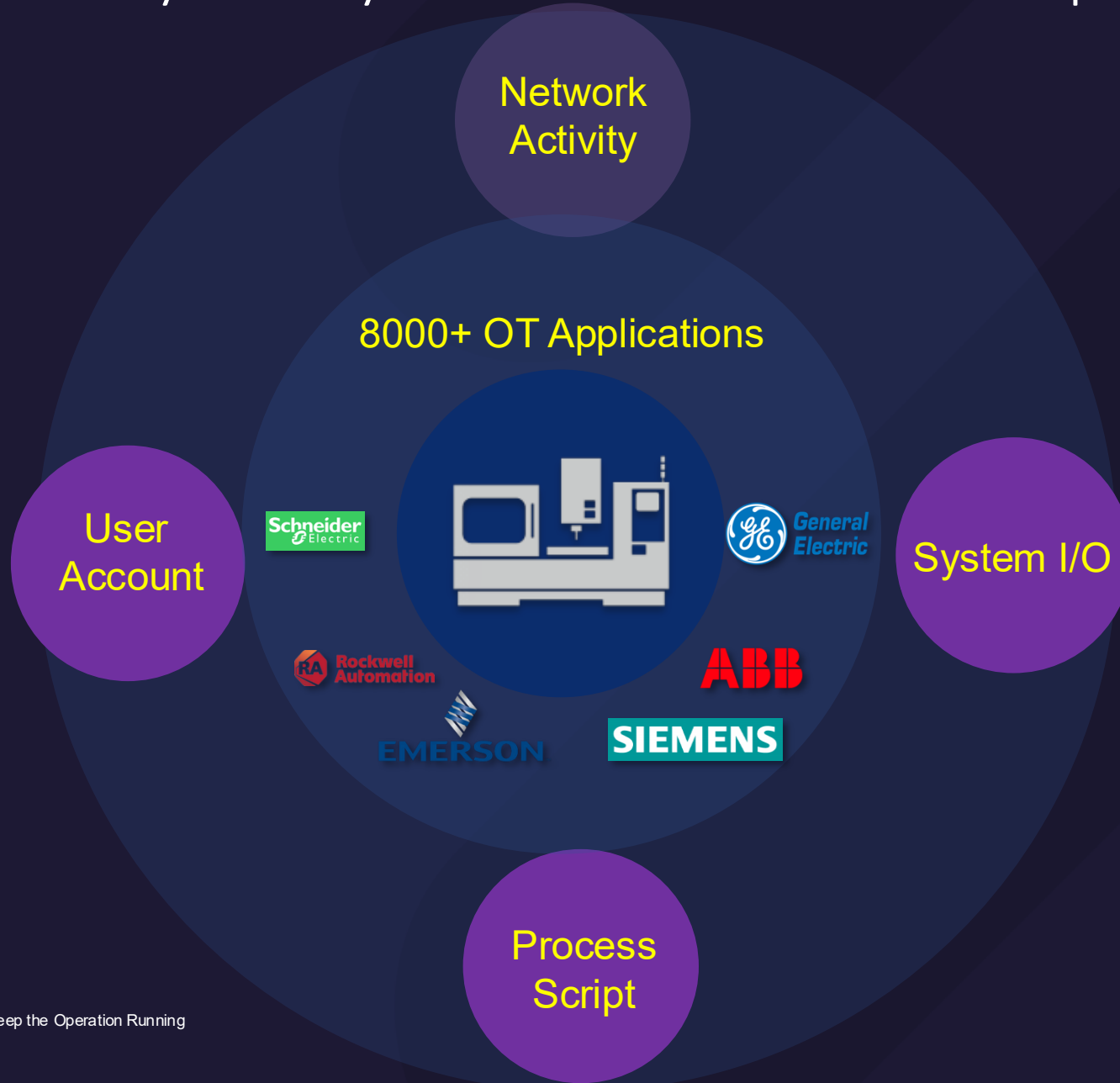
Security Controls: 5/8 Anomaly Detection



Anomaly Detection

- **OBAD**-Operations Behavior Anomaly Detection (Process Chain lockdown)
- Enhanced in Stellar3.0 to include: Script Behavior, User Login and Application Behavior
- Now also called **CPSDR**: Cyber-Physical System Detection and Response (an OT-optimized EDR/XDR)
- Requires a Learning period; then automatically switches to Enforce or Detect

Cyber-Physical System Detection and Response (CPSDR)



- CPSDR is implemented by OBAD
- Reduce the noise by comparing detections with known behaviors of 8000+ OT Applications (versions)
- Preserve OT Endpoint availability
- Ensure integrity by building up the unique asset *fingerprint* by collecting, Network Activity, System IO, User Account and Process Script

OBAD Baselines

OBAD
baselines can
only be seen
in StellarOne;
not on the
agents

stellarOne

Dashboard

Agents

Logs

Administration

About

← CHRIVTEST

StellarProtect

Policy Inheritance: ☐ Inherit from parent group: ChrisV

Self-management: ☐

General Info

Policy

Baseline

Situational Awareness

OT Applications5

OT Certificates1

Approved Script Behaviors576

Approved Login Accounts1

Approved Applications69

Approved Network Access0

Baseline Toggle

The baseline below shows the approved script behaviors for the agent. You can boost your security protection by manually adding commonly-abused applications to the Watchlist on the Policy page.

Monitored Process / Script

☒

"C:\Windows\System32\WindowsPowerShell\v1.0\powershell.exe" [Console]::OutputEncoding = [Sys

☒

"C:\Windows\System32\WindowsPowerShell\v1.0\powershell.exe" [Console]::OutputEncoding = [Sys

☒

"C:\Windows\System32\WindowsPowerShell\v1.0\powershell.exe" \$wmi_proc = Get-WmiObject -Cla

☒

"C:\Windows\System32\WindowsPowerShell\v1.0\powershell.exe" [Console]::OutputEncoding = [Sys

☒

"C:\Windows\System32\WindowsPowerShell\v1.0\powershell.exe" [Console]::OutputEncoding = [Sys

☒

"C:\Windows\System32\WindowsPowerShell\v1.0\powershell.exe" [Console]::OutputEncoding = [Sys☒☒☒☒

Gray detections for Applications

- Behavior monitoring rules created by TXOne threat researchers
- Capture suspicious/malicious behaviors performed by ransomware
 - Scheduled task added
 - Memory dumping
 - ...

OBAD Modes

Operations Behavior Anomaly Detection ⓘ

NEW

- ☐ Learn: Collect behavioral patterns from the monitored agent-devices to establish baseline fingerprints
- ☒ Detect: Identify and send alerts for any unexpected changes and security threats by analyzing current behaviors against the fingerprints at the agent-device and central management levels
 - ☐ Strict mode ⓘ
- ☐ Enforce: Take preventative action on detected fingerprint deviations to defend operation stability and security

Learn

**Builds baseline
(now a minimum Learn period
of at least 1 day is mandatory)**

Detect

**Generates log events
(log-only; no blocking)**

Enforce

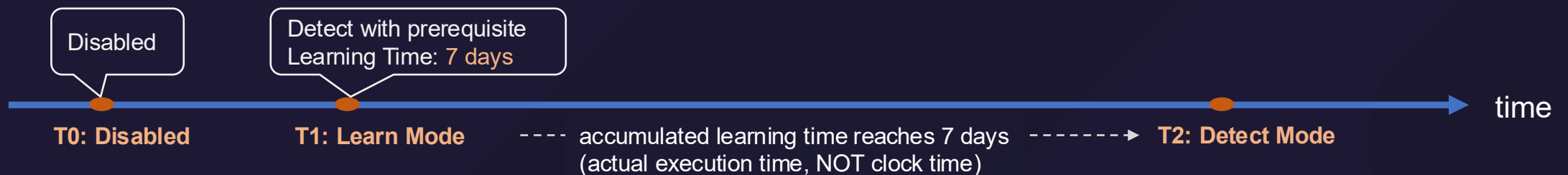
**Block abnormal behaviors
(In Stellar 3.0, *User* and
Application Behavior are
"Detect" only)**

Learning Mode is a prerequisite for Detect/Enforce

⚠ All agent-devices must start with the "Learn" mode to establish their own baseline fingerprints before entering the "Detect" or "Enforce" mode. The agents (e.g. newly-added agents) will learn first and then automatically switch to the specified mode according to the policy.

Learning time

- When a policy is deployed with OBAD in Detect or Enforce and the agent has not completed a Learning mode, then the Learning mode will first be enabled
- When the Learning period has completed, the Agent will automatically switch to Detect or Enforce mode



- This prerequisite learning time doesn't apply when the agent is upgraded from previous versions with Script Behavior enabled in Detect or Enforce mode

Baseline *Building* (during *Learning* Mode)

OBAD Feature	Example Action	Added to the baseline	
1. Script Behavior	A script " sample.ps1 " is started from Powershell which is started from CMD	cmd.exe -> powershell.exe -File sample.ps1 Process chain and parameters of executed scripts (the name of the script, but not the hash)	
2. User Login	User Alice from the TxOne Domain remotely logs-in from IP 10.8.100.244	TXOne, Alice, RemoteInteractive, 10.8.100.244 Domain, username, login type (local/remote) and IP address of successful login activities	
3. Application Behavior	Application "sample.exe" creates a scheduled task	sample.exe, 499DCD0E640A3BC5C54F5CA58A86A61CD3477D40 Hash of executed applications and application behaviors that hit gray detection rules	
4. Network Access	User execute Internet explorer to access a web server 192.168.23.196 over port 80	IP: 192.168.23.196 Direction: Outbound Remote/Local Port: 80	L4 protocol: TCP L7 protocol: HTTP Application: C:\Program...iexplorer.exe

Baseline *Matching* (during Detect / Enforce modes)

Operations Behavior Anomaly Detection ⓘ

NEW

☐

 Learn: Collect behavioral patterns from the monitored agent-devices to establish baseline fingerprints

☒

 Detect: Identify and send alerts for any unexpected changes and security threats by analyzing current behaviors against the fingerprints at the agent-device and central management levels

☐

 Strict mode ⓘ

1. Script Behavior baseline matching

Baseline
cmd.exe -> powershell.exe -File sample.ps1

Not-strict Mode (Process chain only)	Strict Mode (Process chain and parameters)

Baseline *Matching* (during Detect / Enforce modes)

Operations Behavior Anomaly Detection ⓘ

NEW

☐

 Learn: Collect behavioral patterns from the monitored agent-devices to establish baseline fingerprints

☒

 Detect: Identify and send alerts for any unexpected changes and security threats by analyzing current behaviors against the fingerprints at the agent-device and central management levels

☐

 Strict mode ⓘ

1. Script Behavior baseline matching

Baseline
cmd.exe -> powershell.exe -File sample.ps1

Not-strict Mode (Process chain only)	Strict Mode (Process chain and parameters)
cmd.exe -> powershell.exe -File other_sample.ps1 ✓ python.exe -> powershell.exe -File sample.ps1 ✗	

Baseline *Matching* (during Detect / Enforce modes)

Operations Behavior Anomaly Detection ⓘ

NEW

☐ Learn: Collect behavioral patterns from the monitored agent-devices to establish baseline fingerprints

☒ Detect: Identify and send alerts for any unexpected changes and security threats by analyzing current behaviors against the fingerprints at the agent-device and central management levels

☐ Strict mode ⓘ

1. Script Behavior baseline matching

Baseline
cmd.exe -> powershell.exe -File sample.ps1

Not-strict Mode (Process chain only)	Strict Mode (Process chain and parameters)
cmd.exe -> powershell.exe -File other_sample.ps1 ✓ python.exe -> powershell.exe -File sample.ps1 ✗	cmd.exe -> powershell.exe -File other_sample.ps1 ✗ python.exe -> powershell.exe -File sample.ps1 ✗

Baseline *Matching* (during Detect / Enforce modes)

2. *User* login behavior

Baseline
TXOne, Alice, RemoteInteractive, 10.8.100.244

Not-strict Mode (Login domain and username only)	Strict Mode (Login domain, username, type and IP address)

Baseline *Matching* (during Detect / Enforce modes)

2. *User* login behavior

Baseline
TXOne, Alice, RemoteInteractive, 10.8.100.244

Not-strict Mode (Login domain and username only)	Strict Mode (Login domain, username, type and IP address)
TXOne, Alice, RemoteInteractive, 2.2.2.2 ✓ TXOne, Eve, RemoteInteractive, 10.8.100.244 ✗	

Baseline *Matching* (during Detect / Enforce modes)

2. *User* login behavior

Baseline
TXOne, Alice, RemoteInteractive, 10.8.100.244

Not-strict Mode (Login domain and username only)	Strict Mode (Login domain, username, type and IP address)
TXOne, Alice, RemoteInteractive, 2.2.2.2 ✓ TXOne, Eve, RemoteInteractive, 10.8.100.244 ✗	TXOne, Alice, RemoteInteractive, 2.2.2.2 ✗ TXOne, Eve, RemoteInteractive, 10.8.100.244 ✗

Baseline *Matching* (during Detect / Enforce modes)

3. Application Behavior baseline matching

Apps (those Apps are installed on the endpoint)

notepad.exe, (just a normal app, no suspicious behavior)

calculator.exe (just the calculator, but it will not be run during LEARN period; app will not be in baseline)

sneaky.exe, (app does not show suspicious behavior during Learn, but it will do credential dumping AFTER Learn period)

Baseline (those Apps were executed during OBAD Learn period)

notepad.exe, (no suspicious behavior)

sneaky.exe, (with no suspicious behavior during Learn period)

Strict Mode disabled (OBAD in Enforce or Monitor mode with Strict disabled)

(Unrecognized Applications are allowed + Unrecognized behavior is NOT allowed)

Strict Mode

(Unrecognized applications are NOT allowed + Unrecognized behavior is NOT allowed)

Application Behavior Monitoring (some examples)

3. Application Behavior baseline matching

Apps (those Apps are installed on the endpoint)

notepad.exe, (just a normal app, no suspicious behavior)

calculator.exe (just the calculator, but it will not be run during LEARN period; app will not be in baseline)

sneaky.exe, (app does not show suspicious behavior during Learn, but it will do credential dumping AFTER Learn period)

Baseline (those Apps were executed during OBAD Learn period)

notepad.exe, (no suspicious behavior)

sneaky.exe, (with no suspicious behavior during Learn period)

Even though calculator is not in the baseline. In non-strict mode it is OK and it will not trigger an alert

Strict Mode disabled (OBAD in Enforce or Monitor mode with Strict disabled)

(Unrecognized Applications are allowed + Unrecognized behavior will trigger an alert)

- ✓ notepad.exe (normal app. App is in the Baseline: known App + known, unsuspicious, behavior)
- ✓ calculator.exe (was not run during LEARN; is not in baseline; triggers NO alert as there is no suspicious behavior)
- ✗ sneaky.exe (app dumps credentials but did not do that during the LEARN period, known app + **suspicious behavior**)

Strict Mode

(Unrecognized applications are NOT allowed + Unrecognized behavior will trigger an alert)

Application Behavior Monitoring (some examples)

3. Application Behavior baseline matching

Apps (those Apps are installed on the endpoint)

notepad.exe, (just a normal app, no suspicious behavior)

calculator.exe (just the calculator, but it will not be run during LEARN period; app will not be in baseline)

sneaky.exe, (app does not show suspicious behavior during Learn, but it will do credential dumping AFTER Learn period)

Baseline (those Apps were executed during OBAD Learn period)

notepad.exe, (no suspicious behavior)

sneaky.exe, (with no suspicious behavior during Learn period)

Strict Mode disabled (OBAD in Enforce or Monitor mode with Strict disabled)

(Unrecognized Applications are allowed + Unrecognized behavior will trigger an alert)

✓ notepad.exe (normal app. App is in the Baseline: known App + known, unsuspicious, behavior)

✓ calculator.exe (was not run during LEARN; is not in baseline; triggers NO alert as there is no suspicious behavior)

✗ sneaky.exe (app dumps credentials but did not do that during the LEARN period, known app + **suspicious behavior**)

Strict Mode

(Unrecognized applications are NOT allowed + Unrecognized behavior will trigger an alert)

notepad.exe (normal app that is in the Baseline: known app + known, unsuspicious, behavior)

calculator.exe (just the calculator, but it will not be run during LEARN period; app will not be in baseline)

sneaky.exe (app dumps credentials but did not do that during the LEARN period: known app + **suspicious behavior**)

Application Behavior Monitoring (some examples)

3. Application Behavior baseline matching

Apps (those Apps are installed on the endpoint)

notepad.exe, (just a normal app, no suspicious behavior)

calculator.exe (just the calculator, but it will not be run during LEARN period; app will not be in baseline)

sneaky.exe, (app does not show suspicious behavior during Learn, but it will do credential dumping during Enforce)

Baseline (those Apps were executed during OBAD Learn period)

notepad.exe, (no suspicious behavior)

sneaky.exe, (with no suspicious behavior during Learn period)

Strict Mode disabled (OBAD in Enforce or Monitor mode with Strict disabled)

(Unrecognized Applications are allowed + Unrecognized behavior will trigger an alert)

✓ notepad.exe (normal app. App is in the Baseline: known App + known, unsuspicious, behavior)

✓ calculator.exe (was not run during LEARN; is not in baseline; triggers NO alert as there is no suspicious behavior)

✗ sneaky.exe (app dumps credentials but did not do that during the LEARN period, known app + **suspicious behavior**)

Strict Mode

(Unrecognized applications are NOT allowed + Unrecognized behavior will trigger an alert)

notepad.exe (normal app that is in the Baseline: known app + known, unsuspicious, behavior)

calculator.exe (just the calculator, but it will not be run during LEARN period; app will not be in baseline)

sneaky.exe (app dumps credentials but did not do that during the LEARN period: known app + **suspicious behavior**)

Even though "sneaky.exe" is in the baseline, it will raise an Alert even in "not-Strict" mode because during LEARNING it behaved OK and now it behaves Suspiciously

Enforce Mode:

Operations Behavior Anomaly Detection ⓘ

- ☐ **Learn:** Establish the Baseline to enable behavior analysis in Detect and Enforce modes.
- ☐ **Detect:** Log events when abnormal activities are detected.
- ☒ **Enforce:** Log events when abnormal activities are detected, and automatically block unexpected **Script Behavior** changes. To block activities against **Application Behavior**, the agent's Suspicious Objects Blocklist must be created first.
 - ☒ **Strict mode:** Less tolerance for deviation between running behaviors and the Baseline. This may generate more events.
- ☐ **Disable**

[> Learn more](#)

Enforce mode can block only script behaviors
Other behaviors are only logged/detected, even in Enforce mode

Operations Behavior Anomaly Detection ⓘ

☐ **Learn:** Establish the Baseline to enable behavioral analysis in Detect and Enforce modes.

☐ **Detect:** Log events when baseline deviations are detected in Detect and Enforce modes.

☒ **Enforce:** Log events when baseline deviations are detected in Detect and Enforce modes. For events detected in **Enforce** mode, the Event Details window will show a button to block the detected file.

☒ Still Baseline

☐ **Disable**

> Learn more

⌵ ⚙ Application Behavior



Log events when baseline deviations are detected in **Detect** and **Enforce** modes. For events detected in **Enforce** mode, the Event Details window will show a button to block the detected file.

Action: Manually add detected applications to the Suspicious Objects Blocklist.

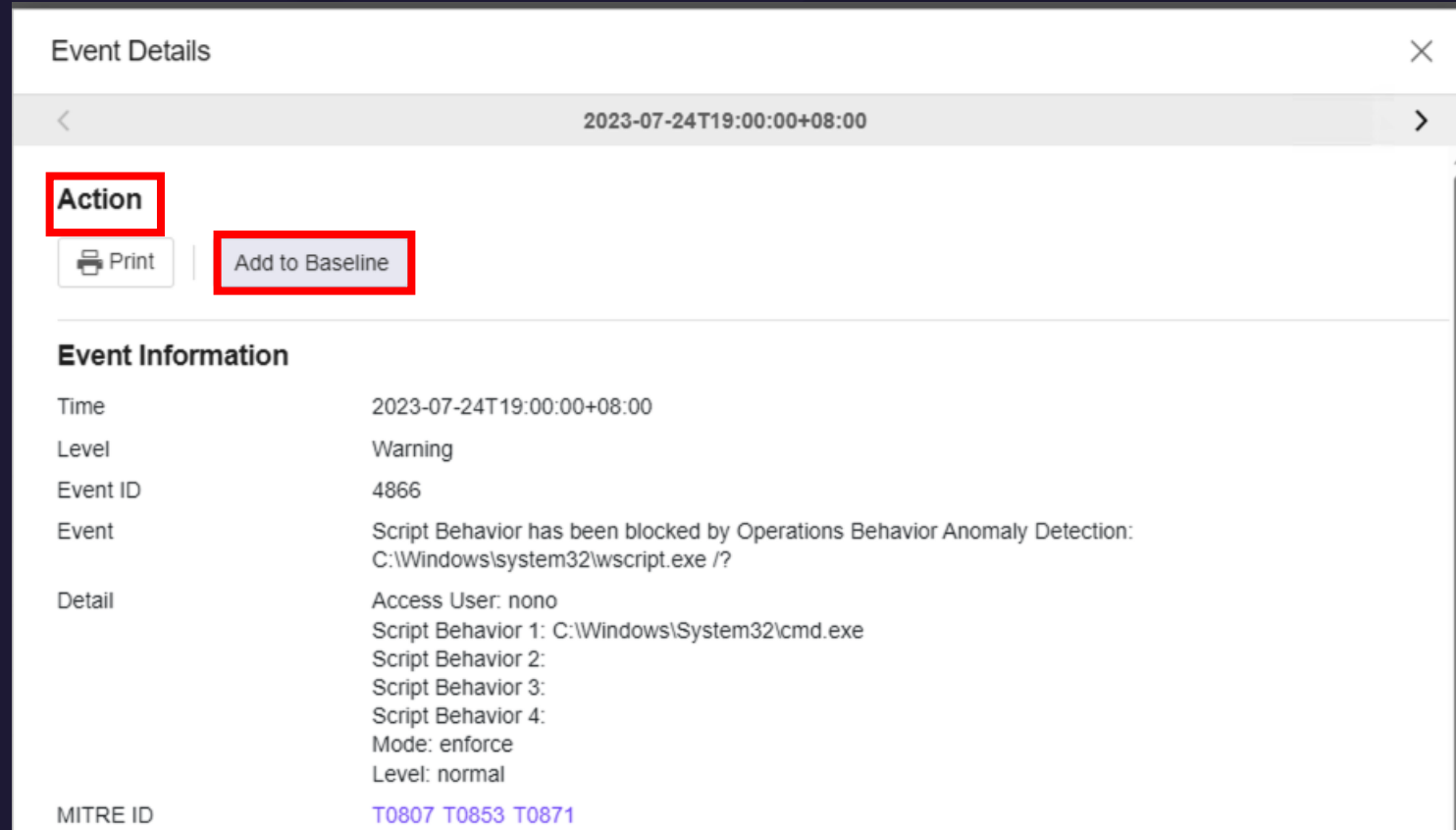
Policy-based Approved Applications (0)

Manually add approved applications to avoid false alert.

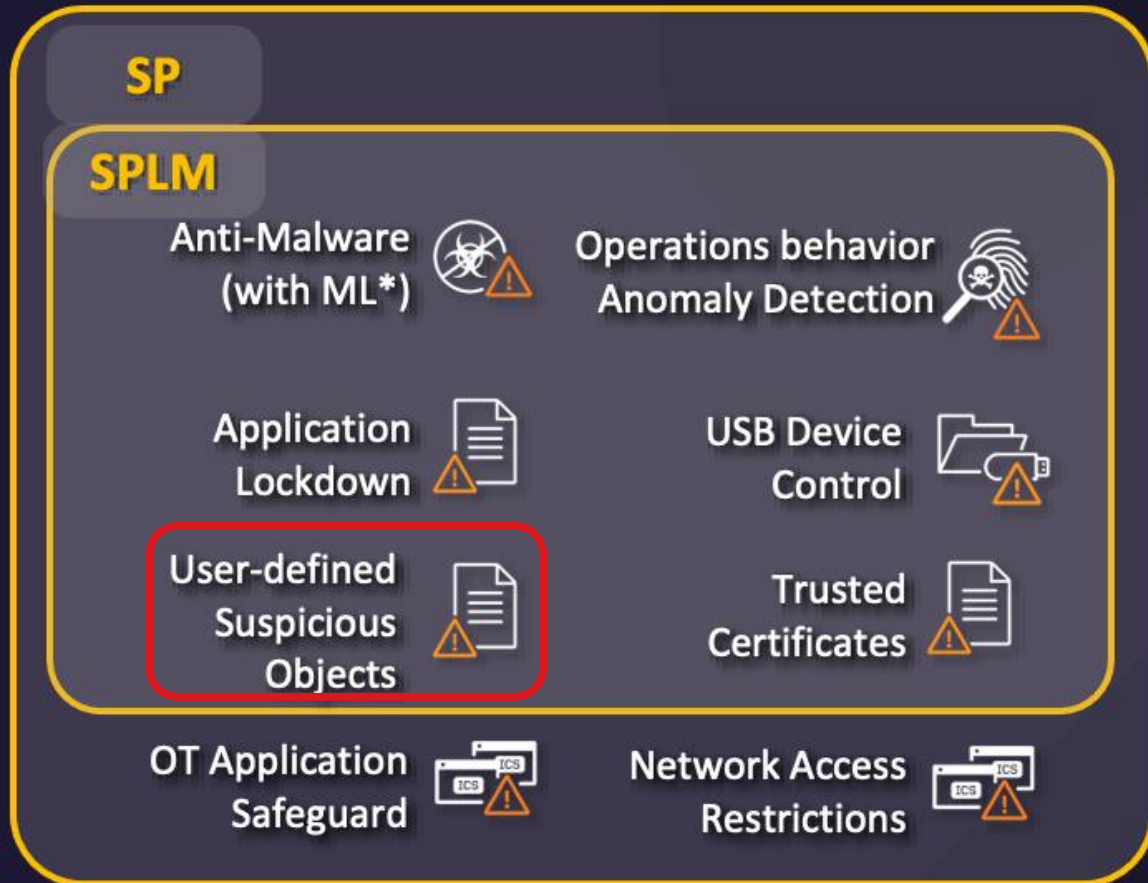
If you want to block behaviors of Applications, you have to manually add those to the Suspicious Objects list

Manually add files to the Baseline (Operations)

- When Operation Behavior Anomaly Detection mode is enabled with Detect or Enforce mode, unknown behaviors will trigger warning events.
- If those behaviors are considered valid, users can add them to baseline by performing Event Actions. (actionable log events)



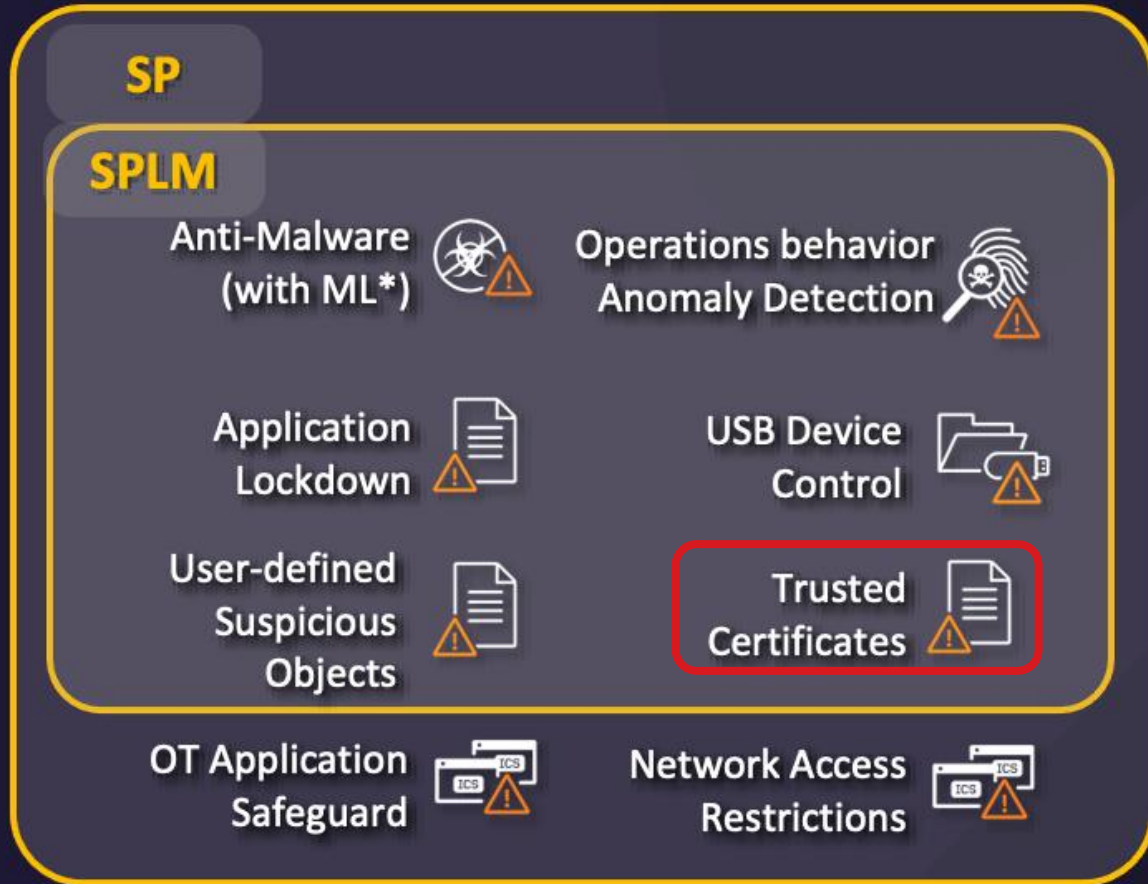
Security Controls: 6/8 User Defined Suspicious objects



- Pro-actively add known Indicators of Compromise (IOCs) to the Agents
- During runtime, this is the FASTLANE for protection/blocking of a file
- It overrules all other controls

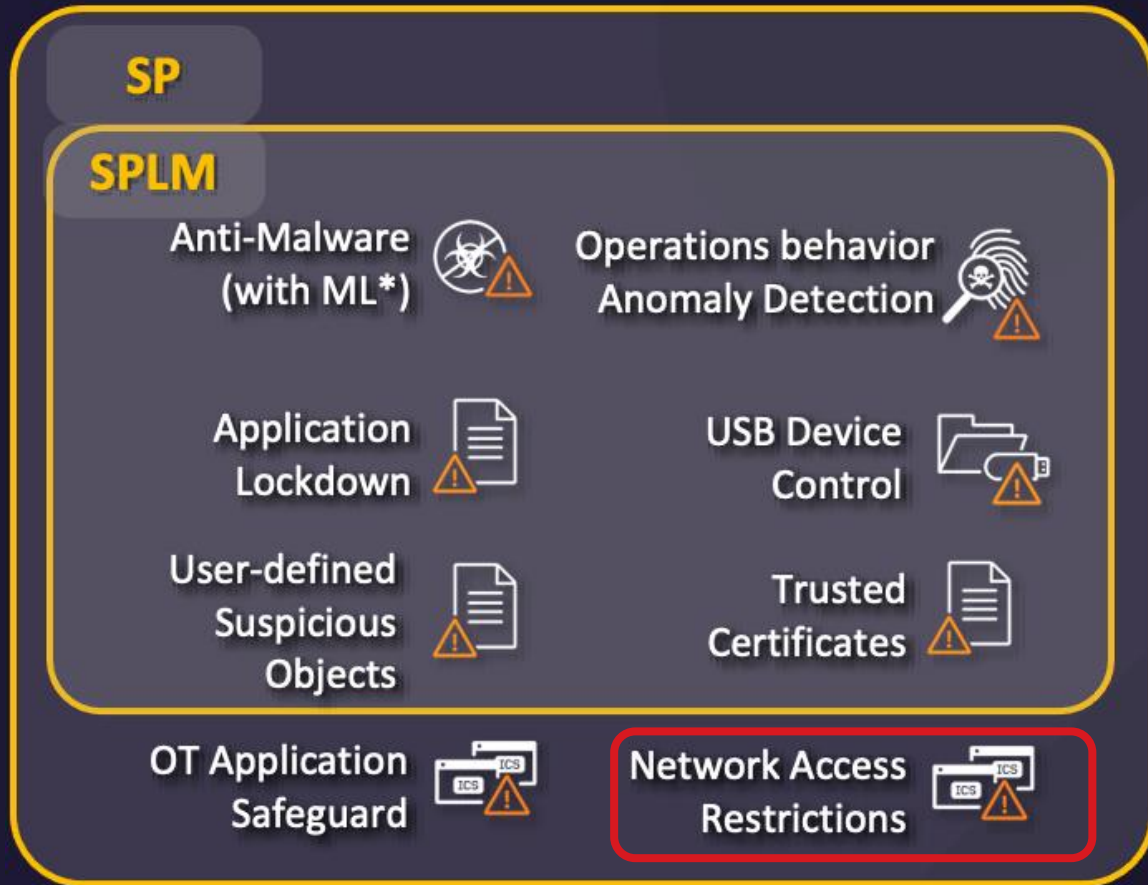
Lab16: User-defined Suspicious Objects

Security Controls: 6/8 Trusted, Signed Code ("Trusted Certificates")



- Simply checking the signature of a file against a trusted CA has a much lower performance impact than other checks. So the other security controls first check for a trusted signature before they process.
- Trusted Certificates is the FASTLANE to ensure an application will run
- Application Lockdown and OBAD/CPSDR Process allow trusted signed file even if those are not in the baseline.
- We only trust the immediate signer certificate on the file instead of entire chain of signer.
For example, DigCert > TXOne > good.exe, customer has to add TXOne certificate.

Security Controls: 8/8 Network Access Restrictions



- Prevents selected endpoints from connecting to specified IP addresses, ports or network.
- Prevents selected endpoints from accepting connections from specified IP addresses, ports or network
- (ports needed for DNS lookup and DHCP will always be permitted)

Network Access Restrictions

The managed endpoints will be blocked from connecting to the specified network, but the ports needed for DNS lookup and DHCP will be permitted. Implementing the network access restriction may impact the operational stability.

Traffic Direction* ☐ Inbound ☐ Outbound

IP Address

Port

Description

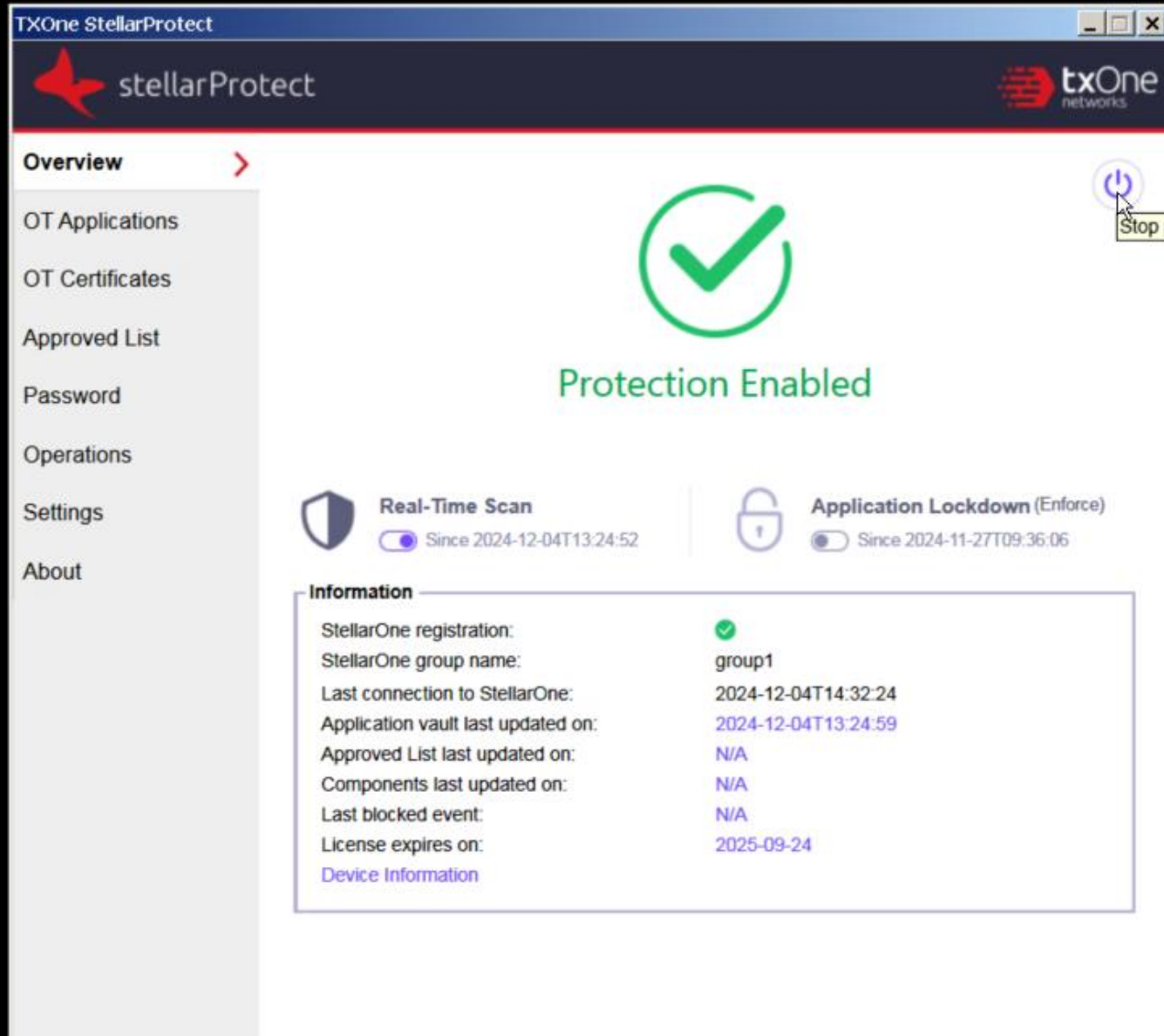
☐ I am aware that blocking network access may impact the operations.*

Add Cancel

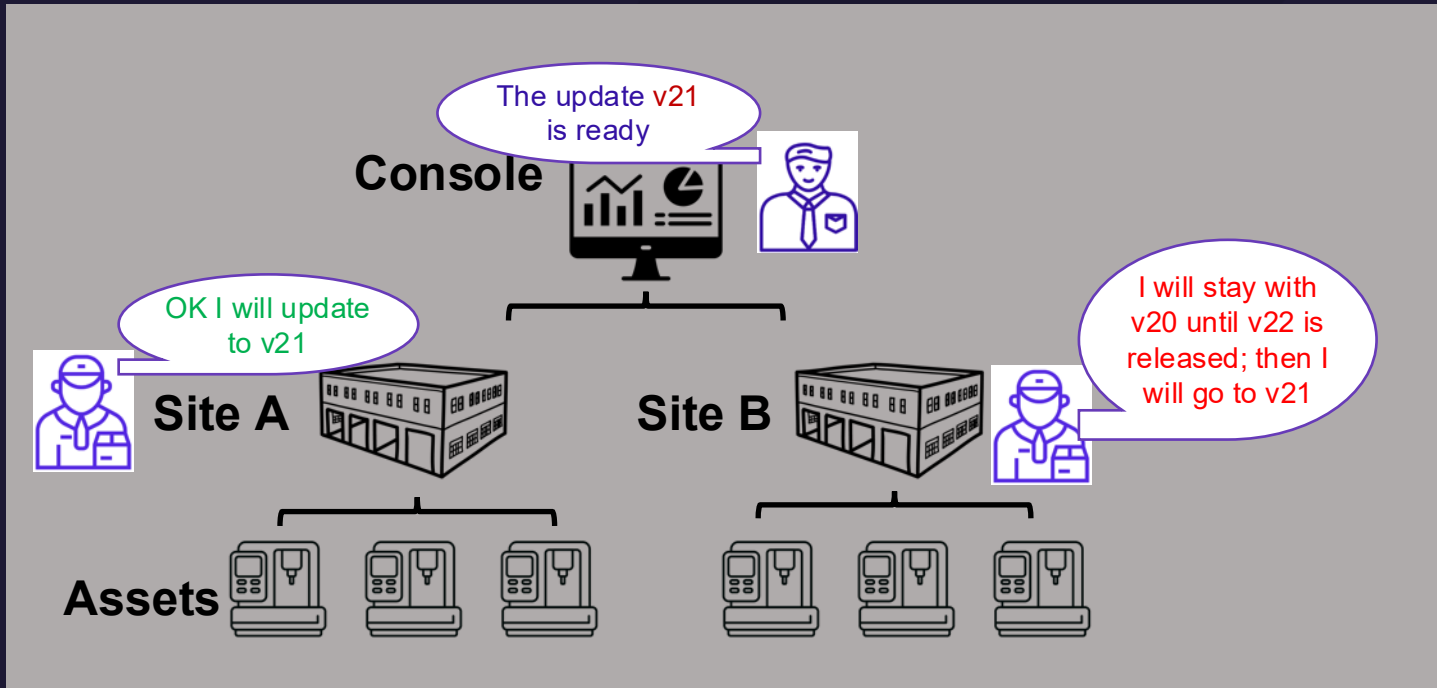


Operations

OT-native: Agent Emergency Stop



OT-native agent – Deferred update



2024 CrowdStrike incident

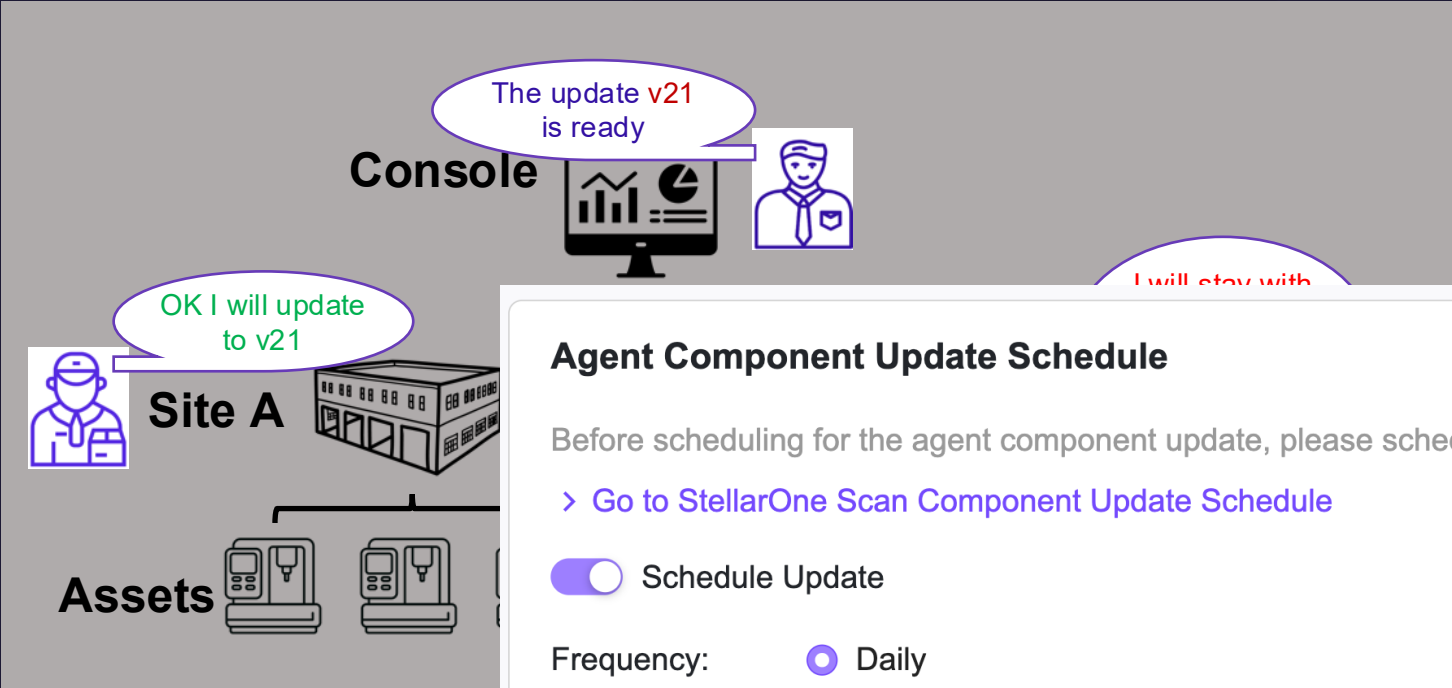


Multiple blue screens of death caused by a faulty software update on baggage carousels at LaGuardia Airport, New York City

Date	19 July 2024
Location	Worldwide
Type	IT outage, computer crash
Cause	Faulty CrowdStrike software update
Outcome	~8.5 million Microsoft Windows operating systems crash worldwide, causing global disruption of critical services

- A global policy is centrally defined
- Plant managers can customize the plant policy.
- At the same time, agent should have the flexibility of self-managemnt (decentralized).
- For urgent maintenance, the team leader in front of a system can temporarily modify settings.

OT-native agent – Deferred update



Agent Component Update Schedule

Before scheduling for the agent component update, please schedule for StellarOne's component update first.

> [Go to StellarOne Scan Component Update Schedule](#)

☒ Schedule Update

Frequency: ☒ Daily
☐ Weekly, every Sunday ▼
☐ Monthly, on day 01 ▼

Start Time: 11 ▼ : 30 ▼

☒ **Defer Component Update**
Instead of automatically deploying the latest version to the agent/group, StellarOne will manage to deploy only the previous version.

2024 CrowdStrike incident

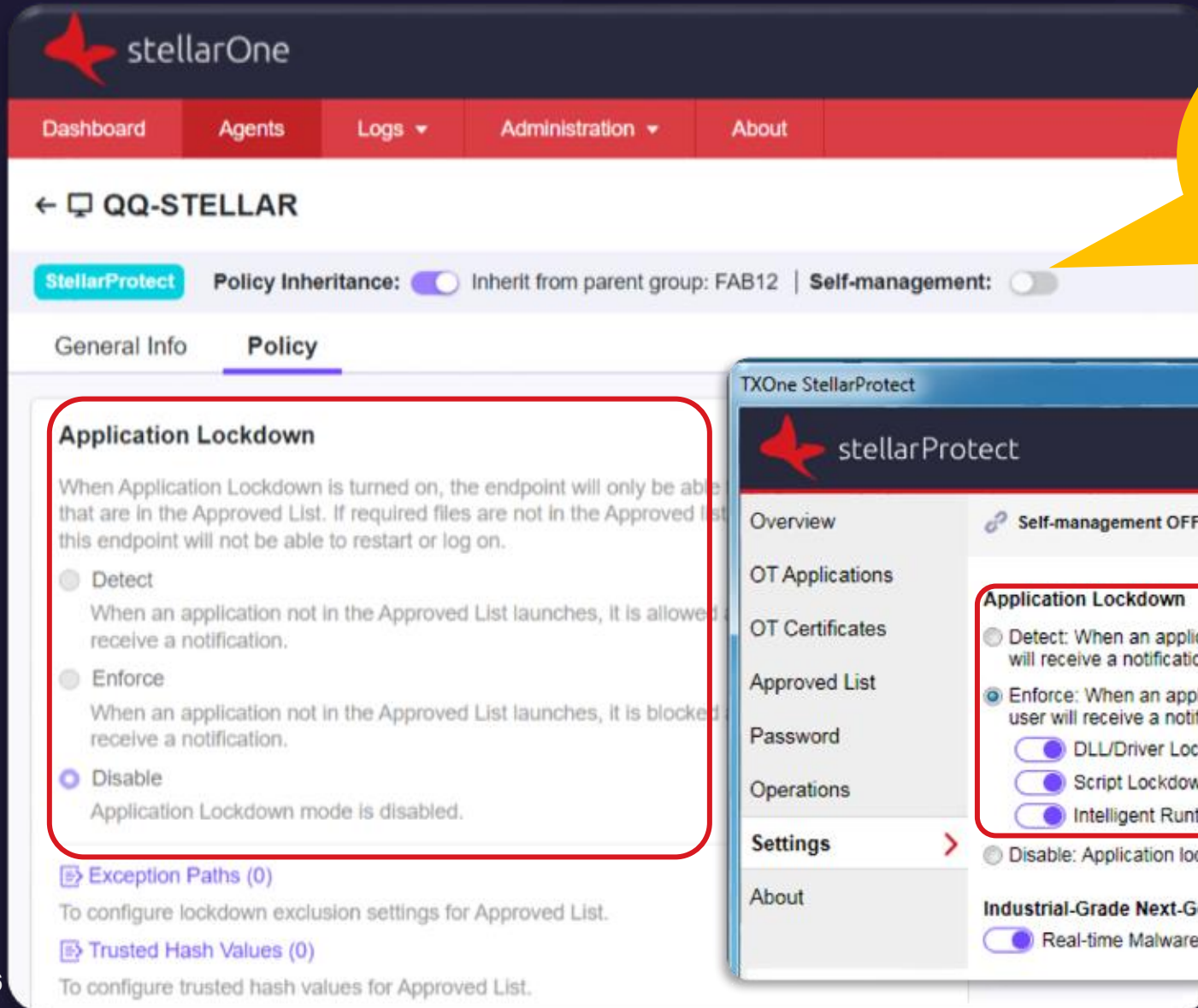


Multiple blue screens of death caused by a CrowdStrike update at New York City's LGA airport.

h
are update
ndows
worldwide,
of critical

- A global policy is centered on the agent
- Plant managers can control the update process
- At the same time, agents can be updated
- For urgent maintenance, agents can be updated

OT-native: Agents can be configured from StellarOne or from the Agent Console



stellarOne

Dashboard Agents Logs Administration About

← QQ-STELLAR

StellarProtect Policy Inheritance: ☒ Inherit from parent group: FAB12 Self-management: ☐

General Info Policy

Application Lockdown

When Application Lockdown is turned on, the endpoint will only be able to run applications that are in the Approved List. If required files are not in the Approved List, this endpoint will not be able to restart or log on.

☐ Detect
When an application not in the Approved List launches, it is allowed to run and the user will receive a notification.

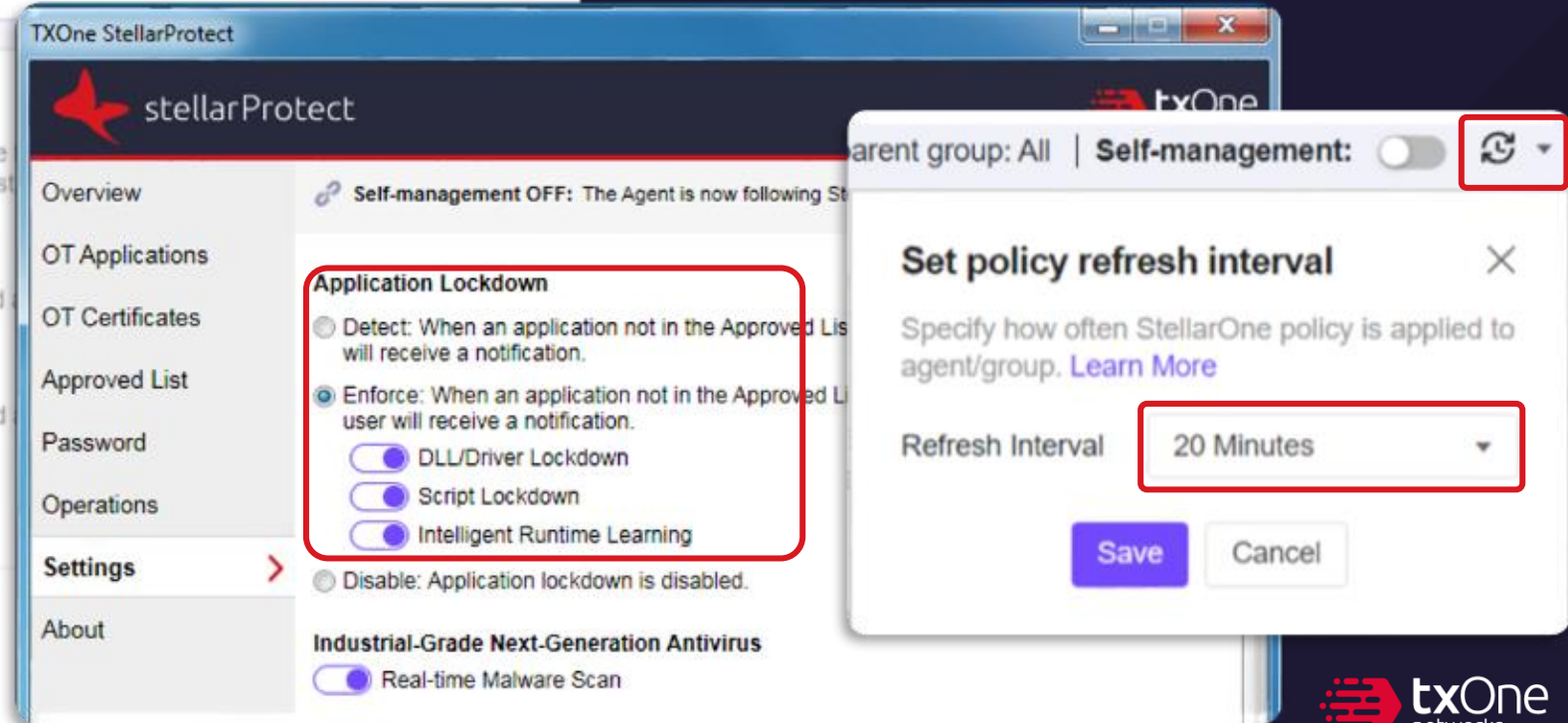
☐ Enforce
When an application not in the Approved List launches, it is blocked and the user will receive a notification.

☒ Disable
Application Lockdown mode is disabled.

[Exception Paths \(0\)](#)
To configure lockdown exclusion settings for Approved List.

[Trusted Hash Values \(0\)](#)
To configure trusted hash values for Approved List.

If this is not turned on, then any changes made **LOCALLY** on the agent **WILL BE REVERTED** at the next agent synchronization



TXOne StellarProtect

stellarProtect

parent group: All Self-management: ☐ 

Overview OT Applications OT Certificates Approved List Password Operations Settings About

Self-management OFF: The Agent is now following StellarOne policy

Application Lockdown

☐ Detect: When an application not in the Approved List launches, it is allowed to run and the user will receive a notification.

☒ Enforce: When an application not in the Approved List launches, it is blocked and the user will receive a notification.

☐ Disable: Application lockdown is disabled.

☒ DLL/Driver Lockdown

☒ Script Lockdown

☒ Intelligent Runtime Learning

Set policy refresh interval

Specify how often StellarOne policy is applied to agent/group. [Learn More](#)

Refresh Interval 20 Minutes

Save Cancel

Enable / Disable Agent Notifications

stellarOne

chriv (chriv)

txOne networks

Dashboard

Agents

Logs

Administration

About

← CHRIVTEST

StellarProtect

Policy Inheritance: ☐ Inherit from parent group: All

Self-management: ☐

General Info

Policy

Situational Awareness

Application Lockdown

When Application Lockdown is turned on, the endpoint... the Approved List. If required files are not in the Appro... able to restart or log on.

☒ Detect

When an application not in the Approved List launch... notification.

☐ DLL/Driver Lockdown

☒ Script Lockdown

☐ Intelligent Runtime Learning

☐ Enforce

When an application not in the Approved List launch... notification.

☐ Disable

Application Lockdown mode is disabled.

Excluded Paths (2)

To configure lockdown exclusion settings for Approved...

Trusted Hash Values (1)

To configure trusted hash values for Approved List.

Multi-Method Threat Prevention

OT root of trust and advanced threat scan secure the assets without the interruption on the operations.

☒ Real-Time Scan

☒ Predictive Machine Learning

Agent User Interface Preferences

Customize user interface settings on managed endpoints at the agent or group level.

Notifications

☐ Display a pop-up notification for a detected malware

Allow a notification to alert the agent user when a malware is detected.

☒ Display a pop-up notification for a blocked USB device

Allow a notification that enables the agent administrator to grant one-time access to a blocked USB device.

Accessibility

☒ Add the [Scan with TXOne Stellar...] option to the Windows context menu

Add a scan option to the right-click context menu that enables agent users to quickly run a scan for the selected file, folder, or drive.

Save

Cancel

Set Policy Refresh Interval

Agent User Interface Preferences

ADMINISTRATOR

USER

New Password*

Confirm Password*



Licensing

Stellar Licenses

Bundled features	ICS
Targeted users	Complete edition for Critical Infra, Manufacturing, Pharma
Industrial application repository	✓
OT application safeguard	✓
Intelligent runtime learning	✓
Next-generation antivirus	✓
Operations lockdown	✓
Operations behavior anomaly detection	✓
Trusted USB device control	✓
Support legacy systems	✓

What happens if the license expires?

1. 60days before license expiration: a yellow warning banner pops-up in StellarOne
2. 30day grace period
3. License expires:
security fails-open: (no more ApplicationLockdown, no more OT Application Safeguard, no more OBAD, no more Device Control, AV still works but no more updates)

Roles-based Access Control (RBAC) and Single Sign On (SSO using SAML)

RBAC: Create Account -> Assign Role -> Set Scope

RBAC: Create Account -> Assign Role -> Set Scope

The image displays a web interface for Role-Based Access Control (RBAC) management, divided into three main sections: Account Management (Users), Account Management (Roles), and Edit User Account.

Account Management (Users): This section shows a table of existing users. The table has columns for ID, Name, Role, Authentication Source, and Group Control. The users listed are admin, admin01, and admin02.

ID	Name	Role	Authentication Source	Group Control
admin	admin	Master Admin	Local	All, Australia, Cannington, FA...
admin01	admin_Hillside	Operator	Local	Hillside Aluminium
admin02	admin_GEMCO	Op		

Account Management (Roles): This section shows a table of existing roles. The table has columns for Role and Description. The roles listed are Admin, Operator, and Viewer. The 'Operator' role is highlighted with a red box.

Role	Description
Admin	User
Operator	Asset
Viewer	Read

Edit User Account: This section shows the details for editing a user account. The user selected is admin01. The details include Authentication Source (Local), Role (Operator), ID (admin01), Name (admin_Hillside), Local Password, Confirm Password, and Group Control. The Group Control section shows a list of groups with checkboxes. The 'Hillside Aluminium' group is selected with a red box.

Authentication Source: Local

Role: Operator

ID*: admin01

Name*: admin_Hillside

Local Password:

Confirm Password:

Group Control*:

- ☐ All (2)
- ☐ Australia (0)
 - ☐ Cannington (0)
 - ☐ GEMCO (0)
 - ☐ Perth Head Office (0)
 - ☐ Worsley Alumina (0)
 - ☐ FAB12 (0)
- ☒ Southern Africa (0)
 - ☒ Hillside Aluminium (0)
 - ☐ Johannesburg Office (0)
 - ☐ Mozal Aluminium (0)
 - ☐ South Africa Manganese (0)

SAML SSO

SAML SSO (e.g. with ADFS): Prerequisites

1. Active Directory Federation Services must be installed on Active Directory
2. For ADFS to work, both the user and the workstation need to be member of the domain.
3. StellarOne must be deployed on DNS name with an Enterprise Certificate that is trusted by both parties

SAML SSO (e.g. with ADFS): steps

1. Export Metadata XML file from StellarOne
2. In AD, Add Relying Party Trust and import the Metadata XML file
3. In AD: configure SAML assertions
make Issuance Transform Rules (e.g.: "E-Mail-Address" <-> "Email")
4. In AD export the metadata file
5. In StellarOne: Import the metadata file
6. In StellarOne: Create SSO-groups
 1. Map them to StellarOne roles
 2. Map them to AD Groups
7. Authenticate to StellarOne with corporate email address -> browser checks existing ADFS SAML token and asks to authenticate to AD first if there is no current token -> Browser forwards to SAML token to AD -> Logged in to Stellar

Security Assertion Markup Language (SAML) 2.0

- 1. SAML 1.0: (older Stellar versions)
Outsource Account Management to an Identity Provider
(e.g. Microsoft Active Directory Federation Services; AFDS)
- 2. SAML 2.0: (Stellar 3.3 and onwards)
Assign permissions to users based on their AD groups
and map those groups to roles in StellarOne.

 stellarOne

Dashboard

Agents

Logs ▾

Administration ▾

About

ACCOUNT MANAGEMENT

Users

SSO User Groups

Roles

Password Policy

Locked Users

+ Add Access Group

Group Name ▾	Role	Description	AD Group Name	Group Co
No data to display				

SSO - Azure AD Integration via SAML overview

The image displays two side-by-side screenshots illustrating the SSO integration process. The left screenshot shows the StellarOne 'Single Sign-On' configuration page, and the right screenshot shows the Microsoft Azure 'SAML-based Sign-on' configuration page. Red circles with numbers 1 through 5 highlight specific steps in the process.

StellarOne SAML Configuration (Left):

- 1** Download the StellarOne metadata XML file. Upload the StellarOne XML file to your IdP. (Callout 1 points to the Download button)
- 2** Upload the StellarOne XML file to your IdP. Sign in to your IdP console, upload the StellarOne XML file, and then download the IdP metadata XML file.
- 3** Upload the IdP metadata XML file. Upload the IdP metadata XML file to complete the SAML 2.0 single sign-on configuration. (Callout 5 points to the Upload button)

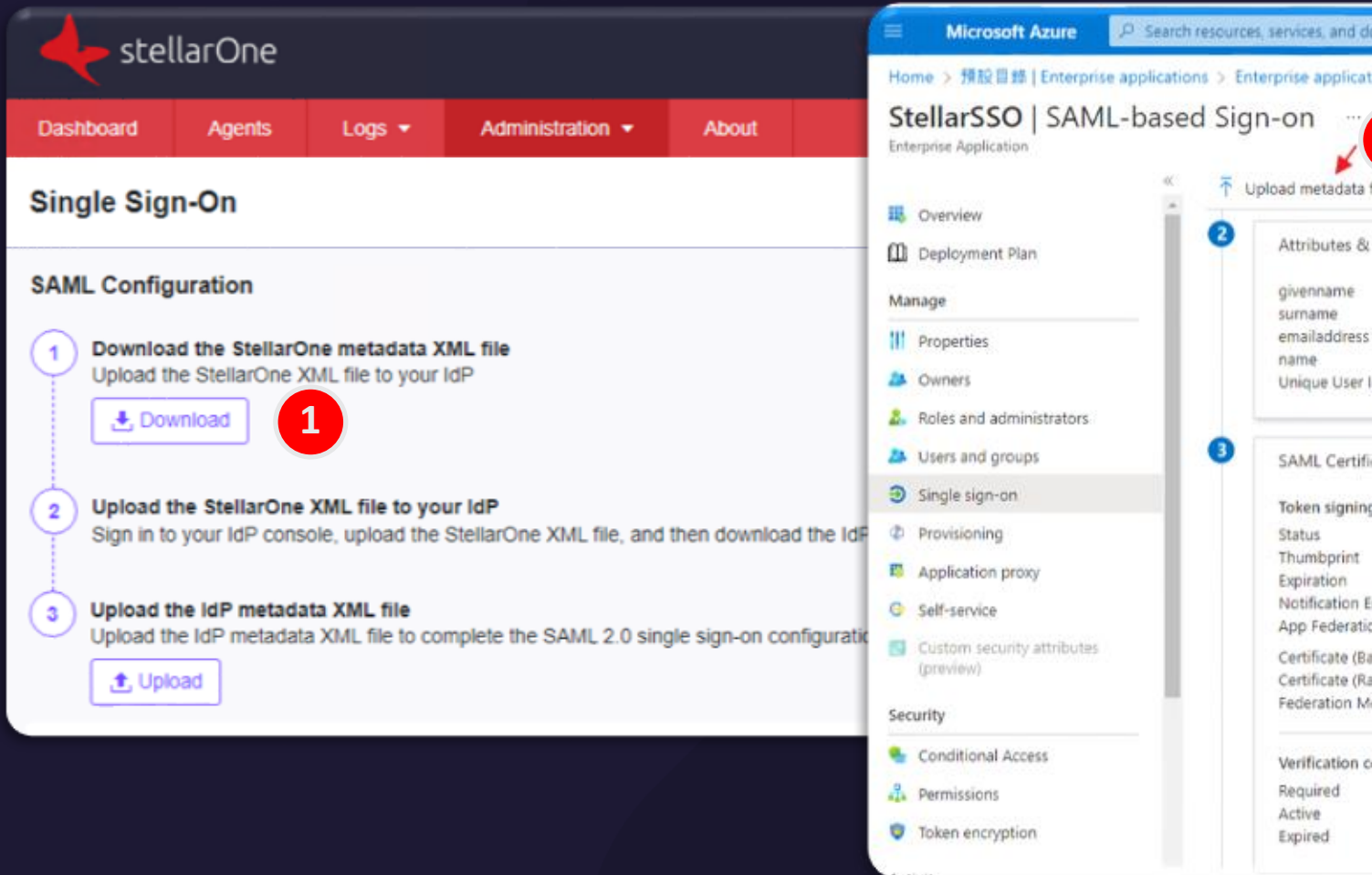
Microsoft Azure SAML-based Sign-on (Right):

- 2** Upload metadata file. (Callout 2 points to the Upload metadata file button)
- 3** Attributes & Claims. (Callout 3 points to the Attributes & Claims section)
- 4** SAML Certificates. (Callout 4 points to the Download button for Federation Metadata XML)

Create: "Issuance Transform Rules"

SSO - Azure AD Integration via SAML

3 Create:
"Issuance Transform Rules"



StellarOne identifies the user in its Assertion (metadata XML) by his email address

Setup an "Issuance Transform Rule" to include the AD user's email address in the Assertion

Rule1:

Claim rule template:

Send LDAP Attributes as Claims

Attribute Store: **Active Directory**

LDAP attr: **E-Mail-Addresses**

Outgoing Claim Type: **E-Mail Address**

Rule2:

Claim rule template:

Transform an Incoming Claim

Incoming claim type: **E-Mail Address**

Outgoing claim type: **Name ID**

Outgoing name ID format: **Email**

Pass through all claim values

Step5: update the IdP metadata XML file in StellarOne

The image displays two side-by-side screenshots. The left screenshot shows the StellarOne 'Single Sign-On' configuration page. It features a 'SAML Configuration' section with three numbered steps: 1. Download the StellarOne metadata XML file, 2. Upload the StellarOne XML file to your IdP, and 3. Upload the IdP metadata XML file. Step 3 is highlighted with a red circle containing the number 5. The right screenshot shows the Microsoft Azure portal for 'StellarSSO | SAML-based Sign-on'. It includes a sidebar with navigation options like Overview, Deployment Plan, and Manage. The main content area shows 'Attributes & Claims' and 'SAML Certificates'. A red arrow points to the 'Download' link for the 'Federation Metadata XML' file under the 'SAML Certificates' section.

StellarOne SAML Configuration Steps:

- 1 Download the StellarOne metadata XML file
Upload the StellarOne XML file to your IdP
[Download](#)
- 2 Upload the StellarOne XML file to your IdP
Sign in to your IdP console, upload the StellarOne XML file, and then download the IdP
- 3 Upload the IdP metadata XML file
Upload the IdP metadata XML file to complete the SAML 2.0 single sign-on configuration
[Upload](#)

Microsoft Azure SAML-based Sign-on Configuration:

Attributes & Claims

givenname	user.givenname
surname	user.surname
emailaddress	user.mail
name	user.userprincipalname
Unique User Identifier	user.userprincipalname

SAML Certificates

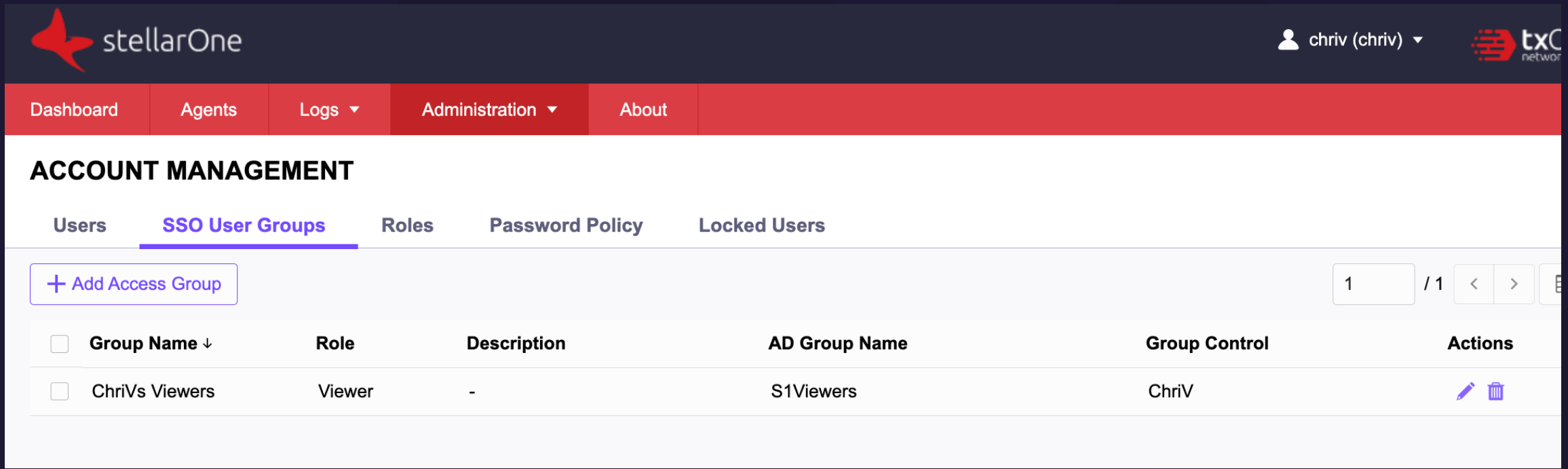
Token signing certificate

Status	Active
Thumbprint	EDF570143CDA201F6CAB16343268BF2452F5FDCE
Expiration	5/3/2025, 1:22:04 PM
Notification Email	DavidHsu0328@hotmail.com
App Federation Metadata Url	https://login.microsoftonline.com/b016e5ba-9b4d...
Certificate (Base64)	Download
Certificate (Raw)	Download
Federation Metadata XML	Download

Verification certificates (optional) (Preview)

Required	No
Active	0
Expired	0

Step6: Create "SSO User Groups" in Stellar



The screenshot shows the StellarOne web interface. At the top, there's a navigation bar with the StellarOne logo, a user profile 'chriv (chriv)', and the txOne network logo. Below this is a red navigation bar with links: Dashboard, Agents, Logs, Administration (selected), and About. The main content area is titled 'ACCOUNT MANAGEMENT' and has tabs for Users, SSO User Groups (selected), Roles, Password Policy, and Locked Users. Under the SSO User Groups tab, there's a '+ Add Access Group' button and a table listing existing groups. The table has columns: Group Name, Role, Description, AD Group Name, Group Control, and Actions. One group is listed: 'ChriVs Viewers' with Role 'Viewer', Description '-', AD Group Name 'S1Viewers', and Group Control 'ChriV'. The Actions column shows edit and delete icons.

☐ Group Name ↓	Role	Description	AD Group Name	Group Control	Actions
☐ ChriVs Viewers	Viewer	-	S1Viewers	ChriV	

Administration > Account Management > SSO User Groups -> Add Access Groups -> Name, Role, AD Group Name

Step7: SAML SSO login proces.

Identity Provider
e.g. ADFS

End User
Browser

Service Provider
e.g. StellarOne

End User accesses StellarOne (the SP)
and enters email address

StellarOne redirects browser to ADFS(the IdP)

StellarOne redirects browser to ADFS

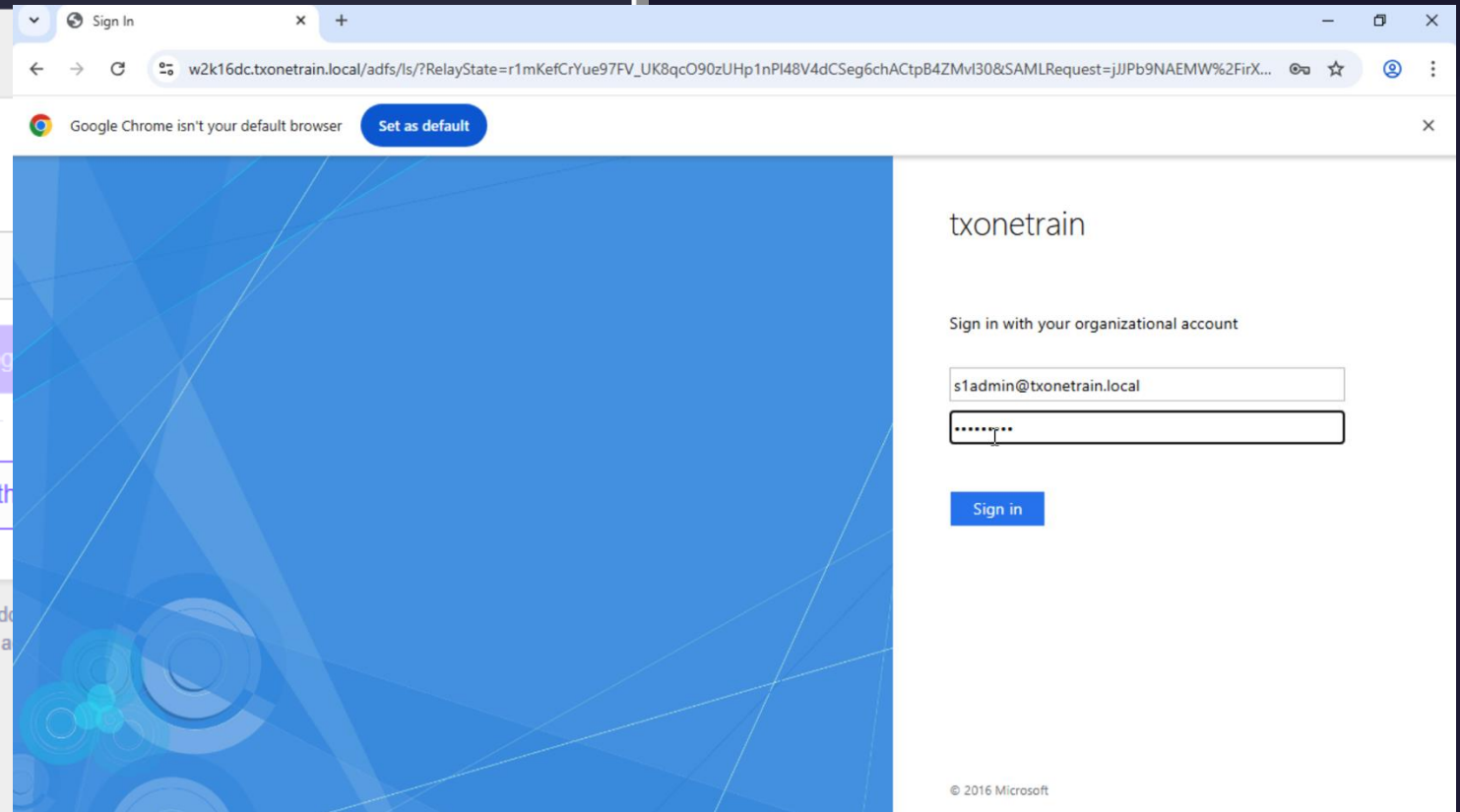
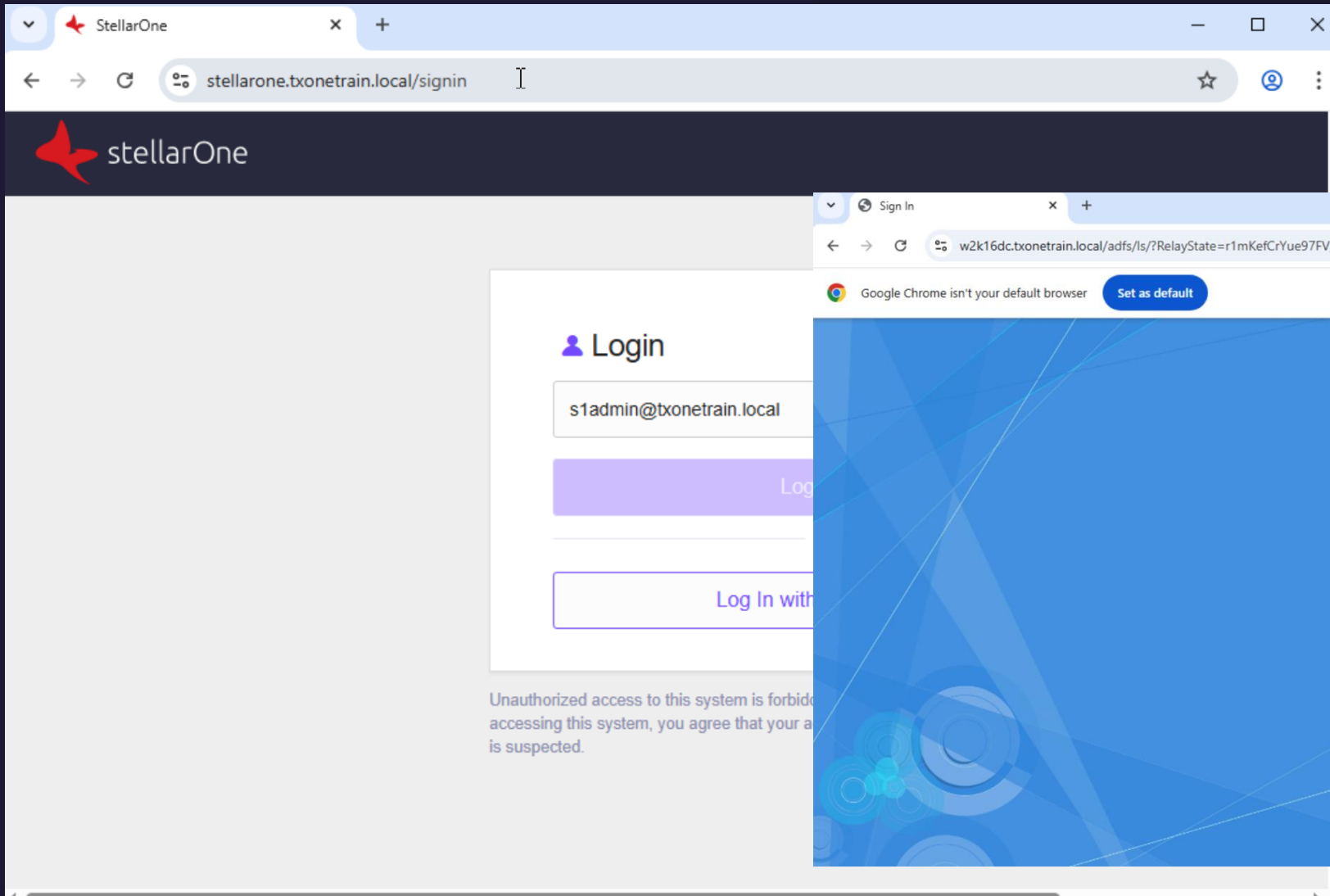
If not already authenticated to AD, the
user authenticates to AD (the IdP)

ADFS generates SAML assertion and
sends it back to the browser

Browser sends SAML assertion to StellarOne

StellarOne checks assertion and
authenticates the user

Login with Domain Account



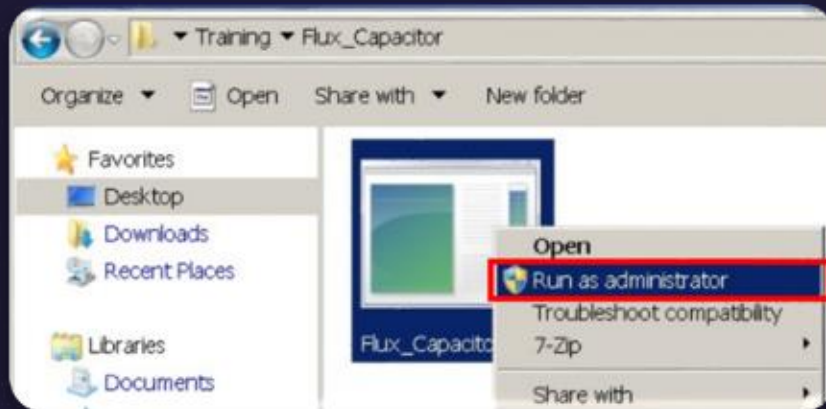
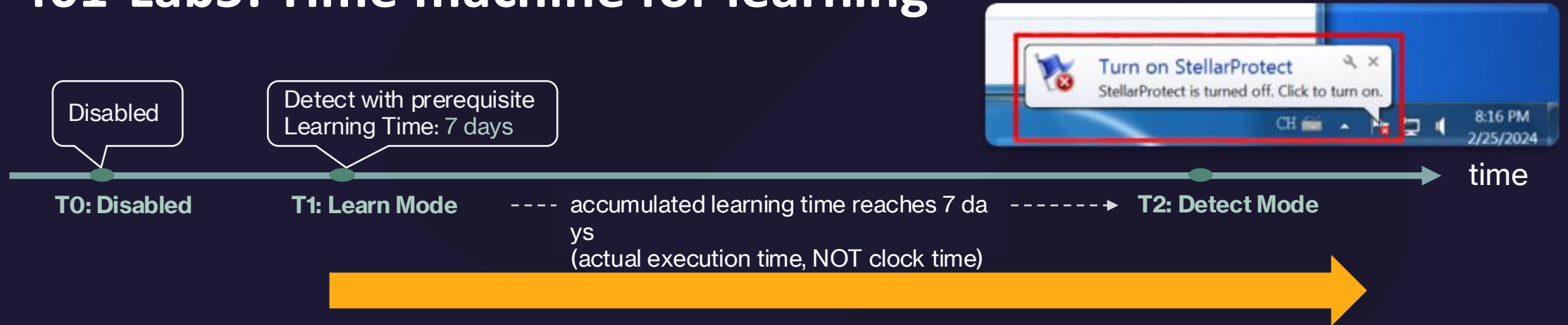
Lab exercises – Stellar - features

Before we begin:
some notes:

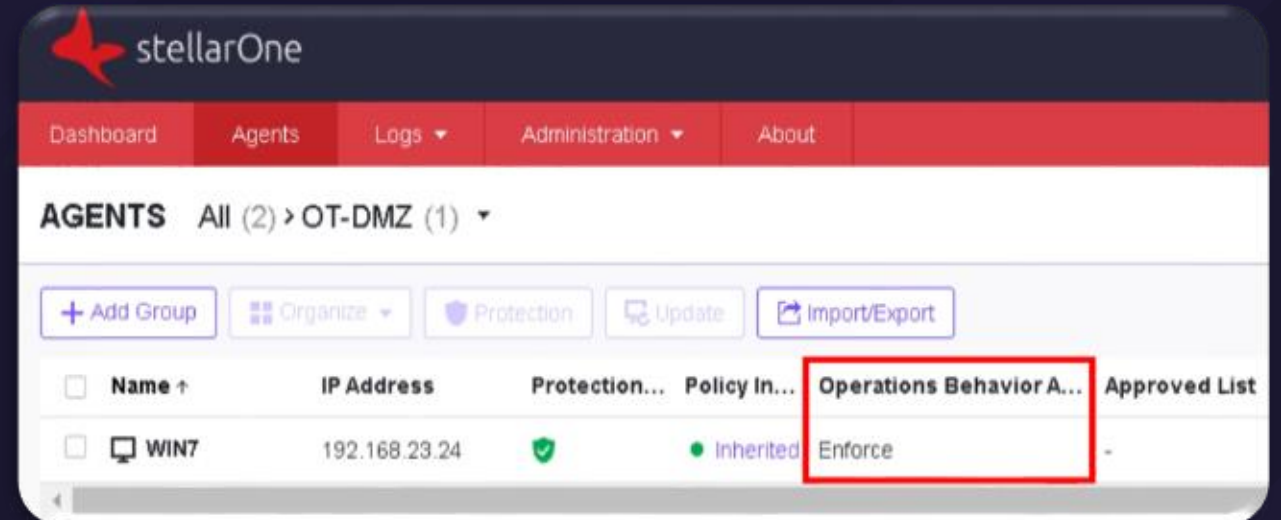
401-lab1: The management group design criteria

Group Name	OT-DMZ	HMI	HVAC
Management Requirement	<u>StellarProtect</u> 1. Customized policy 2. App lockdown disabled 3. Real-Time Scan enabled 4. Predictive ML enabled 5. OBAD Enforced 6. Strict Mode disabled 7. Policy refresh interval set to 5 minutes	<u>StellarProtect</u> 1. Customized policy 2. Agent component update schedule by weekly; Every Sunday, 20:00	<u>StellarProtect Legacy Mode</u> 1. Customized policy 2. Real-Time Scan disabled 3. Schedule Scan disabled 4. App lockdown enabled 5. Agent Component Update Schedule disabled 6. OBAD disabled
Group admin role	Admin	Operator	Operator
Management account	MasterAdmin	Admin_hmi 12345678	Admin_hvac 12345678

401-Lab5: Time machine for learning



Execute the Flux_Capacitor.exe with **Administrator rights**



401-Lab9



Dashboard

Agents

Logs ▾

Administration ▾

About

←  ALL (11)

Function Type

StellarProtect (Legacy Mode) ▾

Policy Inheritance: Customized

Self-management:



Policy ▾

StellarProtect

StellarProtect

✓ StellarProtect (Legacy Mode)

Set as default

Application Lockdown

When Application Lockdown is turned on, the endpoint will only be able to access applications that are in the Approved List. If required files are not in the Approved list, there is a risk that this endpoint will not be able to restart or log on.



Application Lockdown



DLL/Driver Lockdown



Script Lockdown



Write Protection



Also protect Approved List



Fileless Attack Prevention



Intelligent Runtime Learning

Device Control (0)

Block external device s



Device Control



Trusted USB Device

The USB devices add

User-Defined Susp

Update new threat info

Lab exercises – Stellar - features

Stellar Installation Hands-on Labs

301StellarLabGuideTxOneSpecialis.....

- Lab1: StellarOne VM Deployment
Install: **TXOne-S1-3.1.1264.ova**;
it is in the **VMs andVMs_Updates_and_Patches** folder
- Lab2: Basic configuration of StellarOne
- Lab3: Stellar Agent Management using StellarOne
- Lab4: StellarProtect Agent installation
- Lab5: StellarProtect Agent Silent installation
- Lab6: StellarOne and StellarProtect upgrade
(use **TXOne-S1-acus_fw_3.2.4123.acf**
and **txsp-3.2-agent-b1191-patch-en.zi**)
- Files are in the **VMs andVMs_Updates_and_Patches**

You can ignore the other labs in this labguide. They are repeated, but with more detail in the 401 document

401Stellar Configuration Labs (401 series)

- Lab1: RBAC (Role Based Access Control) management
- Lab2: Antivirus
- Lab3: OT Application Safeguard
- Lab4: Upgrade OT Application using the Maintenance Mode
- Lab5: Operations Behavior Anomaly Detection (OBAD)
- Lab6: CPSDR-Cyber Physical System Detection and Response
- Lab7: Application Lockdown
- Lab8: Update the Application Lockdown Approved List
- Lab9: Setup the exception path for the Application Lockdown
- Lab10: User-defined Suspicious Objects
- Lab11: Updating virus patterns in air gapped networks
- Lab12: Update-agents and Update-pools
- Lab13: Using a proxy for StellarOne Updates
- Lab14: Using a proxy for Agent to StellarOne communication (includes Wireshark)
- Lab15: Stellar Agent transfer to a new StellarOne
- Lab16: StellarProtect Stand alone agent installation and management

Quiz: Can you name the 8 Security Controls in Stellar ?

Security Controls

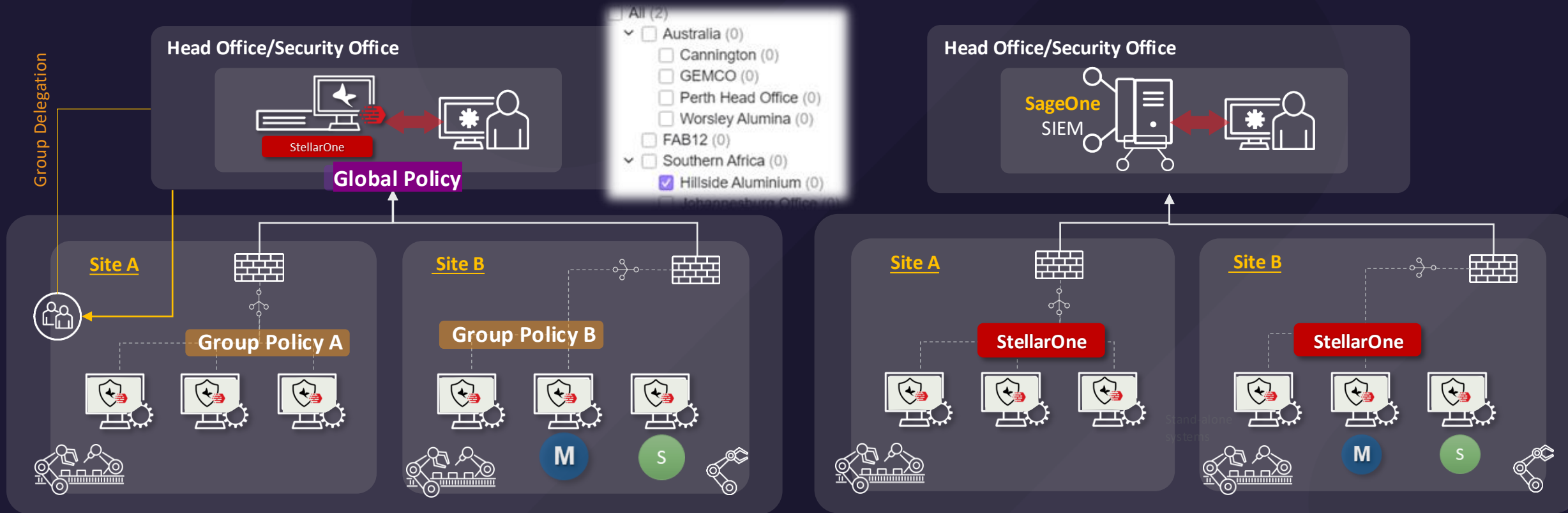
1. Anti-Virus: (Multi-Method Treat Prevention)
2. Lockdown: (Application Lockdown + OT Application Safeguard)
3. Maintenance Mode (Operations)
4. Behavior Monitoring (OBAD: Operations Behavior Anomaly Detection)
 - Scripts, Applications, , User Login, Network Activity
5. USB Storage Device Control
6. Trusted Code (code-signed by trusted certificates)
7. User-defined Suspicious Objects
8. Network Access Restrictions

Deployment best practices

Reference Deployment model

"Single" management console at head office with RBAC

"Multiple" management consoles at each site

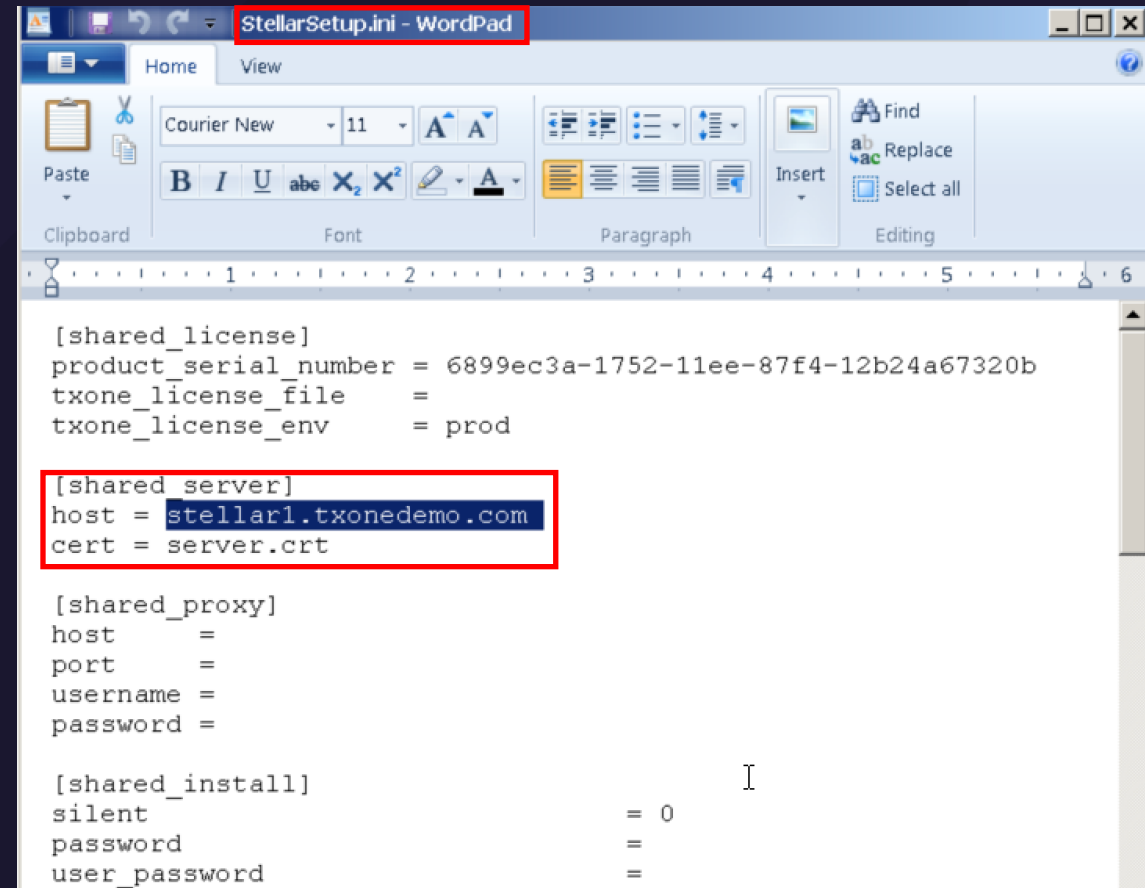


S Standalone installation

M Agent managed by StellarOne

Deploying Stellar by DNS name ("advertise-addr")

```
(root) $ env advertise-addr mystellarone.local
vShell: unable to resolve address mystellarone.local or it's an invalid IP. Try to add --force after
the address to force the setup.
(root) $ env advertise-addr mystellarone.local --force
Successfully set up advertise address: mystellarone.local.
Type 'service reload' to restart the service for reloading the specified configuration.
(root) $ service reload
Start to reload services...
(root) $ env ls
Hostname:          ODC
Status:            RUNNING
```



```
StellarSetup.ini - WordPad

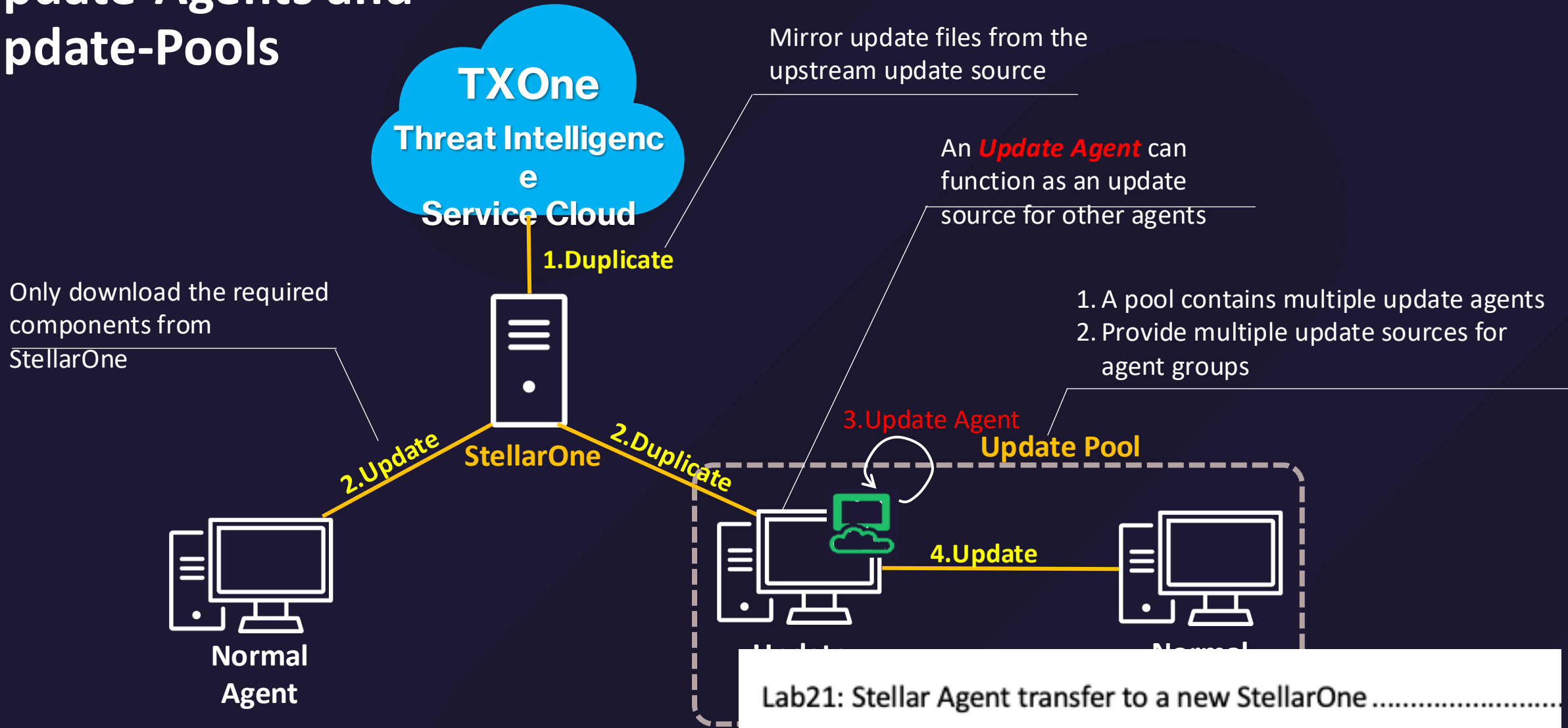
[shared_license]
product_serial_number = 6899ec3a-1752-11ee-87f4-12b24a67320b
txone_license_file    =
txone_license_env     = prod

[shared_server]
host = stellar1.txonedemo.com
cert = server.crt

[shared_proxy]
host =
port =
username =
password =

[shared_install]
silent = 0
password =
user_password =
```

Update-Agents and Update-Pools



Lab21: Stellar Agent transfer to a new StellarOne

nginx:
tcp 14337



Troubleshooting

<https://help.txone.com/>

Web interface not coming online

Web interface not coming online

Forgot to add 2nd disk



```
If you want to exit this shell, please type `exit` or `Ctrl-D`.
$ env ls
Hostname: nnc
Status: FAIL: NO EXTERNAL DISK DETECTED
Product Serial Number: d1368284-2cf8-11ed-ae40-0000c29edfc47
Version: 1.5.8
External IP: Not Set
DPI Engine Version: 2.0.11.33e2e1
DPI Pattern Version: SDP_211215_12
Service Network: config: 169.254.0.0/16, network: [169.254.0.0/24]
$
```

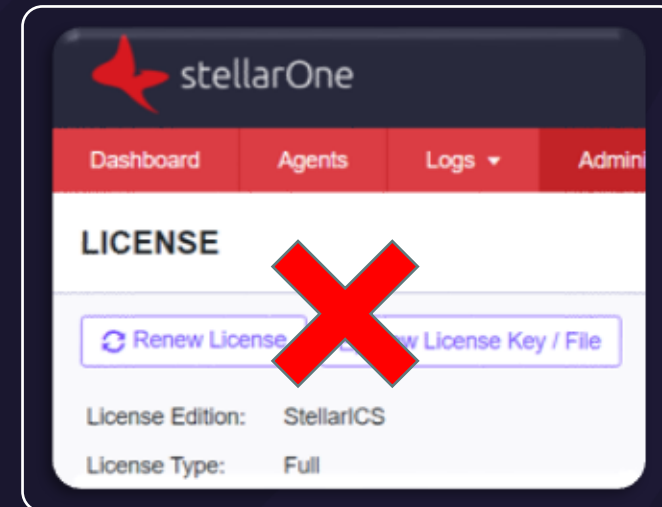
Agents not showing in the StellarOne web UI

Time sync issue



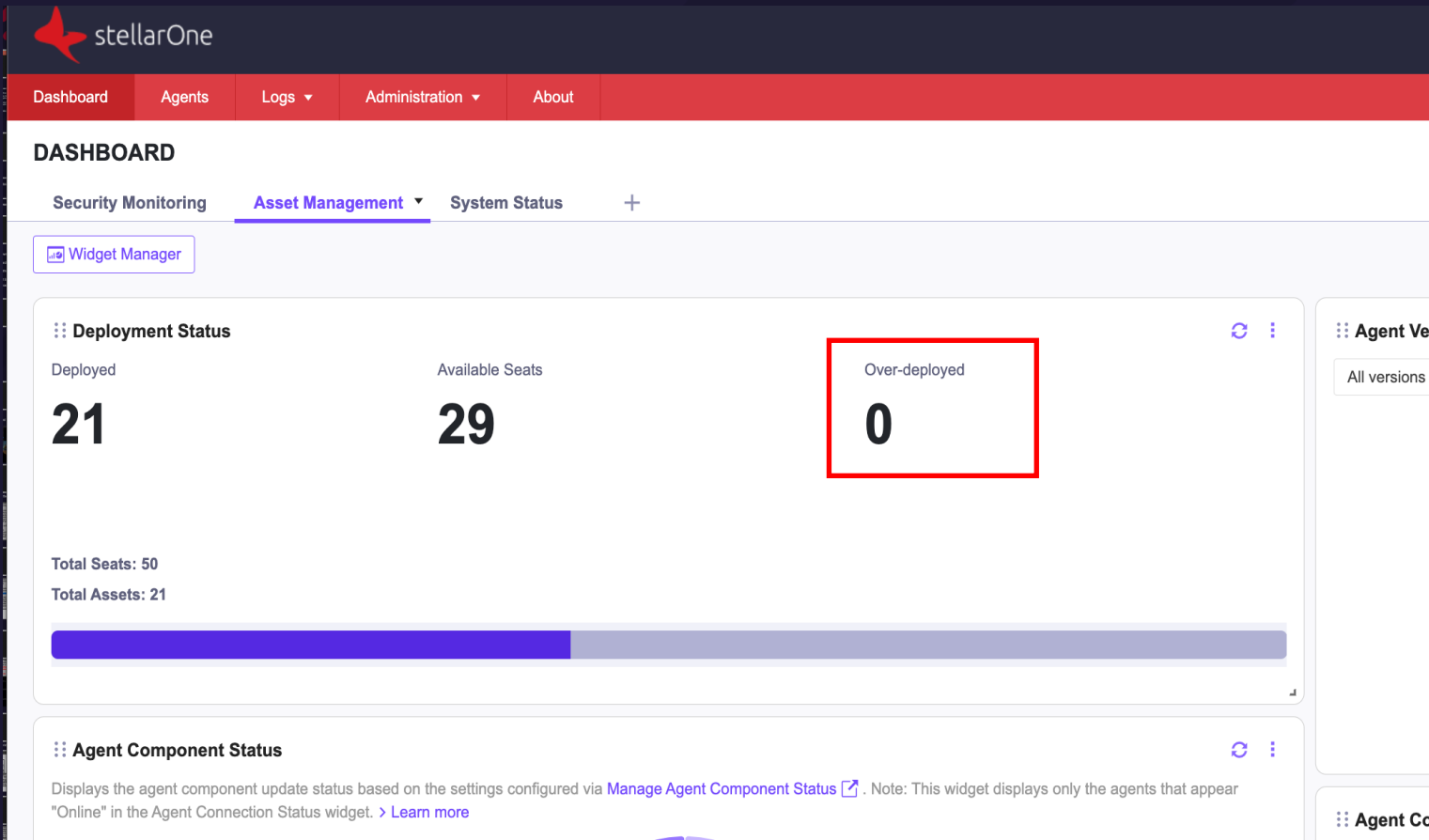
The timestamps not in sync between StellarOne and Agents

License issue



- The license expired
- Wrong license key
- overbooked

Agents not showing in the StellarOne web UI



between StellarOne and
Agents

License issue



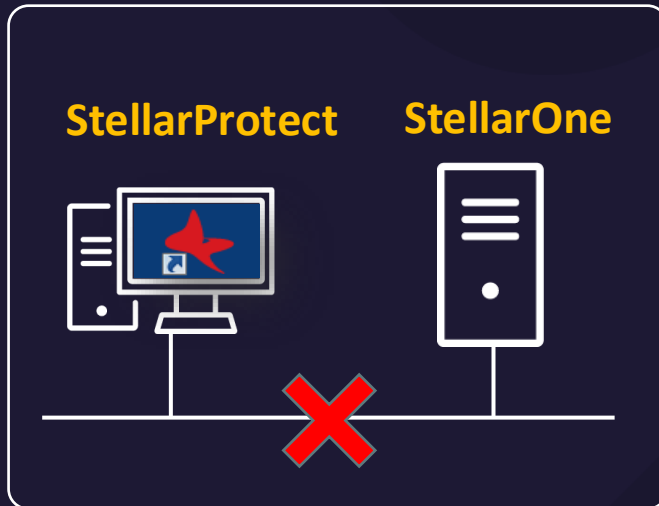
The license expired

- Wrong license key
- overbooked

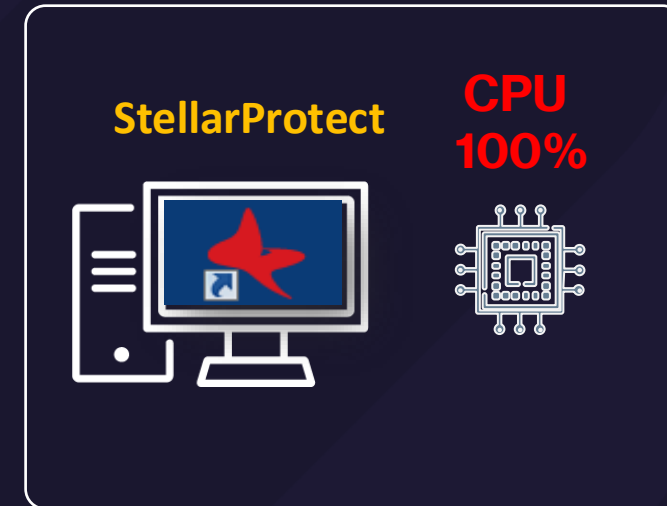
Agents not showing in the StellarOne web UI

Agents not showing in the StellarOne web UI

Network connection issue



Client OS resources exhausted



- Firewall denies access from agents to StellarOne
- StellarOne behind NAT
- Proxy in path but not configured in Agent installer package (StellarProtect.ini)

Forgotten the password for the Web UI

Forgotten the password for the Web UI

Edge

```
If you want to exit this shell, please type `exit` or `Ctrl-D`.  
$ web reset admin  
admin
```

Stellar

```
If you want to exit this shell, please type `exit` or `Ctrl-D`.  
(root) $ web reset admin  
admin
```

Element

```
$ elementone web  
reset  Reset the password of the web master admin's account.  
stpwd  Set up the password of Scanning Tools for registration use. (use "stpwd help" for det...  
help   List all commands.
```

Forgotten the password for the vShell (the StellarOne CLI)

Forgotten the password for the vShell (the StellarOne CLI)

Backup StellarOne disk image included Disk 1 & 2



Install new StellarOne assign same IP address and add "old" Disk2



Logon on to vshell with default password root / txone

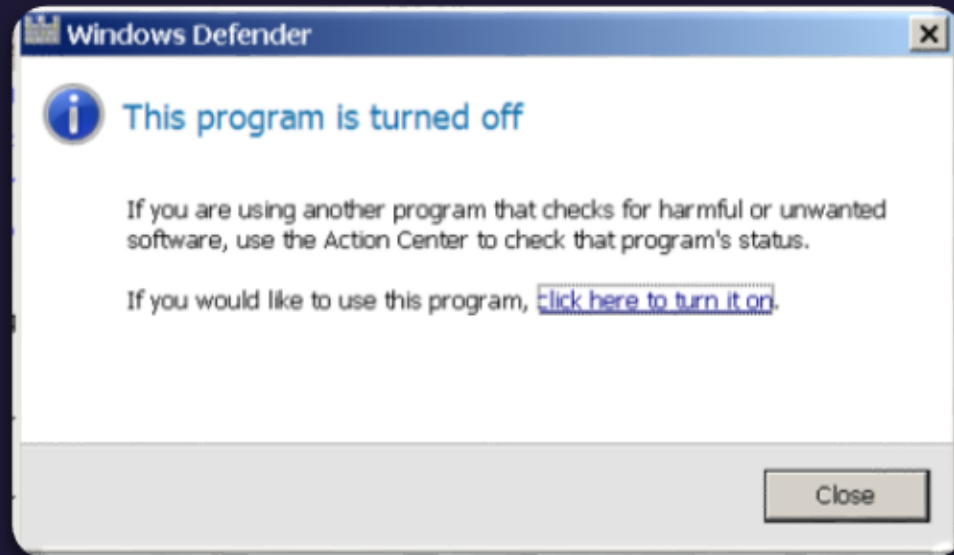
```
===== TXOne Networks StellarOne =====  
  
WARNING: Unauthorized access to this system is forbidden and will  
prosecuted by law. By accessing this system, you agree that your  
may be monitored if unauthorized usage is suspected.  
  
ODC login: root  
Password:  
Linux ODC 5.10.0-8-amd64 #1 SMP Debian 5.10.46-5 (2021-09-23) x86_64  
  
The programs included with the Debian GNU/Linux system are free software;  
the exact distribution terms for each program are described in the  
individual files in /usr/share/doc/*/copyright.  
  
Debian GNU/Linux comes with ABSOLUTELY NO WARRANTY, to the extent  
permitted by applicable law.  
vShell, version ede583c  
  
If you want to exit this shell, please type `exit` or `Ctrl-D`.  
  
Caution: please type the command ``oobe`` to activate the vShell.  
Caution: please type the command ``oobe`` to activate the vShell.  
Caution: please type the command ``oobe`` to activate the vShell.  
Caution: please type the command ``oobe`` to activate the vShell.  
Caution: please type the command ``oobe`` to activate the vShell.  
(root) $ oobe
```

Stellar agents fail to install due to Windows Defender

Stellar agents fail to install due to Windows Defender

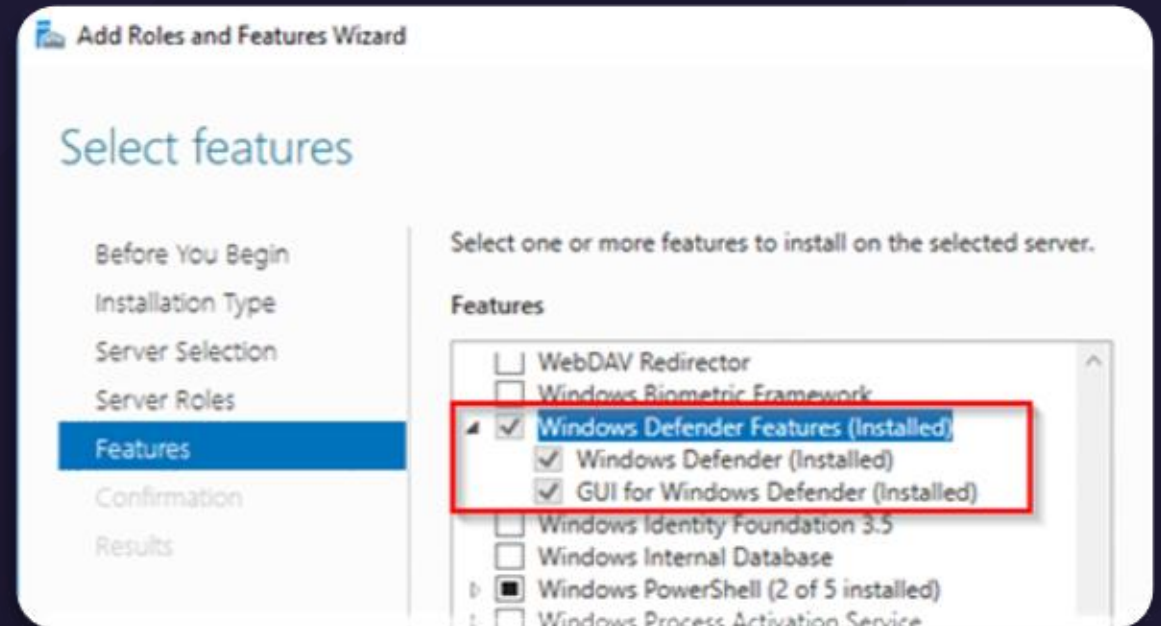
Windows Workstation

Windows Defender is being turned off automatically by the Stellar Agent



Windows Server

Manually *Uninstall* Windows Defender feature (just *disabling it*, is NOT enough)



Stellar Agent Debug Log Collection when installation fails

(Stellar v3.0 SP1 and newer)

StellarProtect / StellarProtect Legacy Mode installation fails

- Modify StellarSetup.ini to enable debug log collection and try the installation again.

Default Value

```
[shared_proxy]
host =
port =
username =
password =

[shared_install]
silent = 0
password =
user_password =

[shared_debug]
enable_debug_log = 0
enable_engine_debug_log = 0

[protect_server]
port = 9443

[protect_listen]
port = 14336
```

Parameters and values

enable_debug_log: 0 (error only log) | 1 (verbose log)

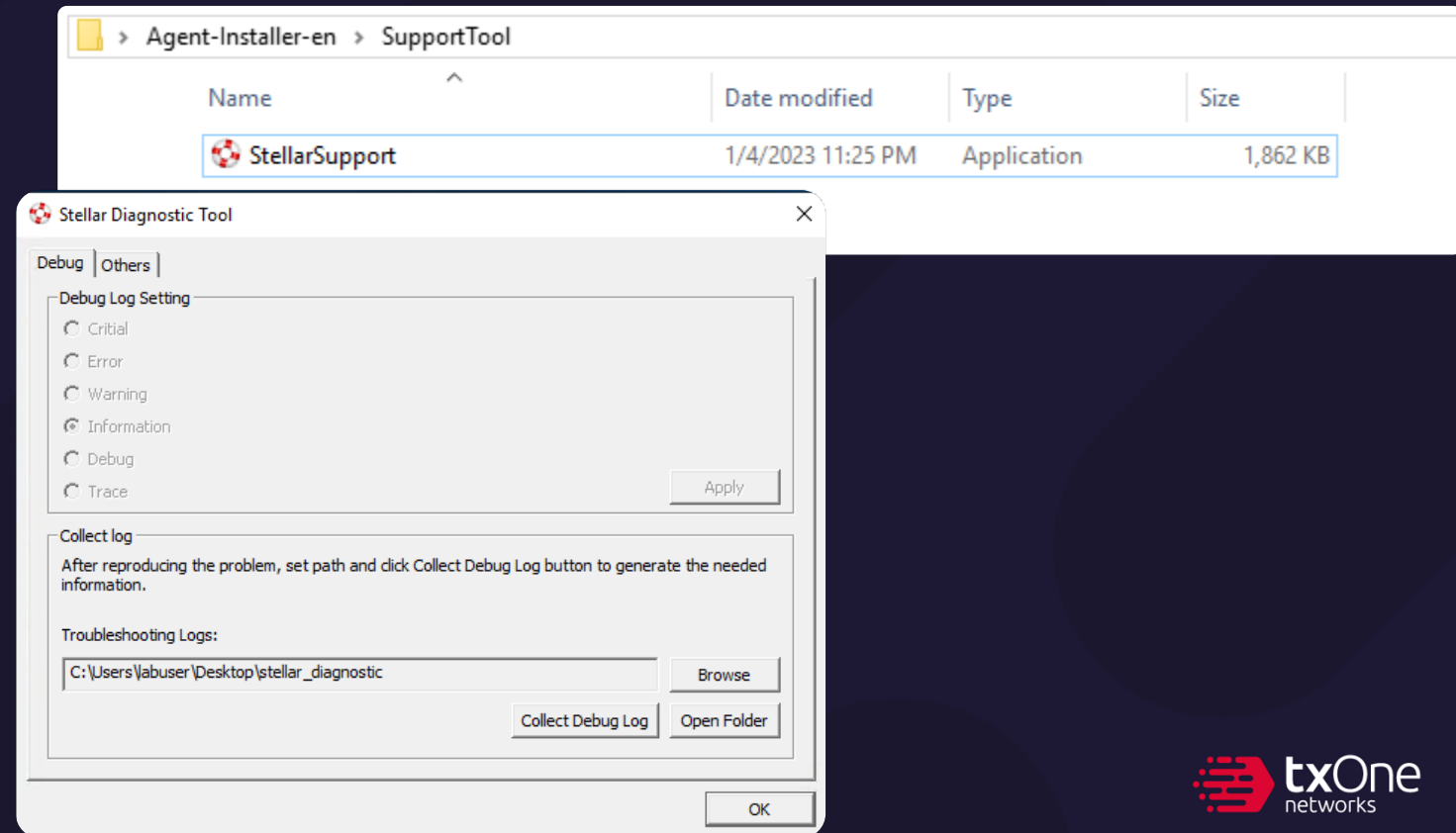
enable_engine_debug_log: 0 (no log) | 1 (verbose log)

enable_debug_log: 1
enable_engine_debug_log: 1

Please only modify these two debug settings for troubleshooting.
DO NOT MODIFY in normal installation to avoid performance issues

StellarProtect / StellarProtect Legacy Mode installation fails

- Collect the StellarProtect logs using **StellarSupport.exe**
 - Modify the StellarSetup.ini (see previous slide)
 - Re-run the installer to reproduce the issue, and do not close any installation windows the installation fails
- Execute StellarSupport.exe (in the Agent Installer Package)
- Click “Collect Debug Log”
- Click “Open Folder” to get the archived zip file

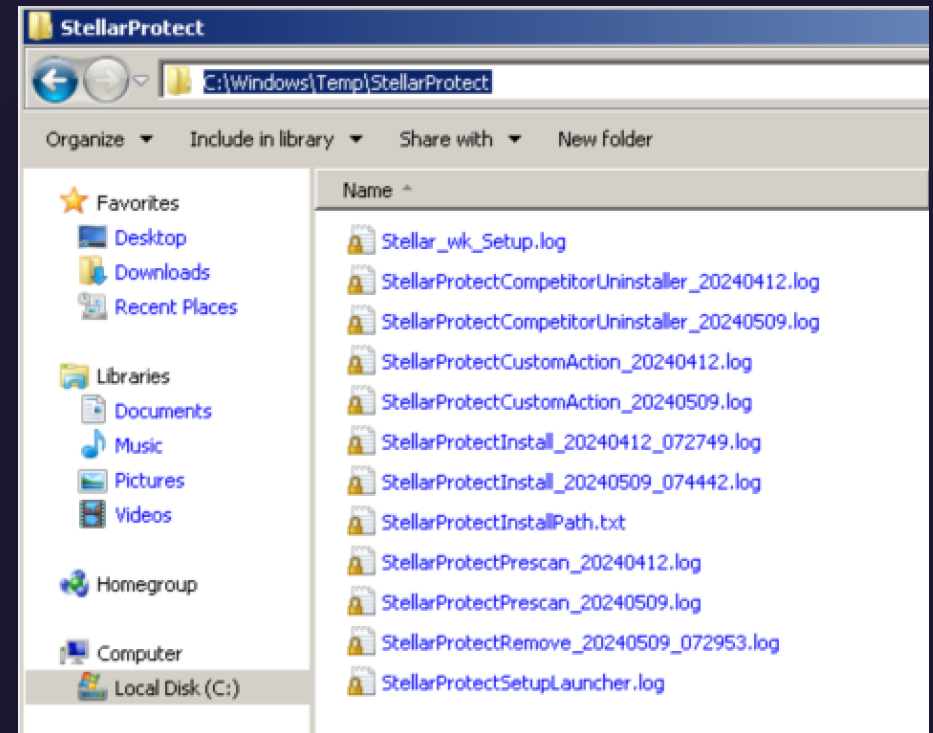


StellarProtect / StellarProtect Legacy Mode installation fails

- Locally, the agent-installation-log files can be found under:
c:\Windows\Temp\StellarProtect

- Pro tip:
to find out in which log file to look for errors,
run the following powershell command:

```
Get-ChildItem "C:\Windows\Temp\StellarProtect" -Filter "*.log" -  
Recurse | Select-String "error"
```

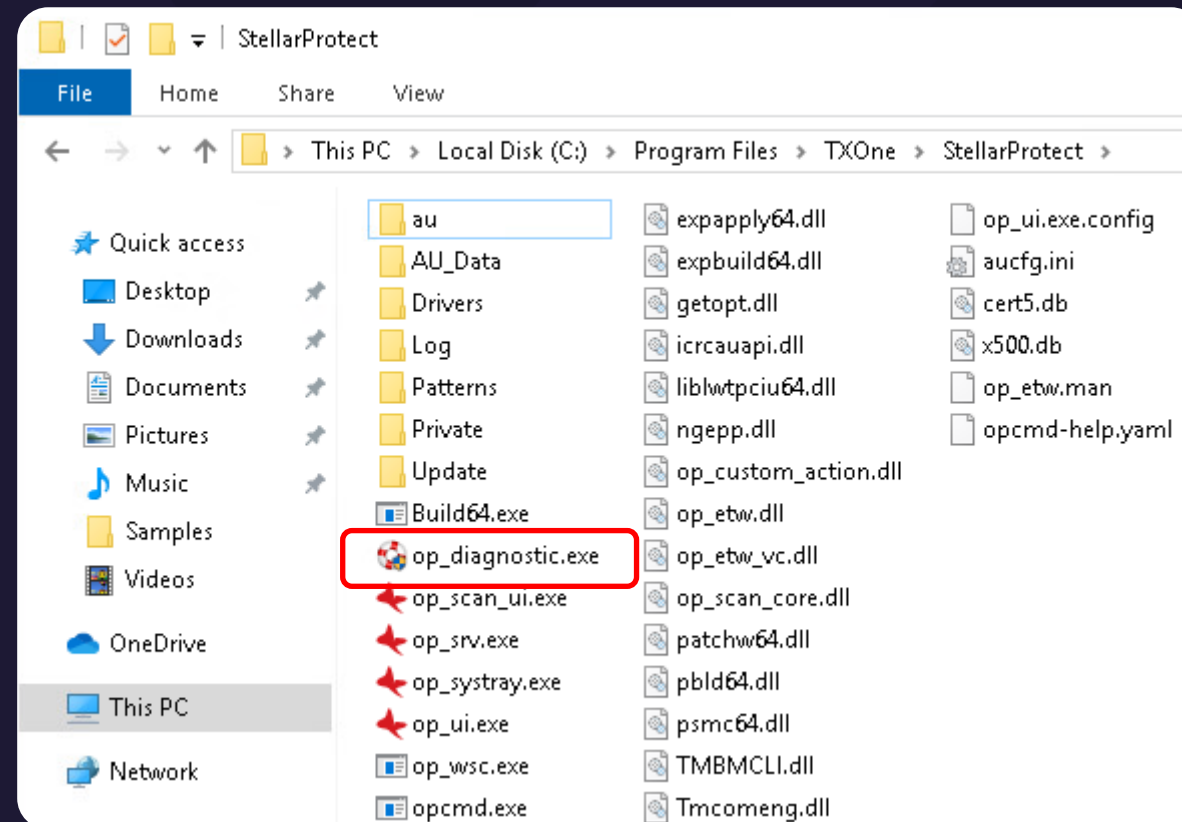


Stellar Agent Debug Log Collection *Post* Installation

Please notice some features in StellarOne are not available in the older version

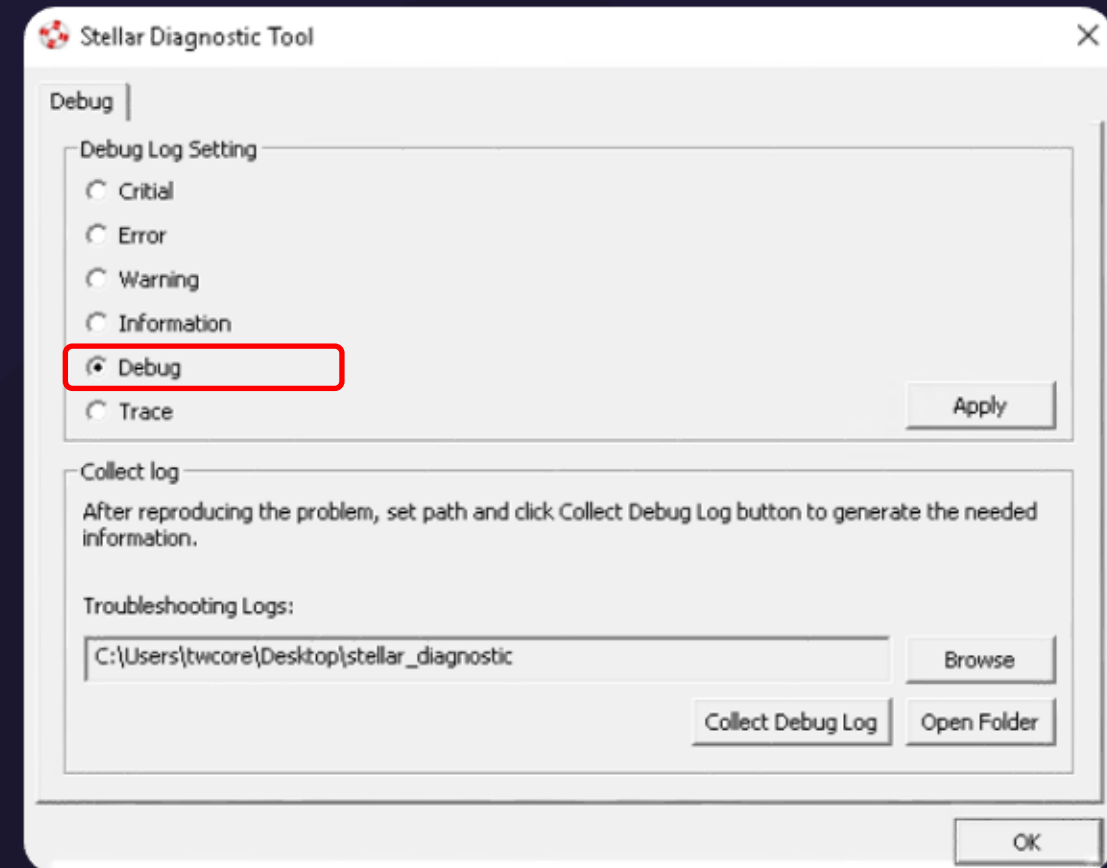
Collect StellarProtect debug log with **op_diagnostic.exe**

- op_diagnostic is the built-in diagnostic tool for StellarProtect
- Path:<installation_dir>\op_diagnostic.exe



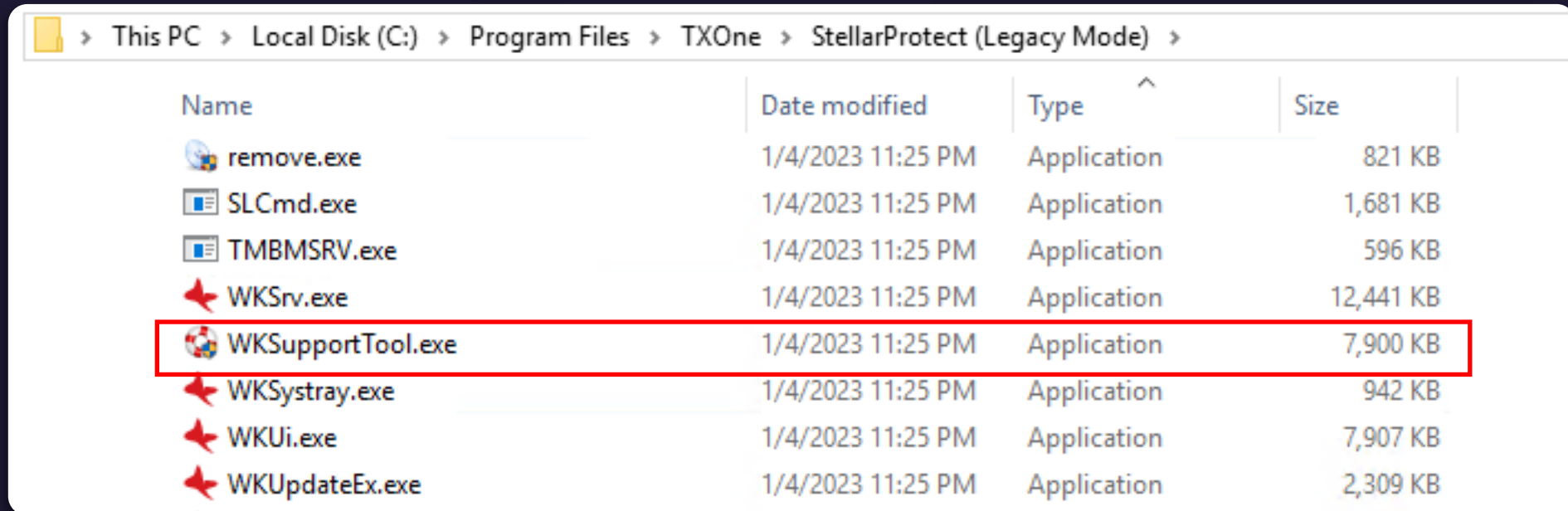
Collect StellarProtect debug log with `op_diagnostic.exe`

- How to collect StellarProtect logs
 1. Select “Debug”
 2. Click “Apply”
 3. Reproduce the issue
 4. Click “Collect Debug Log”
 5. After the “Troubleshooting data collected” message box pop up, click “Open Folder” to get the archived zip file.
 6. Set debug level back to “Error”, click “Apply”



StellarProtect Legacy Mode debug log from **WKSupportTool.exe**

- The WKSupportTool is the built-in diagnostic tool for StellarProtect(Legacy Mode)
 - Path:<installation_dir>\WKSupportTool.exe



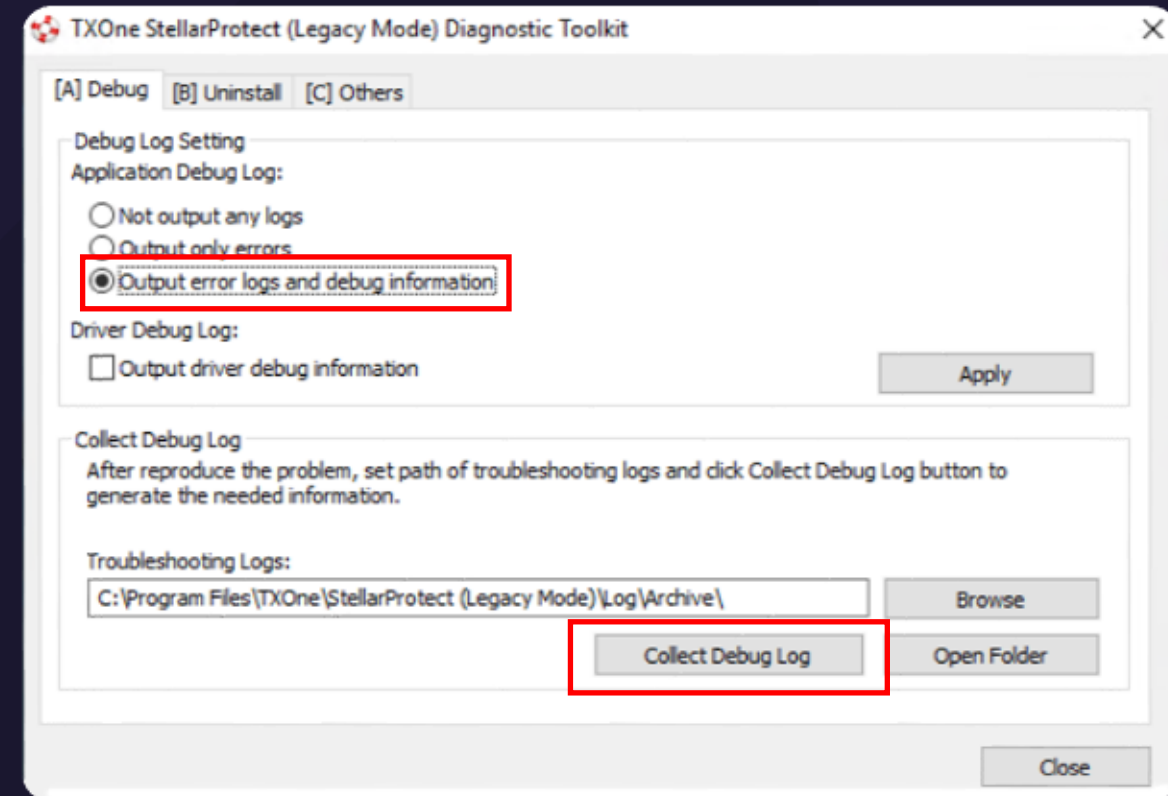
The screenshot shows a Windows File Explorer window with the address bar set to "This PC > Local Disk (C:) > Program Files > TXOne > StellarProtect (Legacy Mode)". The main area displays a list of files and folders. The file "WKSupportTool.exe" is highlighted with a red rectangle. The list includes the following files:

Name	Date modified	Type	Size
remove.exe	1/4/2023 11:25 PM	Application	821 KB
SLCmd.exe	1/4/2023 11:25 PM	Application	1,681 KB
TMBMSRV.exe	1/4/2023 11:25 PM	Application	596 KB
WKSrv.exe	1/4/2023 11:25 PM	Application	12,441 KB
WKSupportTool.exe	1/4/2023 11:25 PM	Application	7,900 KB
WKSystray.exe	1/4/2023 11:25 PM	Application	942 KB
WKUi.exe	1/4/2023 11:25 PM	Application	7,907 KB
WKUpdateEx.exe	1/4/2023 11:25 PM	Application	2,309 KB

StellarProtect Legacy Mode debug log from WKSupportTool.exe

- How to collect StellarProtect(Legacy Mode) logs

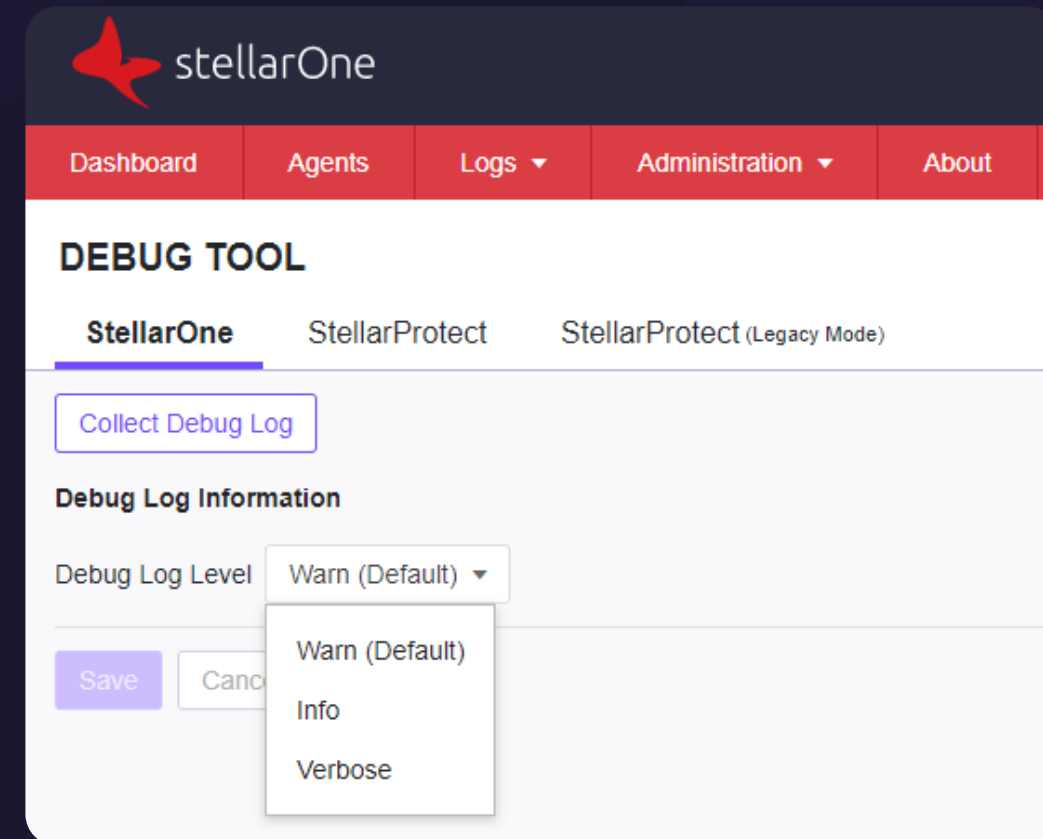
1. Select “Output error logs and debug information”
2. (optional) Check “Output driver debug information”
3. Click “Apply”
4. Reproduce the issue
5. Click “Collect Debug Log”
6. After the “Troubleshooting data collected” message box pop up, click “Open Folder” to get the archived zip file.
7. Set debug level back to “Output only errors”, click “Apply”



Troubleshooting Pages (/rdqa)

StellarOne Troubleshooting Page (/rdqa)

- StellarOne Debug Log can be collected from https://StellarOne_Server/rdqa
 - Select the Debug Log Level: Verbose
 - Click on the “**Save**” button
 - Reproduce the issue (server and/or agent)
 - Click the “**Collect Debug Log**” button
 - Switch the debug log level back to Warn(Default)
 - Click on the “**Save**” button
- Note:
The StellarOne logs are in "journal" (binary) format.



StellarProtect REMOTE agent log collection

- Administrator could also collect StellarProtect debug log remotely via StellarOne management console
 - https://StellarOne_Server/rdqa

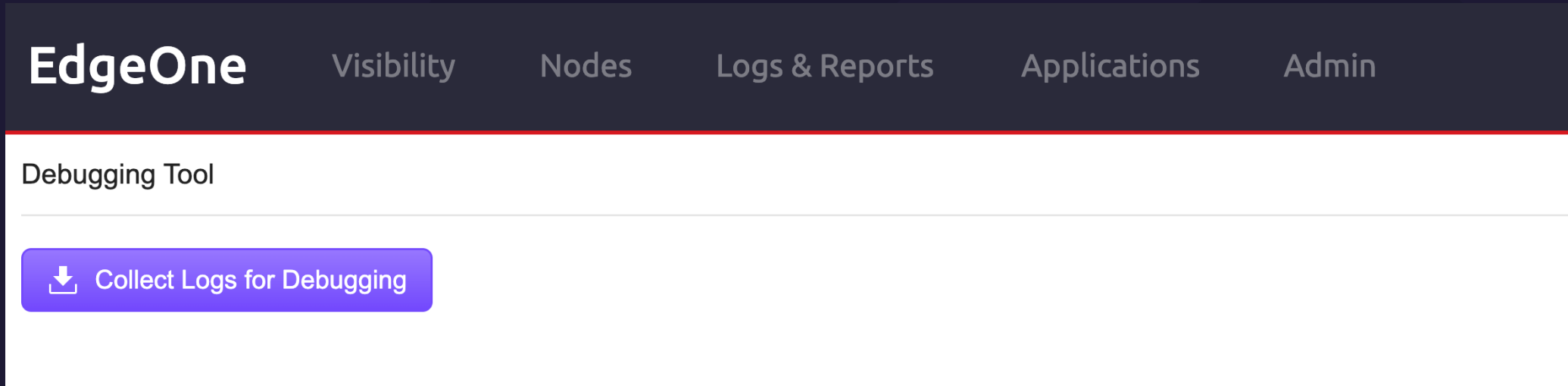
The screenshot shows the StellarOne management console interface. The top navigation bar includes 'Dashboard', 'Agents', 'Logs', 'Administration', and 'About'. The user is logged in as 'admin (admin)'. The 'txOne networks' logo is in the top right corner.

The main section is titled 'DEBUG TOOL'. It features a search bar for 'Endpoint' and a dropdown menu for 'StellarProtect' (Legacy Mode). Below this, there are buttons for 'Check Connection', 'Debug Logging', and 'Purge Collected Information'. The 'Debug Logging' dropdown is open, showing 'Enable' and 'Disable' options. A table of endpoints is displayed below, with columns for 'Endpoint', 'Description', 'IP Address', 'Operating System', 'Last Connection', and 'Debug Logging Status'. The table contains one entry: 'QQ-STELLAR' with IP '192.168.23.24' and status 'Debug logging disabled'. A 'Collect Debug Information' button is located at the bottom right of the table.

Numbered callouts highlight the following elements:

- 1: StellarProtect agent selection
- 2: QQ-STELLAR endpoint selection
- 3: Debug Logging dropdown menu
- 4: Collect Debug Information button

EdgeOne Troubleshooting Page (/#/rdqa)

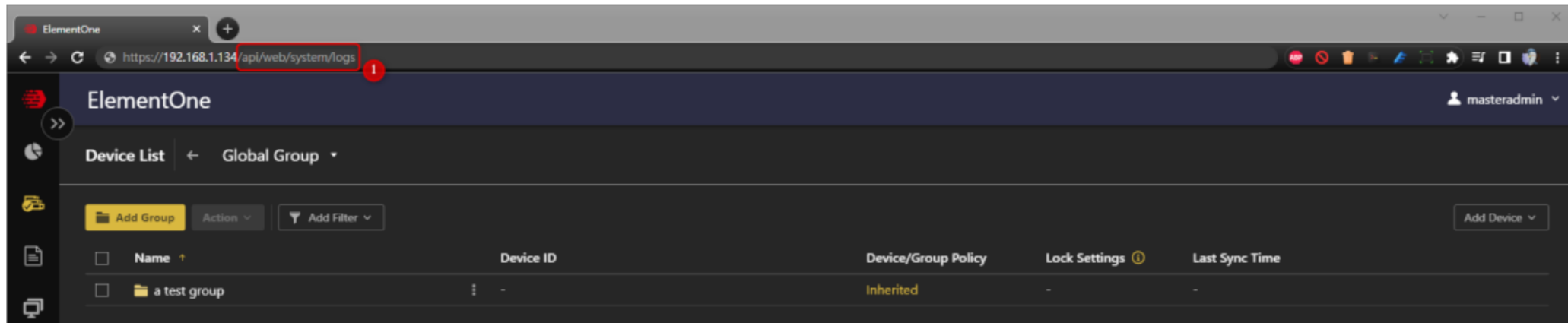


- Note:
The EdgeOne logs are in clear-text.

ElementOne (/api/web/system/logs)

This file can help us determine if the root cause of the issue is caused by ElementOne service.

1. Log in to the ElementOne Web UI with master admin or admin user account.
After login, modify URL to `'https://[ElementOne_IP]/api/web/system/logs'` then press 'Enter'.



Note:
the dlog file is encrypted
and intended for TxOne support

Summary of troubleshooting tools

- **StellarSupport.exe** for troubleshooting when the installation fails.
Can be found in the in the Installer Package.
It is a wrapper around op_diagnostic.exe and WKSupportTool.exe
 - With StellarProtect: you can run this tool when installation fails
 - With StellarProtect LegacyMode: first change parameters in StellarSetup.ini in the installation folder). Then run this StellarSupport.exe tool
- **op_diagnostic.exe** is the built-in diagnostic tool for StellarProtect
Path:<installation_dir>\op_diagnostic.exe
- **WKSupportTool.exe** is the built-in diagnostic tool for StellarProtect Legacy Mode
Path:<installation_dir>\WKSupportTool.exe
- StellarOne troubleshooting:
 - See cli (e.g. env ls -> Status Fail -> No external disk detected
 - [https://\[stellarOne\]/rdqa](https://[stellarOne]/rdqa) -> Debug Log information
 - These logs are encoded in "journal" format (!)



Let's Keep The Business Growing

